

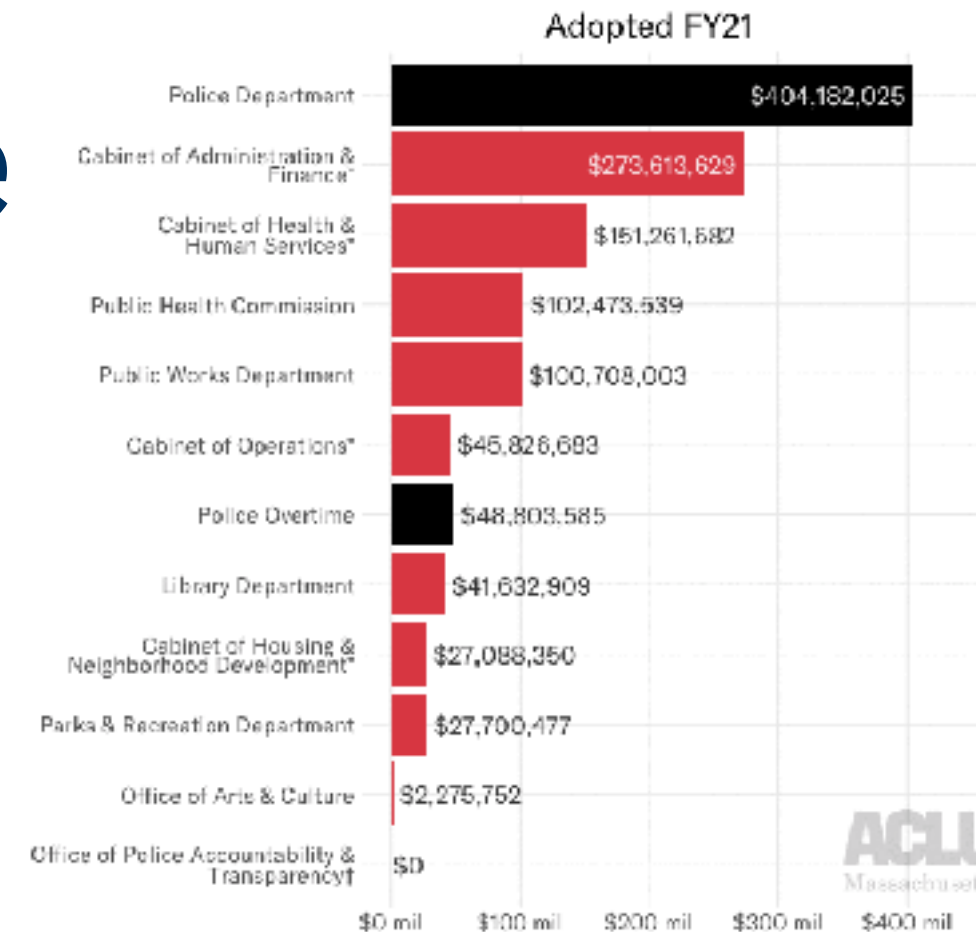
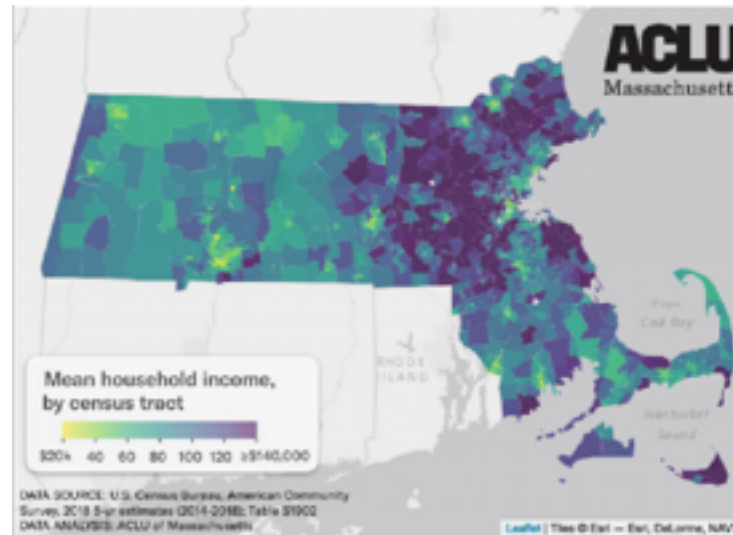
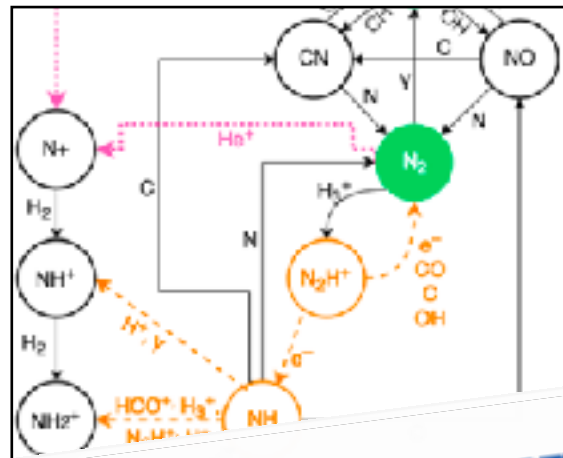
Effective Data Visualization

INFO 290 - Data Products for Public Impact

October 14, 2025

Berkeley SCHOOL OF
INFORMATION

Data viz experience



ACLU
Massachusetts

BOSTON POLICE DEPARTMENT COURT OVERTIME

A report by the ACLU of Massachusetts, Progressive Massachusetts, Councilor Ricardo Lopez, and the EJ Spark Lab!

Submitted in testimony to the Boston City Council Committee on Waged and Minor Employees on October 20, 2020 & #1003 - November 26, 2020

SUMMARY

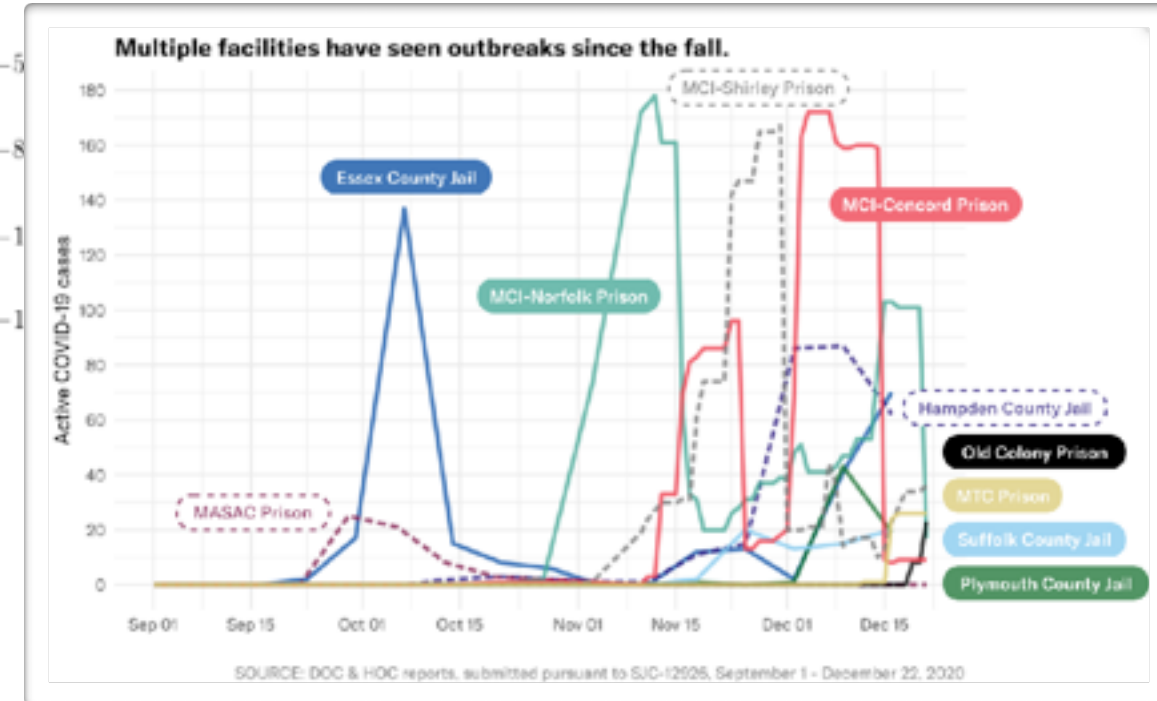
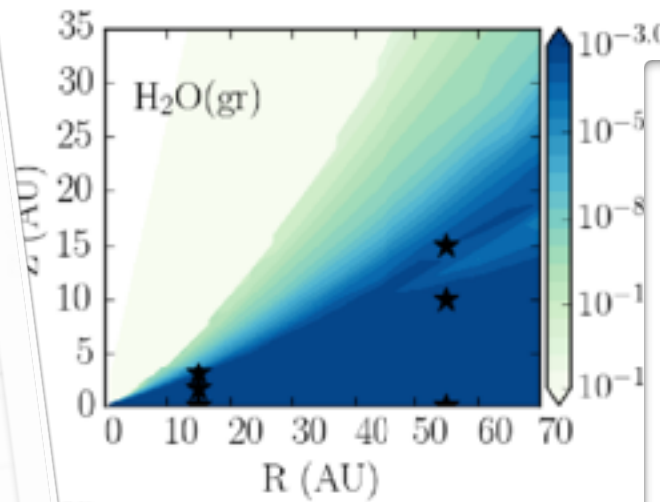
We present findings resulting from data analysis of Boston police court overtime records from 2014 to 2019. These data support five observations about the nature of court overtime practices in the Boston Police Department:

1. Transfers have minimized court appearances paid to warrant.
2. Officers are increasingly exploiting court overtime.
3. Excessive overtime operates as an incentive.
4. Not all court appearances are created equal.
5. BPD accountability requires greater transparency.

While we share these highlights here, the entire portfolio of interactive visualizations is publicly available [here](#).

1. THE 4-HR MINIMUM COURT APPEARANCE POLICY IS WASTEFUL

A major aspect of Boston police overtime policy that has come under scrutiny in recent months is the practice of paying officers four hours of overtime pay for any court appearance lasting four hours or less. This practice is defined in Article 9, Section 1 of the City's Collective Bargaining Agreement with the Boston Police Union. The Association of Boston Police Officers (ABPO) has argued that this practice is wasteful and that officers should be paid for the actual time spent in court, rather than the full four hours. The report argues that this policy is wasteful because it encourages officers to work overtime for minimal pay, and it is not based on the actual time spent in court. The report also argues that the policy is wasteful because it encourages officers to work overtime for minimal pay, and it is not based on the actual time spent in court.



Today we will explore...

- ▶ why data visualization is important
- ▶ how to identify the different components and types of data visualizations
- ▶ how to interpret visualizations critically
- ▶ how to intentionally tell a story by visualizing data

**To learn more,
consider taking
INFO 247:
Information
Visualization &
Presentation**

***All resources shared
today are available at:***

*[https://github.com/
laurenmarietta/
effective-data-viz-2025](https://github.com/laurenmarietta/effective-data-viz-2025)*

**This presentation is
adapted from a
workshop designed
by Kira Tebbe in
February 2020**



Workshop outline

- ▶ What is data visualization 🧠
- ▶ Anatomy of a data visualization
- ▶ Types of data visualization
- ▶ Roast! This! Viz! 🧠
- ▶ ✨ 10 Best Practices ✨
- ▶ Improve! This! Viz!
- ▶ Exercises 🧠
- ▶ Interactivity, inspiration, and learning more

*What does the term
“data visualization”
mean to you? Why is it
important?*



*Why is data
visualization
important?*

data = information

data visualization = communication

good data visualization = effective communication

bad data visualization = miscommunication

What do mediocre visualizations lead to?

- ▶ confusion and frustration
- ▶ loss of attention
- ▶ poor impression of work quality
- ▶ misinterpretation of findings

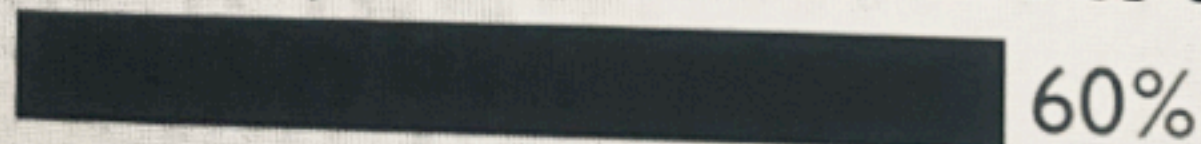
What do Tory voters think?

Q Given the choice, would you prefer that Boris Johnson was still Prime Minister in a year's time, or would you prefer someone else?

Johnson to remain Prime Minister



I would prefer someone else to be PM



Don't know



Source: YouGov, June 22-23.
1,671 adults. Results show
those who voted
Conservative in 2019.

Source: Will Bailey-Watson on [Twitter](#)



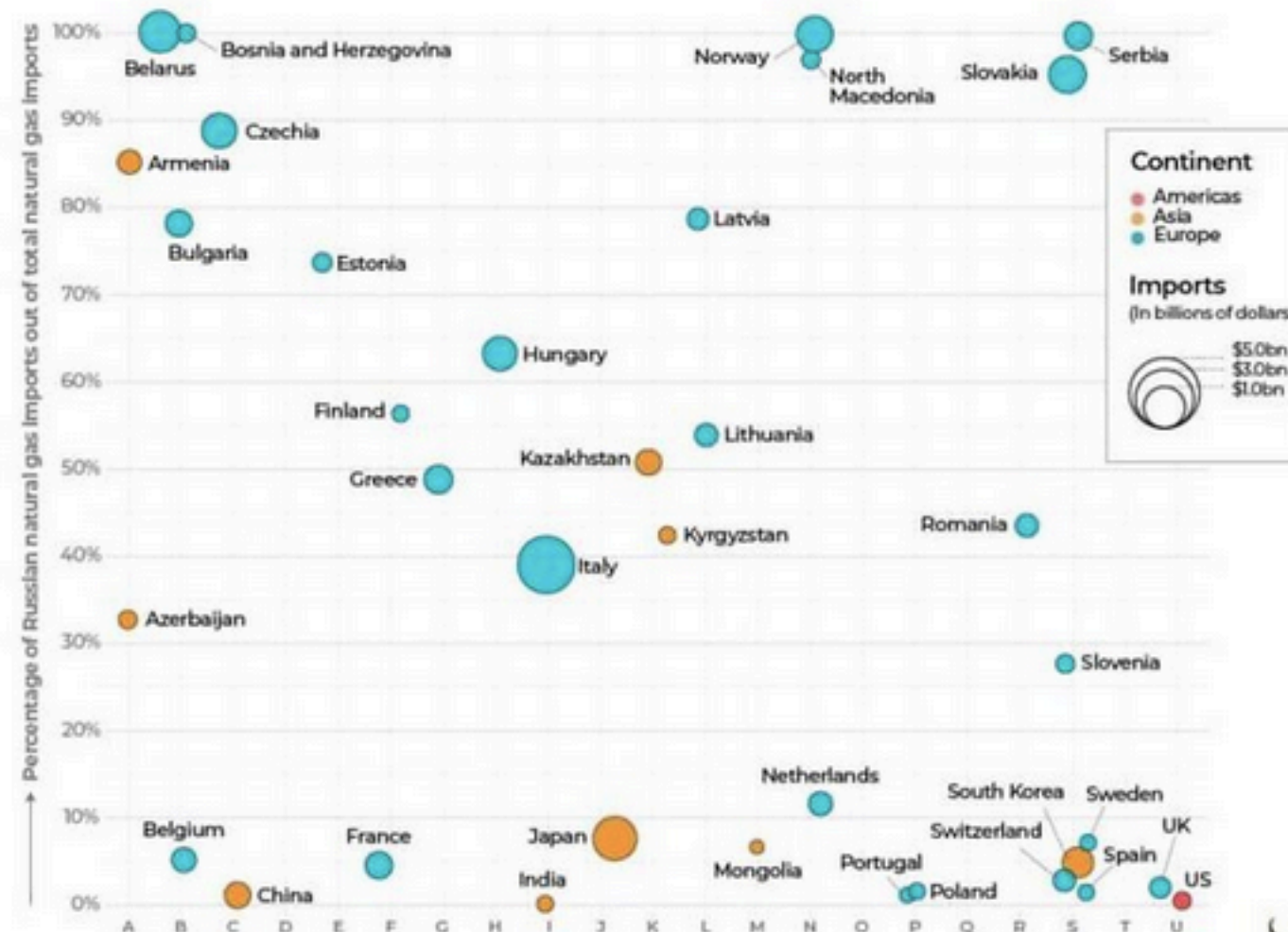
Ben Stanley
@BDStanley

Incredible x-axis here.

ENERGY

Which countries directly import the most Russian natural gas?

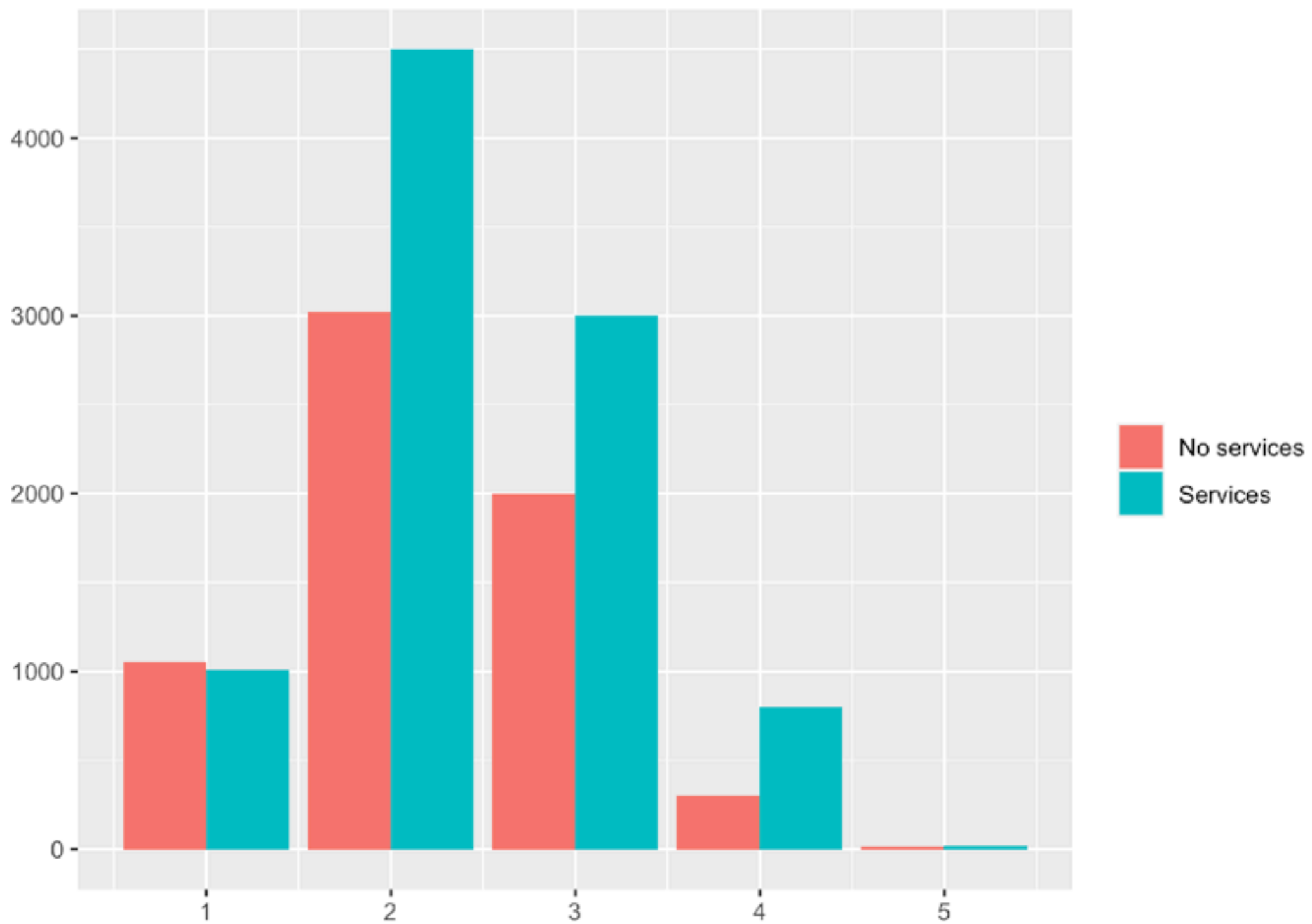
Russia is the world's third highest exporter of natural gas after Australia and Qatar.
In 2019, 37 countries bought Russian gas worth \$24.5bn.



Countries not on this graphic did not import gas from Russia in 2019.
Natural gas includes gas in gaseous state and liquefied.
Source: OEC | 2019

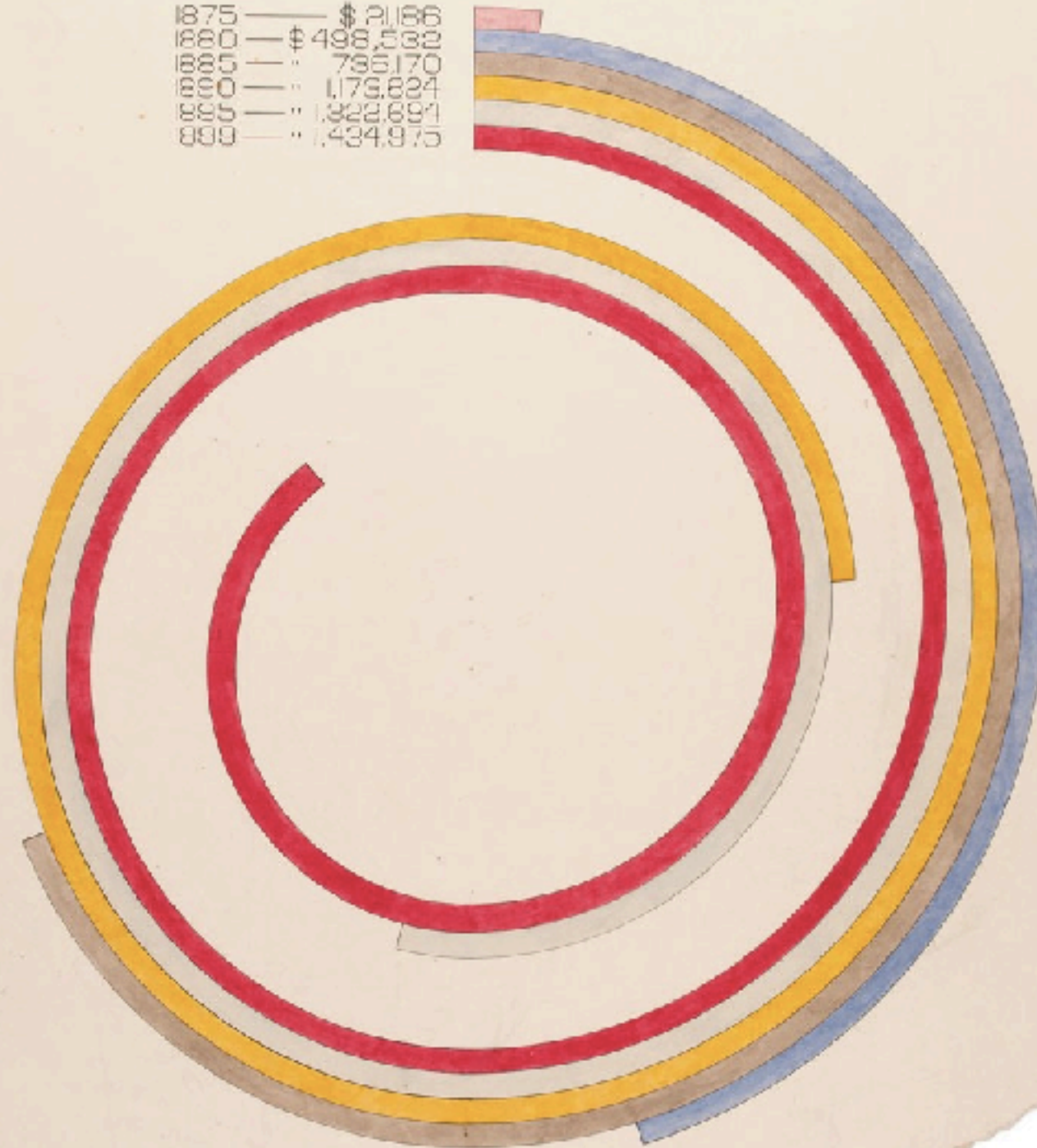
@AJLabs ALJAZEERA

Source: Ben Stanley on [Twitter](#)



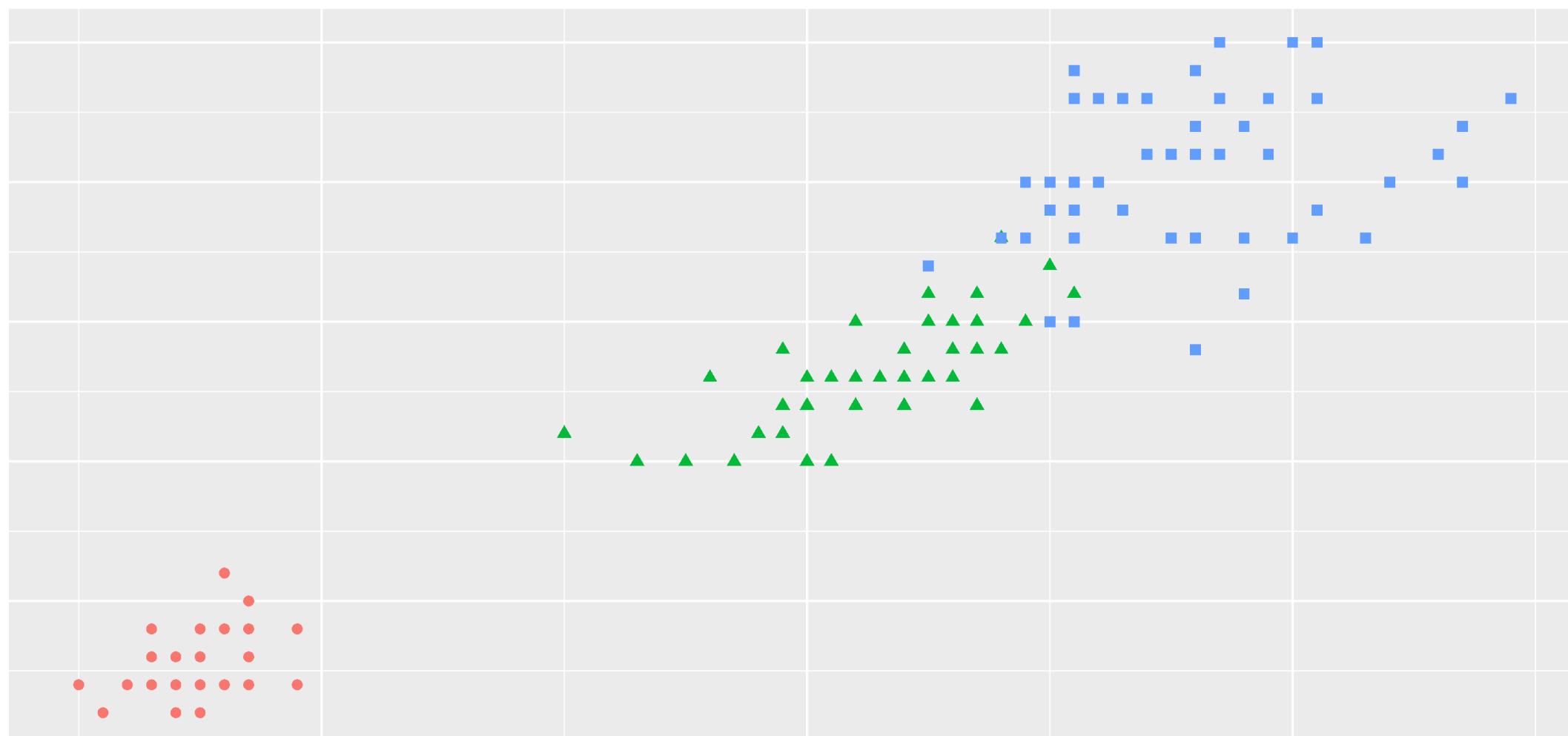
ASSESSED VALUE OF HOUSEHOLD AND KITCHEN FURNITURE OWNED BY GEORGIA NEGROES.

1875	—	\$ 21,186
1880	—	\$ 498,532
1885	—	735,170
1890	—	1,173,624
1895	—	1,322,681
1900	—	1,434,975

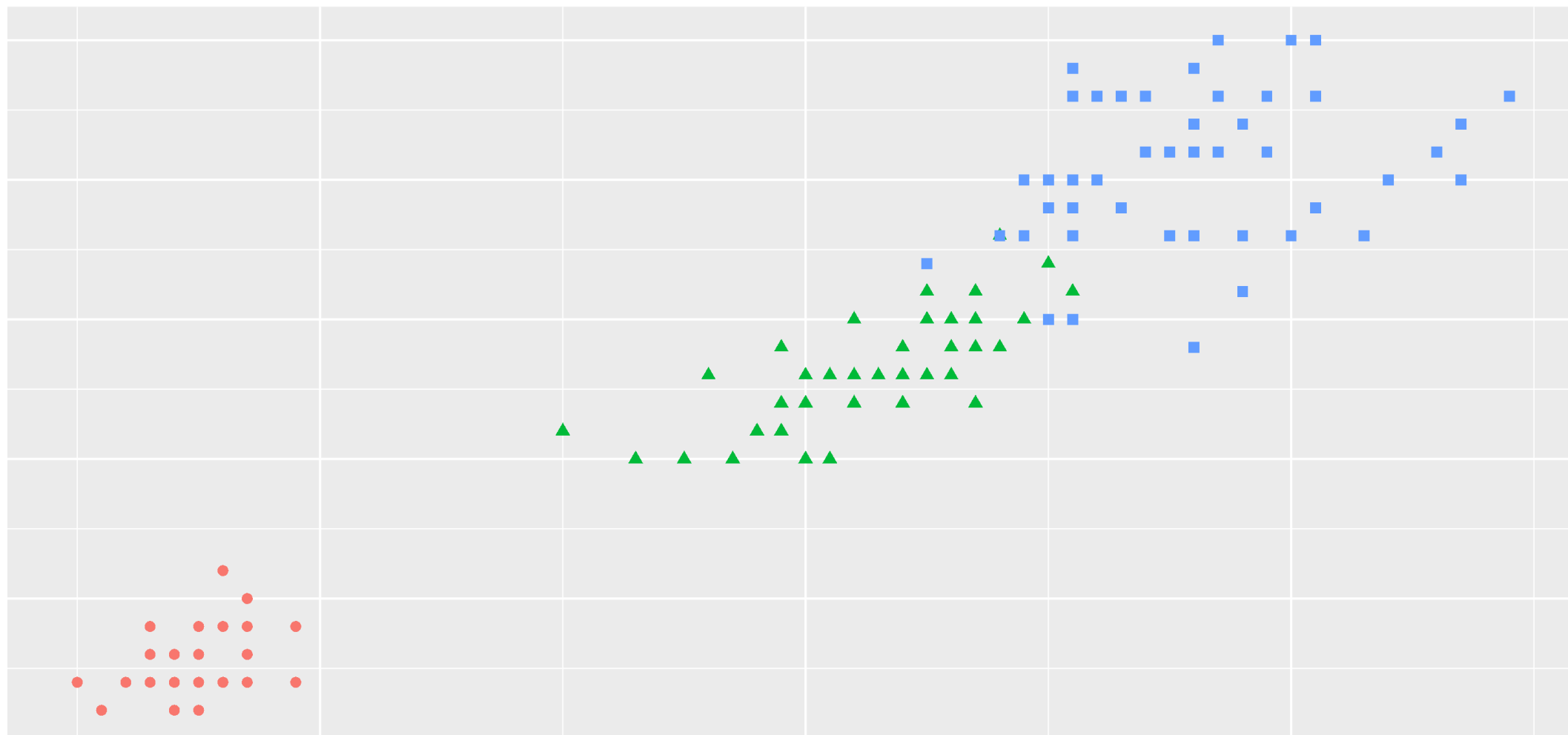


Source: W.E.B. Du Bois, ca. 1900

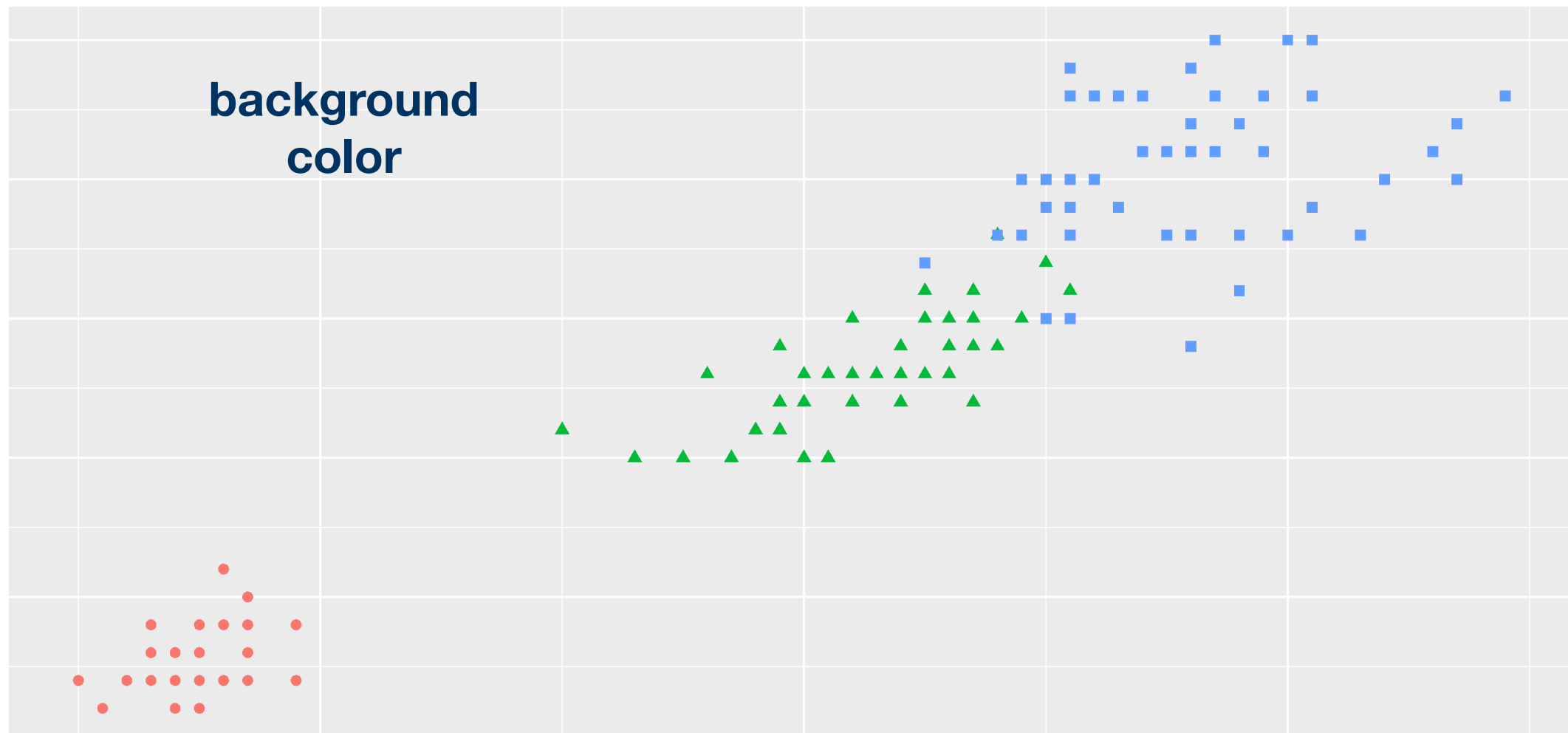
Anatomy of a data visualization



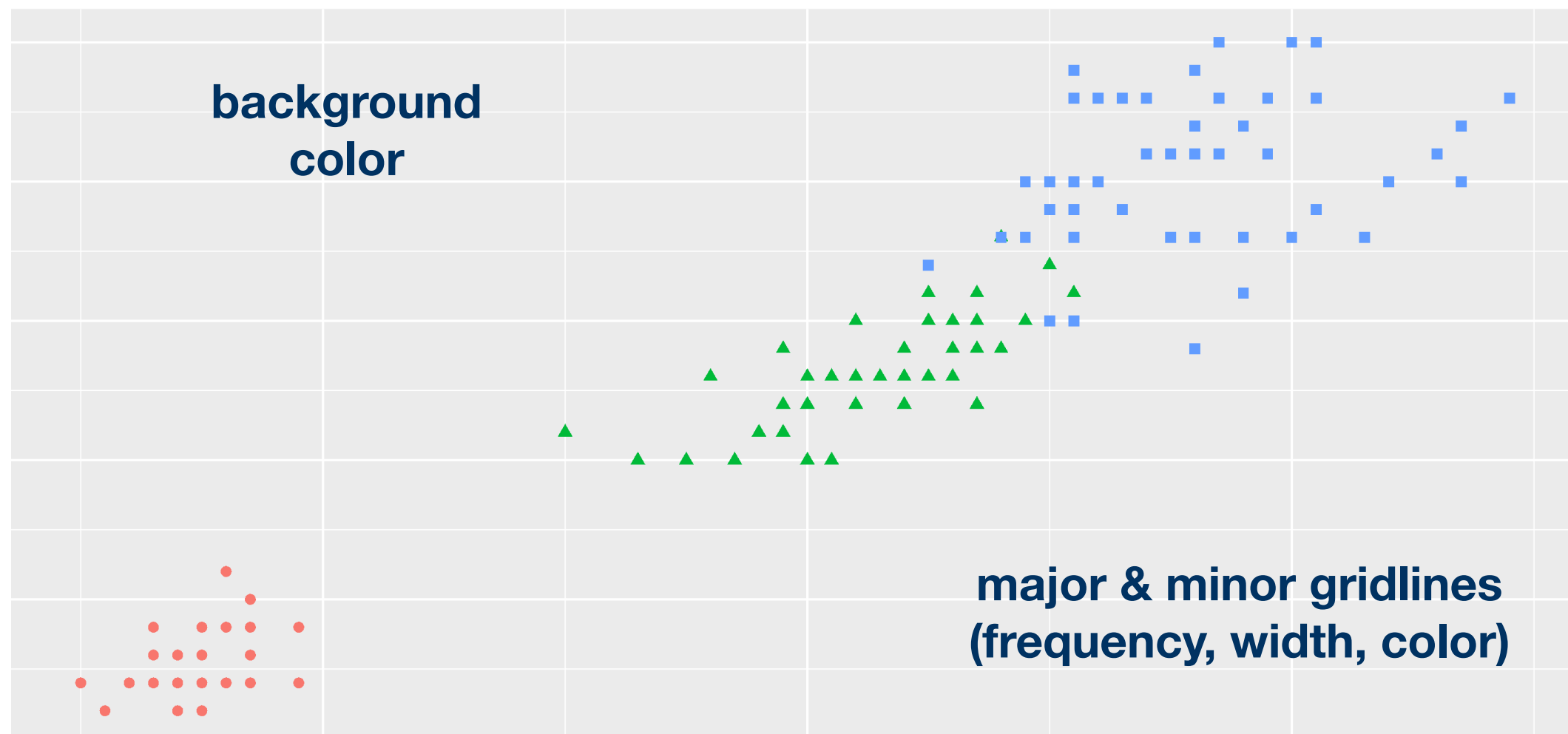
plot type & aesthetics (colors, sizes, shapes, opacity)



plot type & aesthetics (colors, sizes, shapes, opacity)



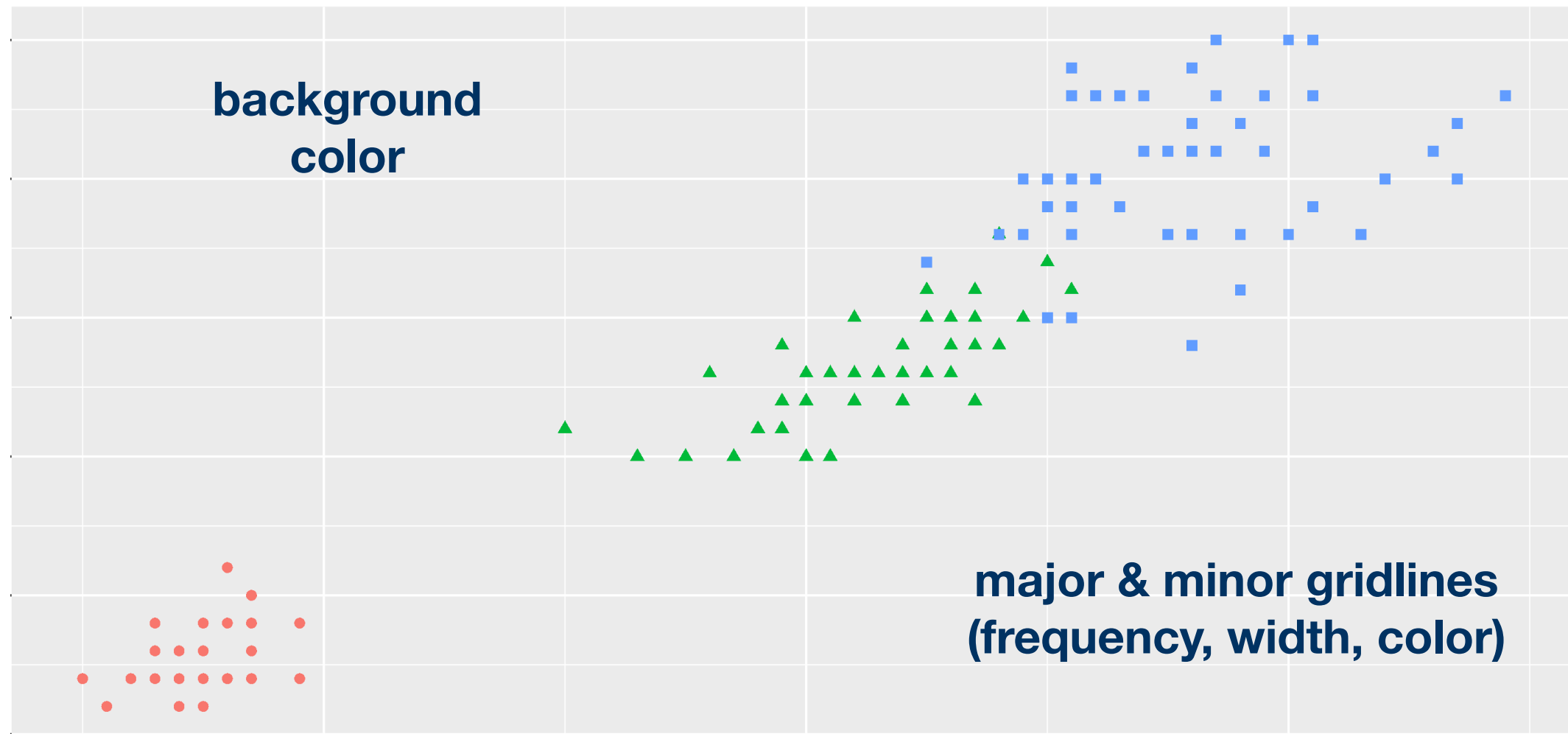
plot type & aesthetics (colors, sizes, shapes, opacity)



title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)

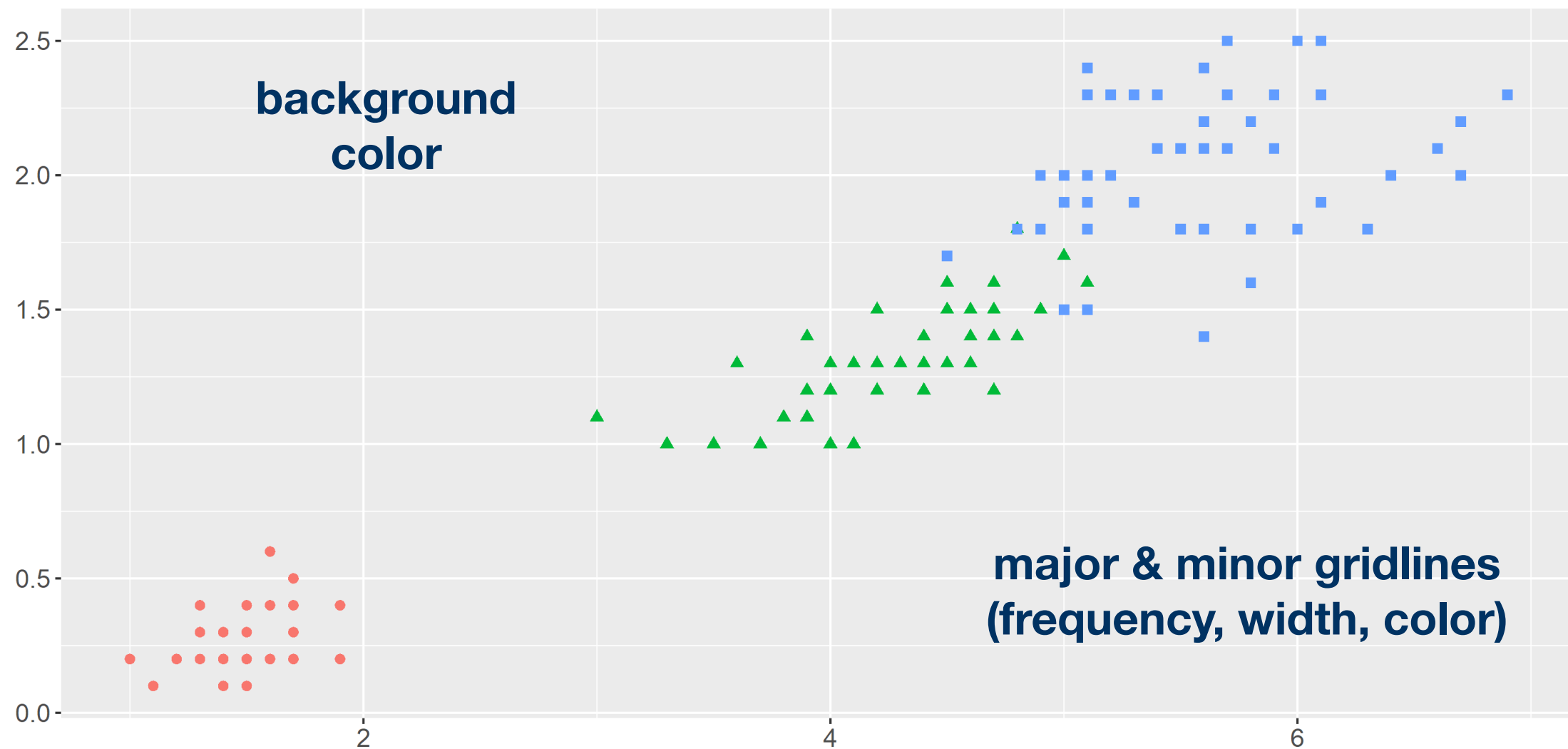
Plant petal sizes are grouped by species
And Virginica are the biggest!



title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)

Plant petal sizes are grouped by species
And Virginica are the biggest!

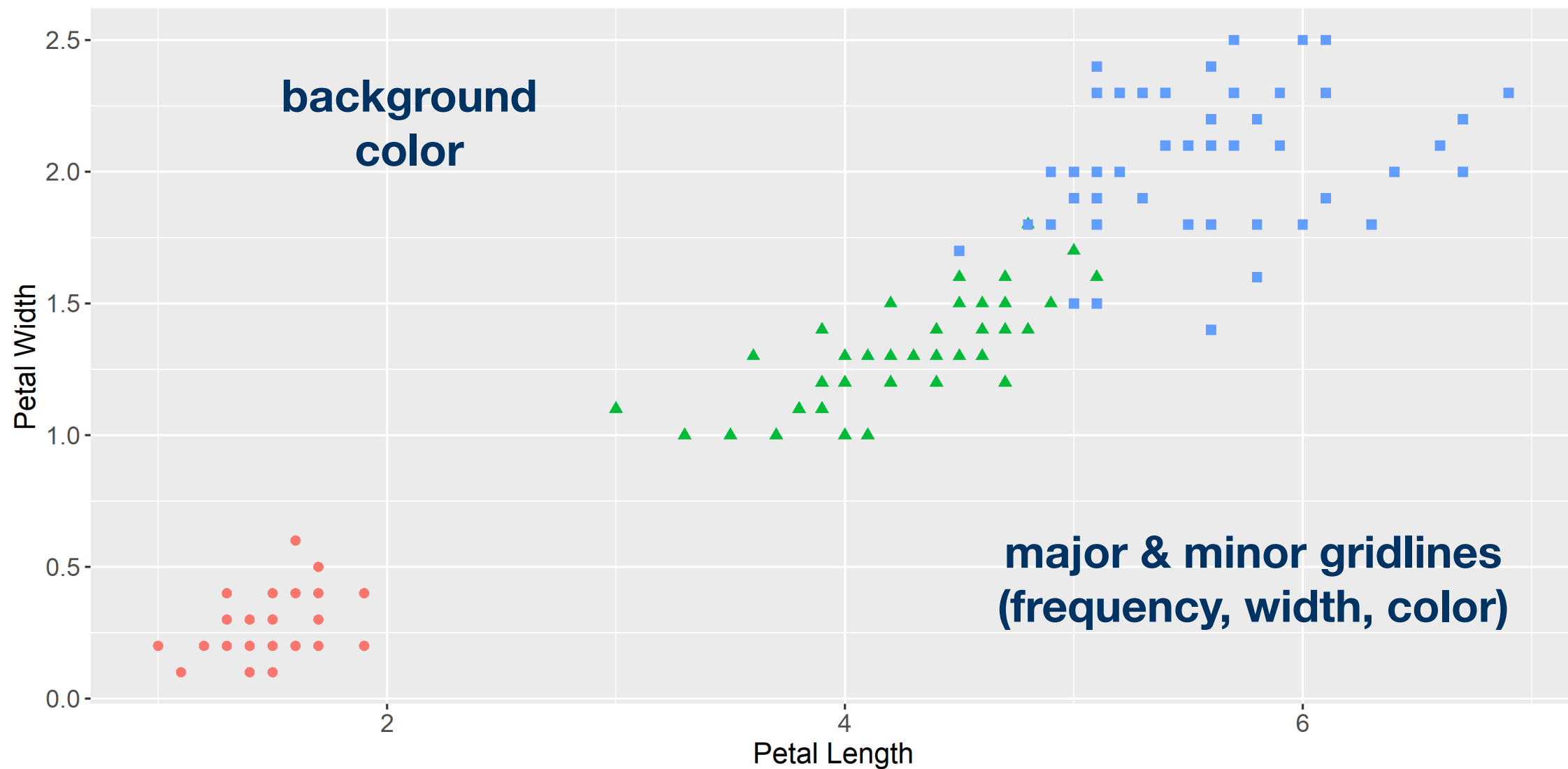


X & Y tick marks
(frequency, size, font, color, angle)

title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)

Plant petal sizes are grouped by species
And Virginica are the biggest!



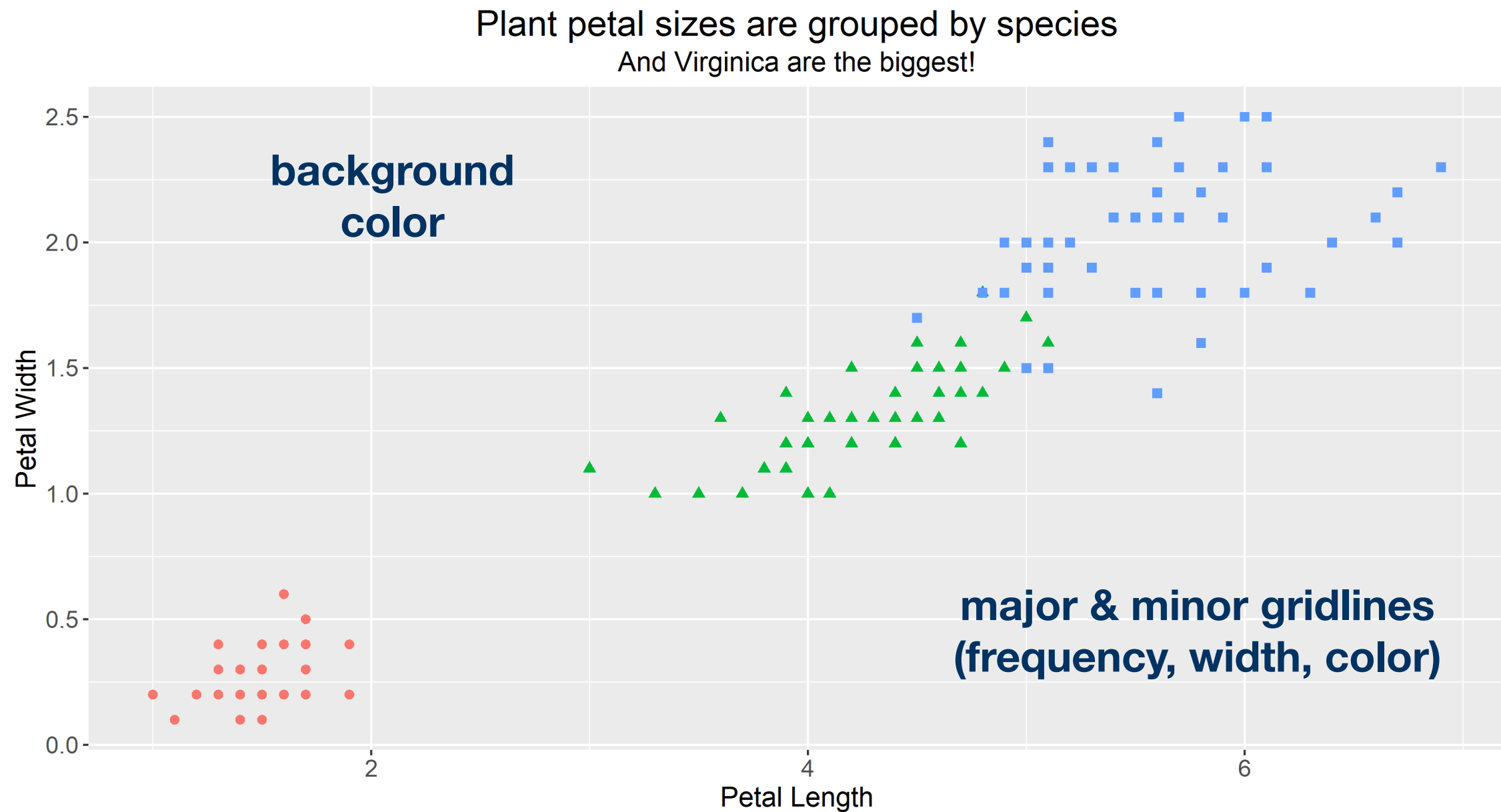
X & Y axis labels
(size, font, color, angle)

X & Y tick marks
(frequency, size, font, color, angle)

title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)

legend & legend title
(location, size, font, color, existence)



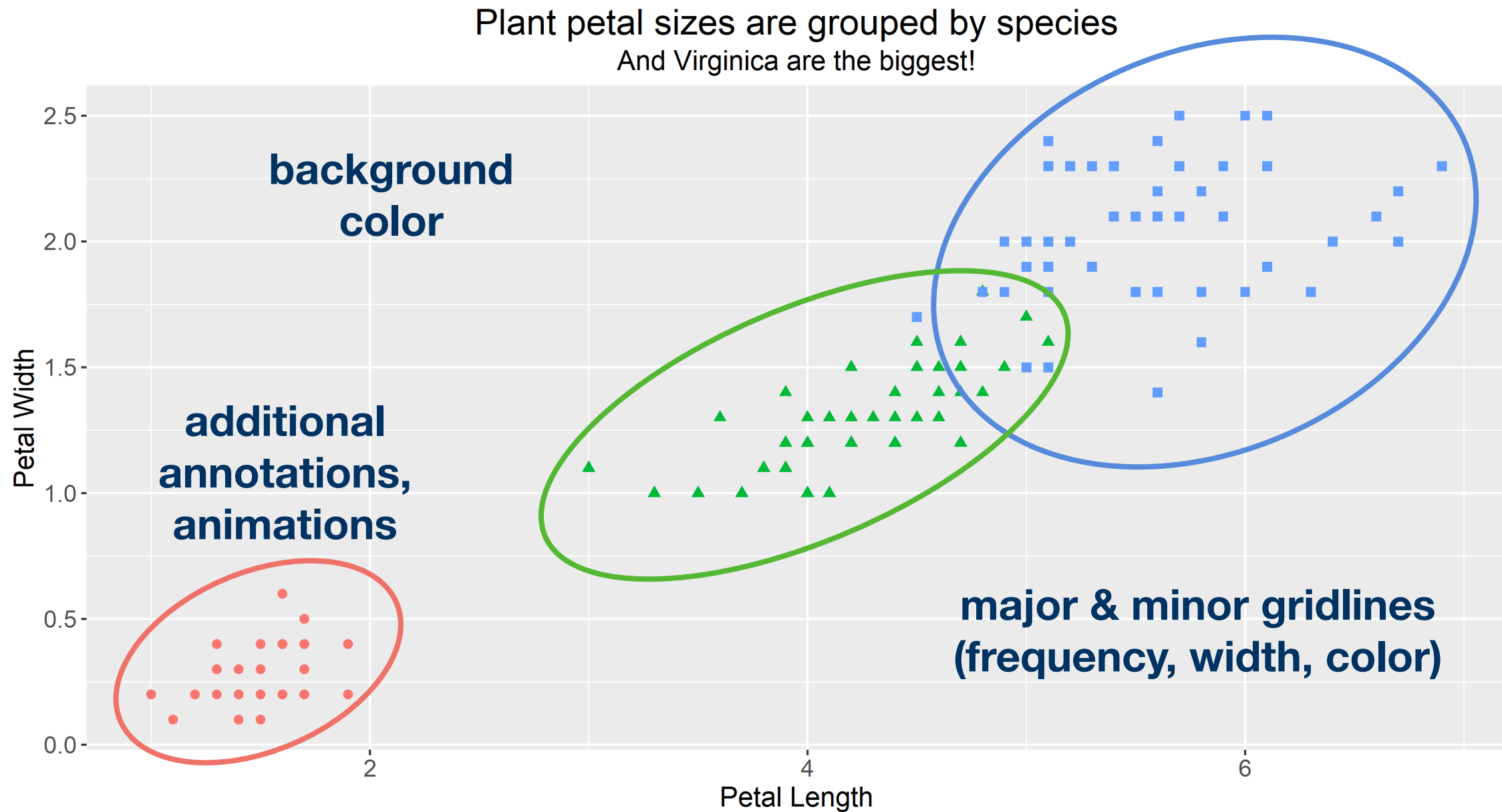
X & Y axis labels
(size, font, color, angle)

X & Y tick marks
(frequency, size, font, color, angle)

title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)

legend & legend title
(location, size, font, color, existence)

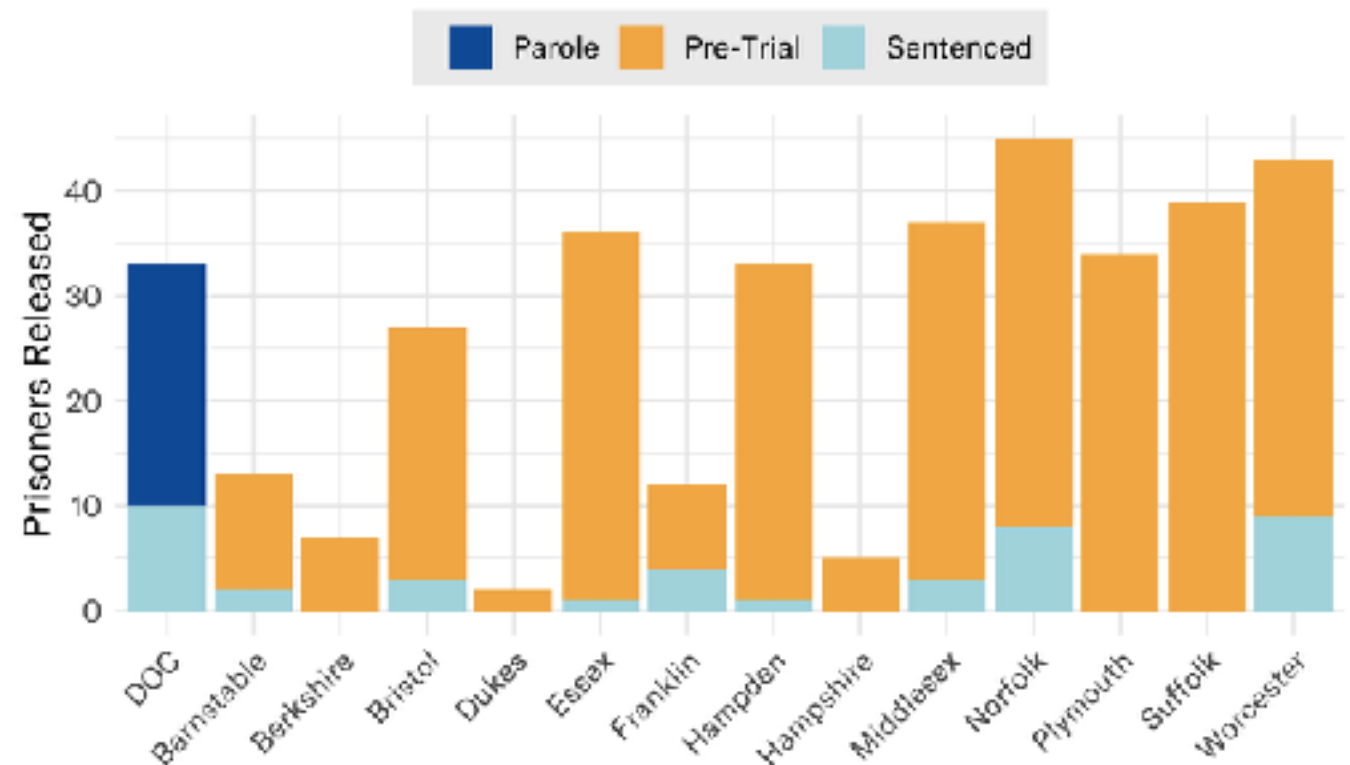


X & Y axis labels
(size, font, color, angle)

X & Y tick marks
(frequency, size, font, color, angle)

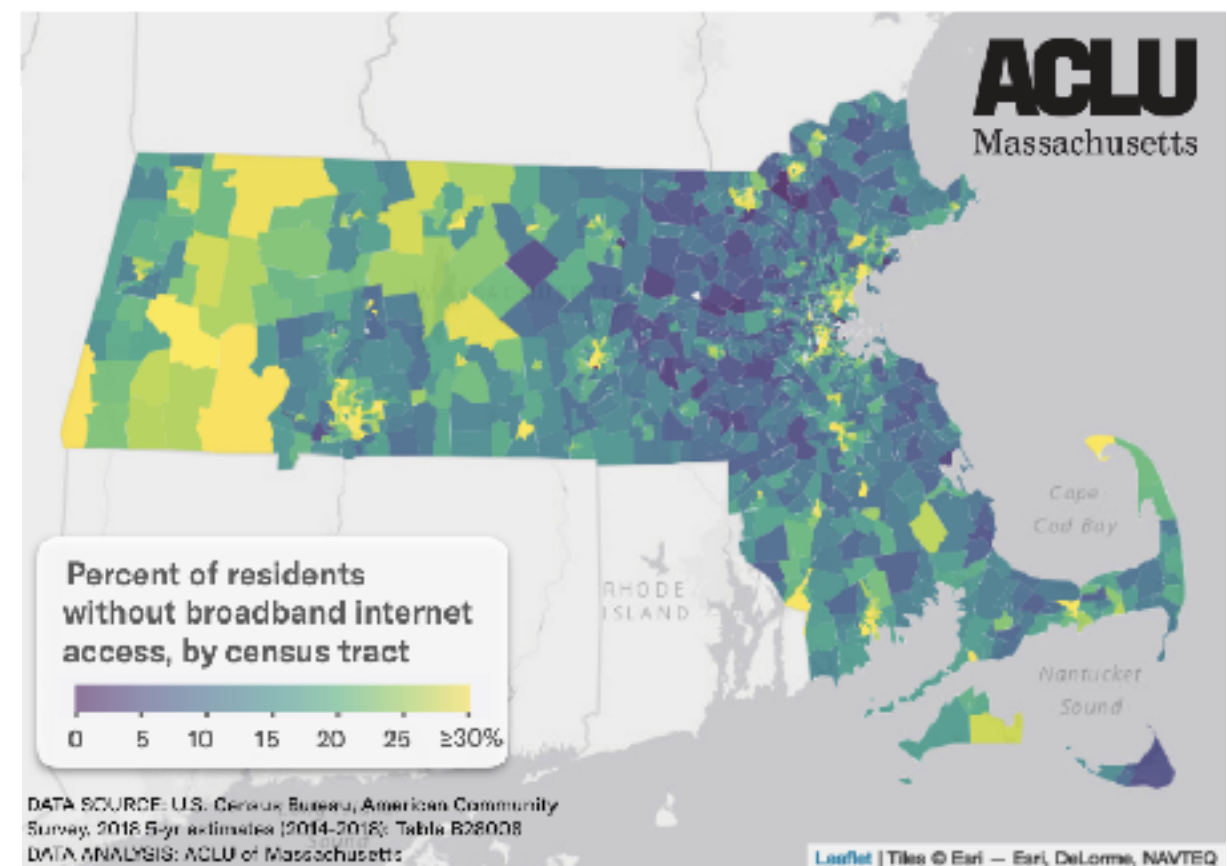
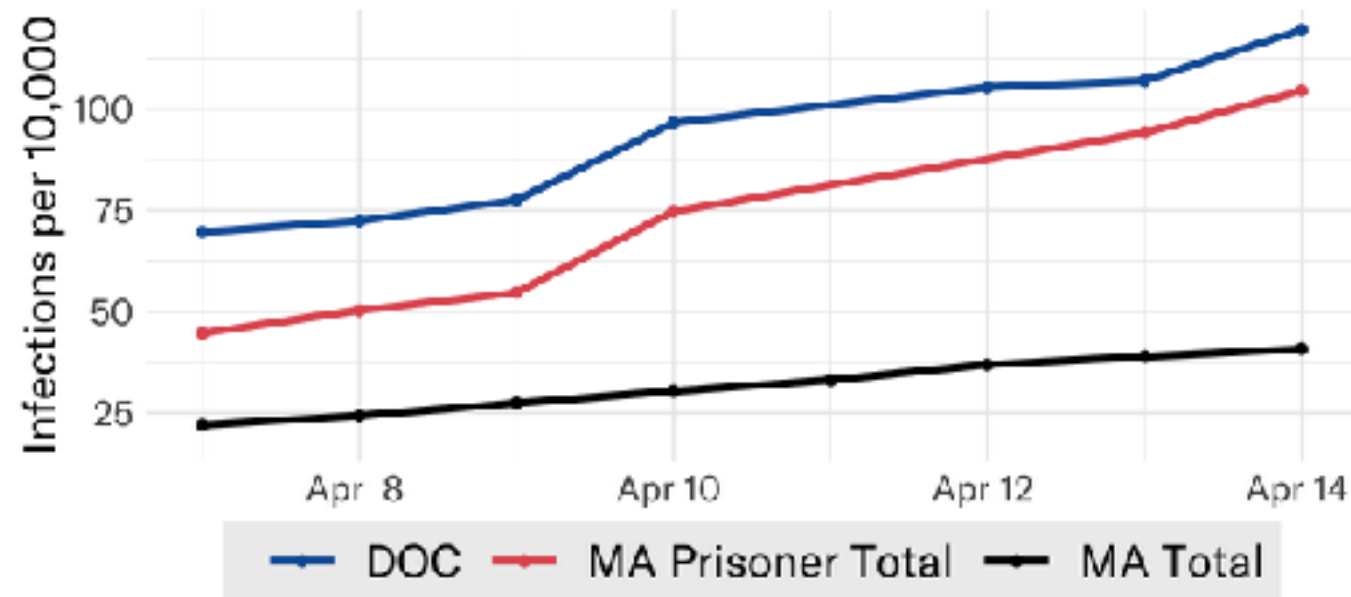
Types of data visualization

Rank	Boston Police Officer	FY20 Earnings	Change From FY19	
			Rank	Earnings
1	Smith, Sean P.	\$365,001	↑ +3	+\$22,378
2	Lee, Waiman	\$360,143	↑ +3	+\$20,628
3	Brown, John M.	\$345,758	↑ +12	+\$28,388
4	Danilecki, John H.	\$339,828	↓ -2	-\$8,127
5	Barrett, Thomas E.	\$339,353	↑ +2	+\$7,275
6	McCormick, Kelley J.	\$336,898	↑ +5	+\$15,922
7	Connolly, Timothy	\$334,077	↑ +18	+\$31,109
8	Sweeney, Steve	\$332,908	↑ +1	+\$3,932
9	Cullity, Patrick J.	\$329,101	↑ +3	+\$9,032
10	Hughes, John C.	\$322,637	↑ +31	+\$31,203
11	Thomas, Terry J.	\$319,001	↑ +9	+\$11,134
12	Pusey, Daniel C.	\$318,886	↑ +24	+\$23,156
13	Webb, Keith A.	\$318,141	↑ +1	-\$200
14	Facay, Lighton B.	\$317,506	↑ +29	+\$28,844
15	Kervin, Timothy M.	\$314,650	↓ -14	-\$40,849
16	Harrington, Mark William	\$314,443	↑ +29	+\$27,098
17	Ciccolo, Robert W.	\$314,026	↑ +9	+\$12,894
18	Kozmiski, Theresa M.	\$313,656	↑ +53	+\$35,552
19	Sullivan, Marc	\$313,437	↓ -2	+\$2,515
20	Gero, Jarrod A.	\$311,210	↑ +19	+\$17,573

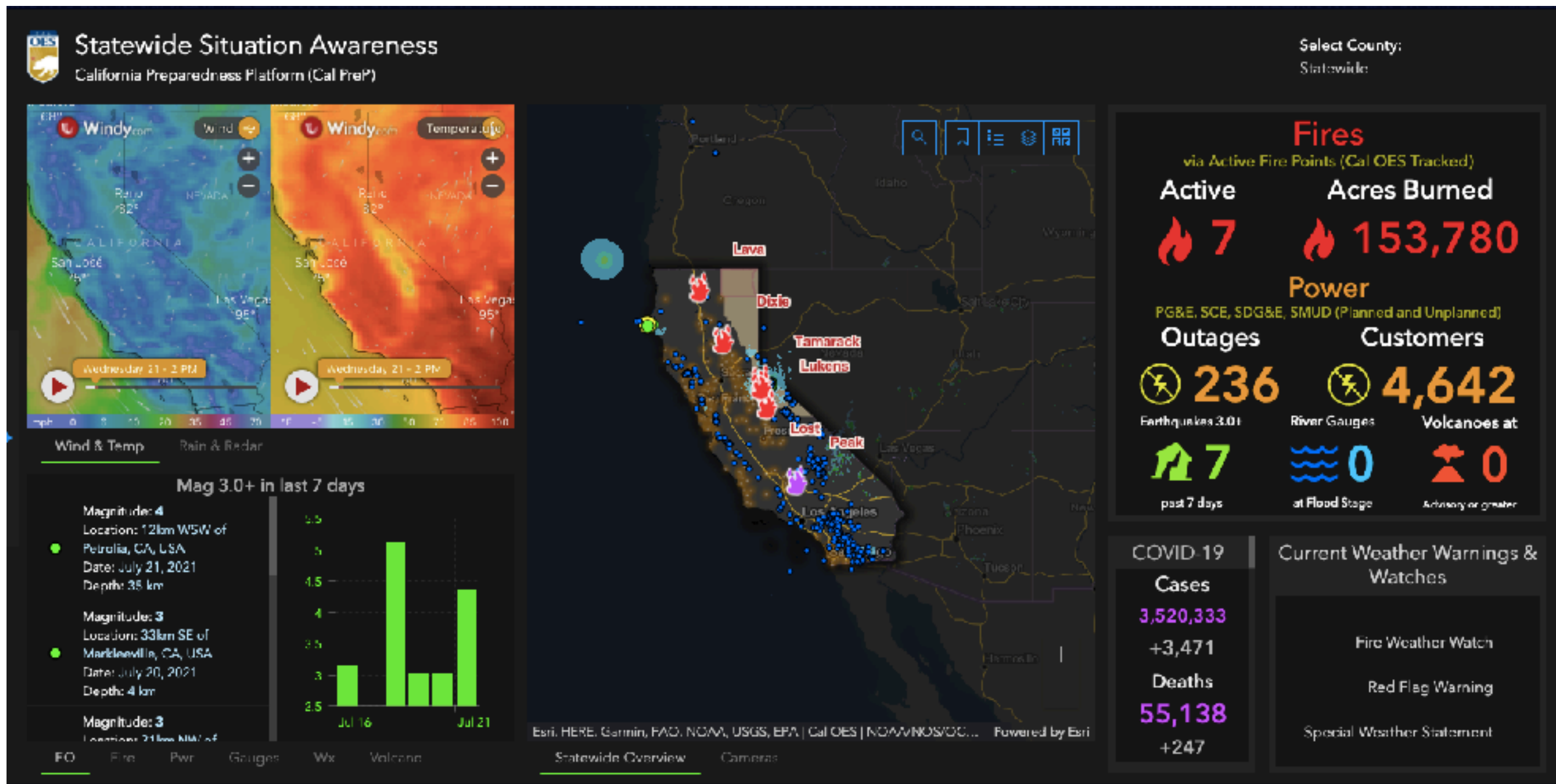


COVID-19 Incidence Rate Over Time

Positive Cases per 10,000 Prisoners

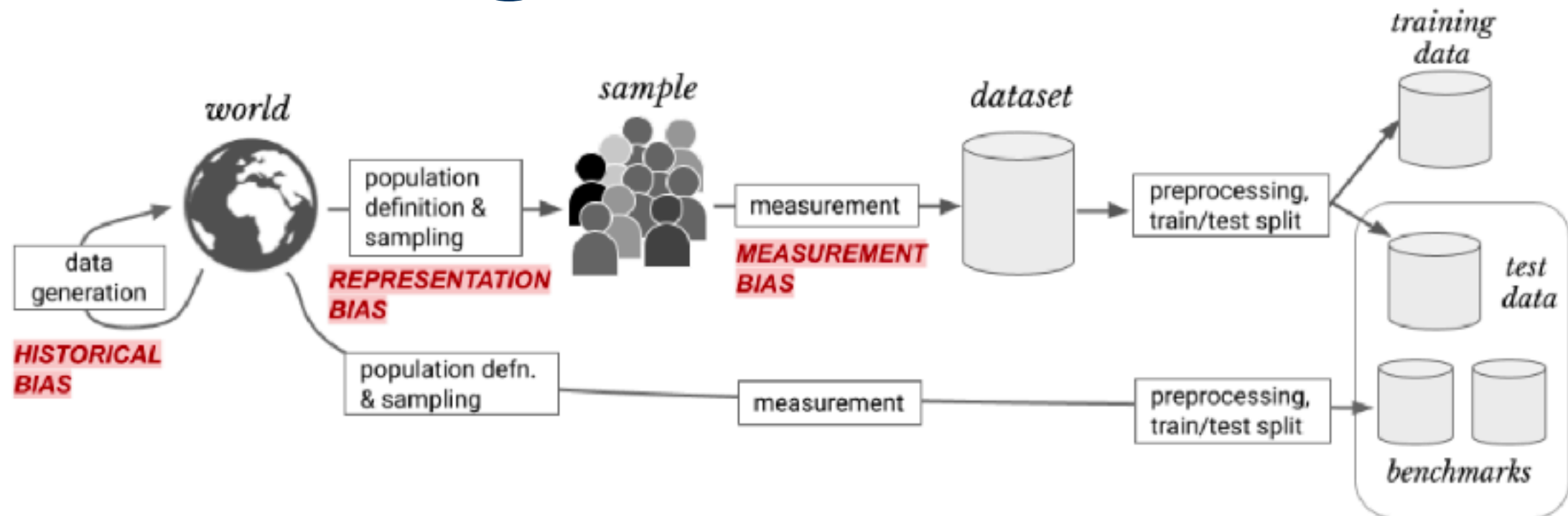


Compound visualizations: dashboards

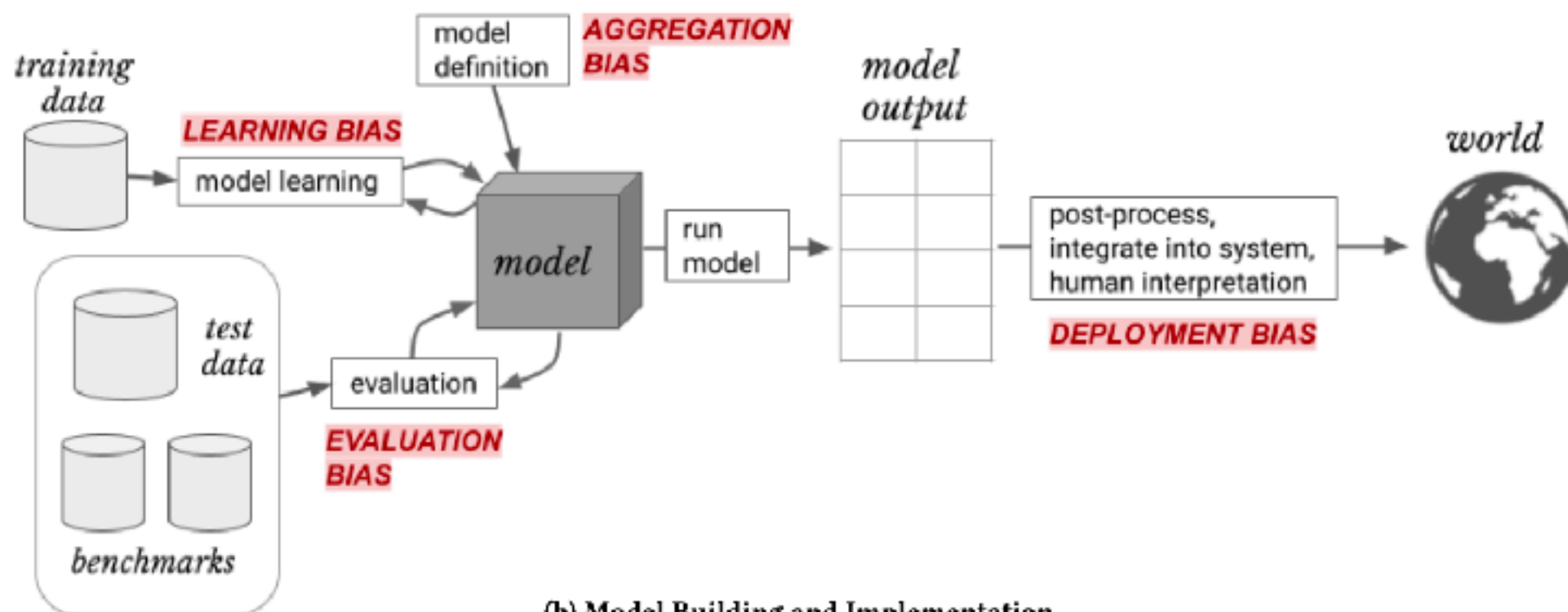


Source: California Governor's Office of Emergency Services Preparedness Platform

Visualizing models: flowcharts



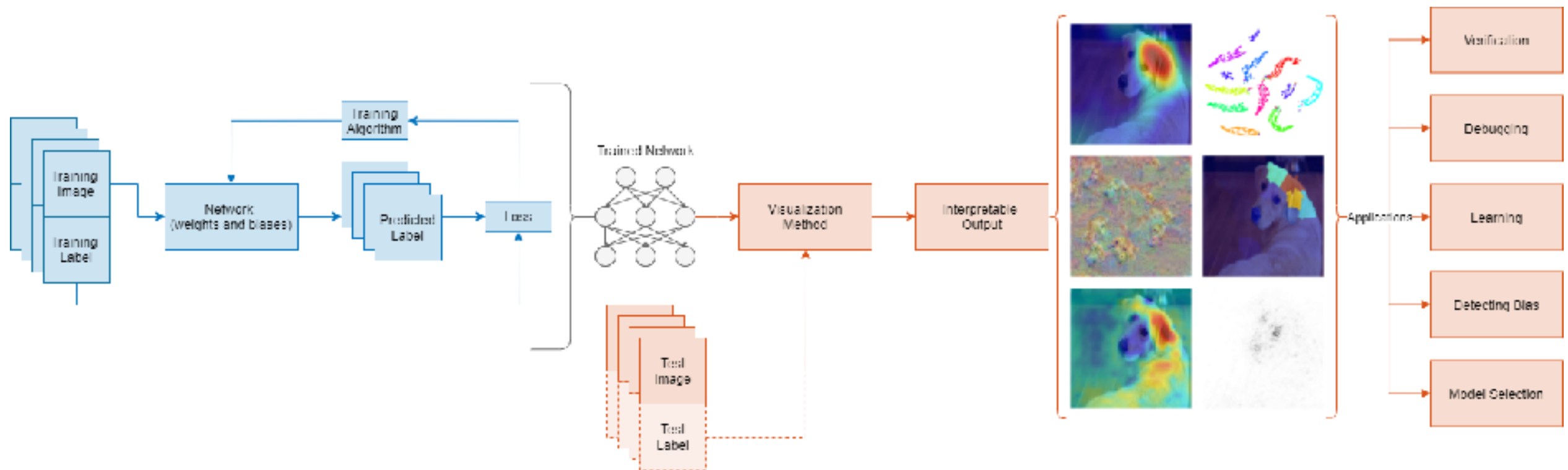
(a) Data Generation



(b) Model Building and Implementation

Source: Suresh & Guttag 2021

Visualizing models: flowcharts



Source: MATLAB

The Science of Visual Data Communication: What Works

Steven L. Franconeri¹, Lace M. Padilla², Priti Shah³,
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DOI: 10.1177/15291006211051956
www.psychologicalscience.org/PSPI
SAGE

Abstract

Effectively designed data visualizations allow viewers to use their powerful visual systems to understand patterns in data across science, education, health, and public policy. But ineffectively designed visualizations can cause confusion, misunderstanding, or even distrust—especially among viewers with low graphical literacy. We review research-backed guidelines for creating effective and intuitive visualizations oriented toward communicating data to students, coworkers, and the general public. We describe how the visual system can quickly extract broad statistics from a display, whereas poorly designed displays can lead to misperceptions and illusions. Extracting global statistics is fast, but comparing between subsets of values is slow. Effective graphics avoid taxing working memory, guide attention, and respect familiar conventions. Data visualizations can play a critical role in teaching and communication, provided that designers tailor those visualizations to their audience.

Keywords

visual communication, graph comprehension, reasoning, statistical cognition, uncertainty communication, data visualization

This report presents research-backed guidelines for creating powerful and intuitive visualizations oriented toward communicating data to students, coworkers, and the general public. We begin by reviewing guide-

visualizations that communicate clearly and suggest resources for readers interested in learning more.

Data visualizations range from simple graphs in elementary school classrooms, to depictions of uncertainty

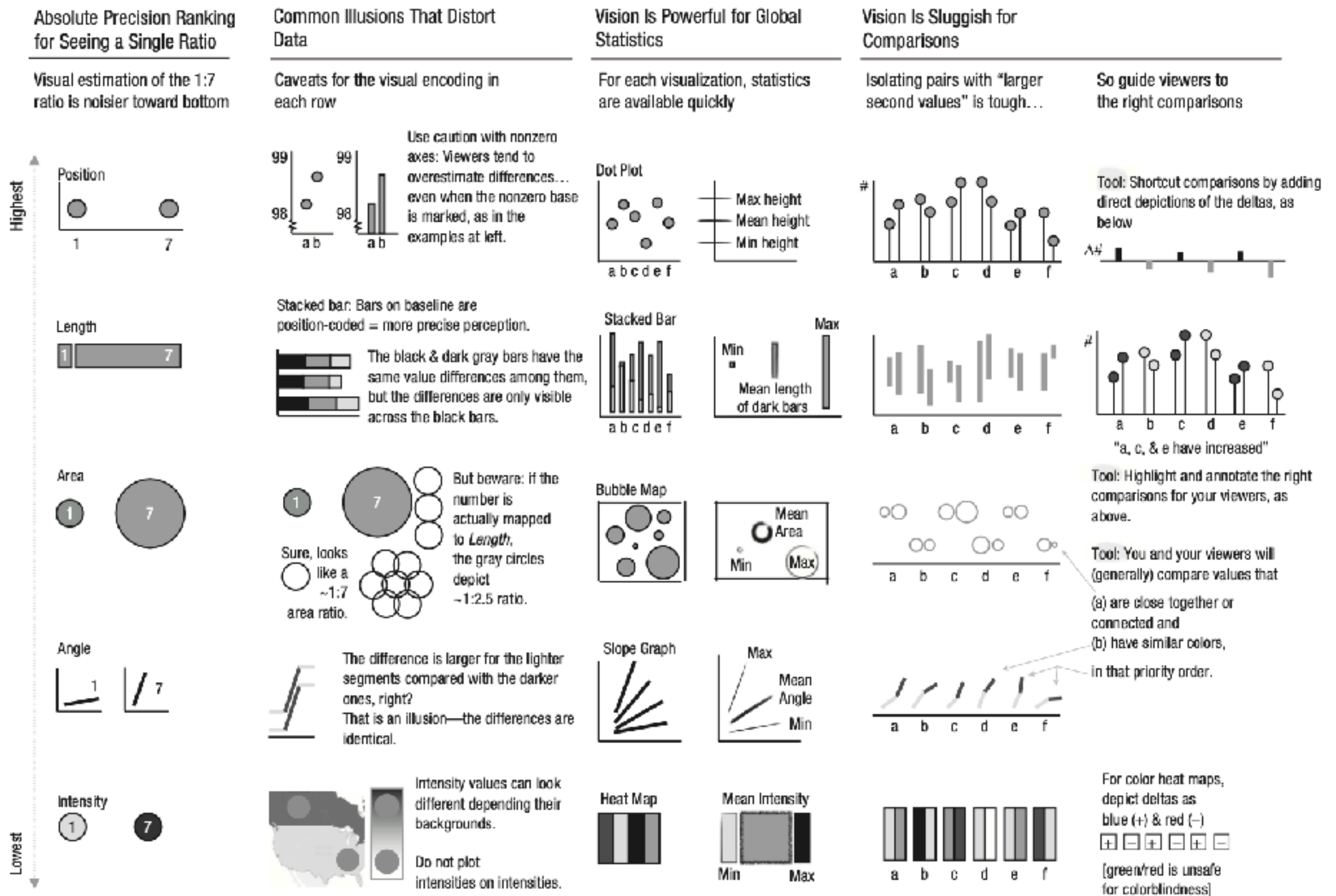
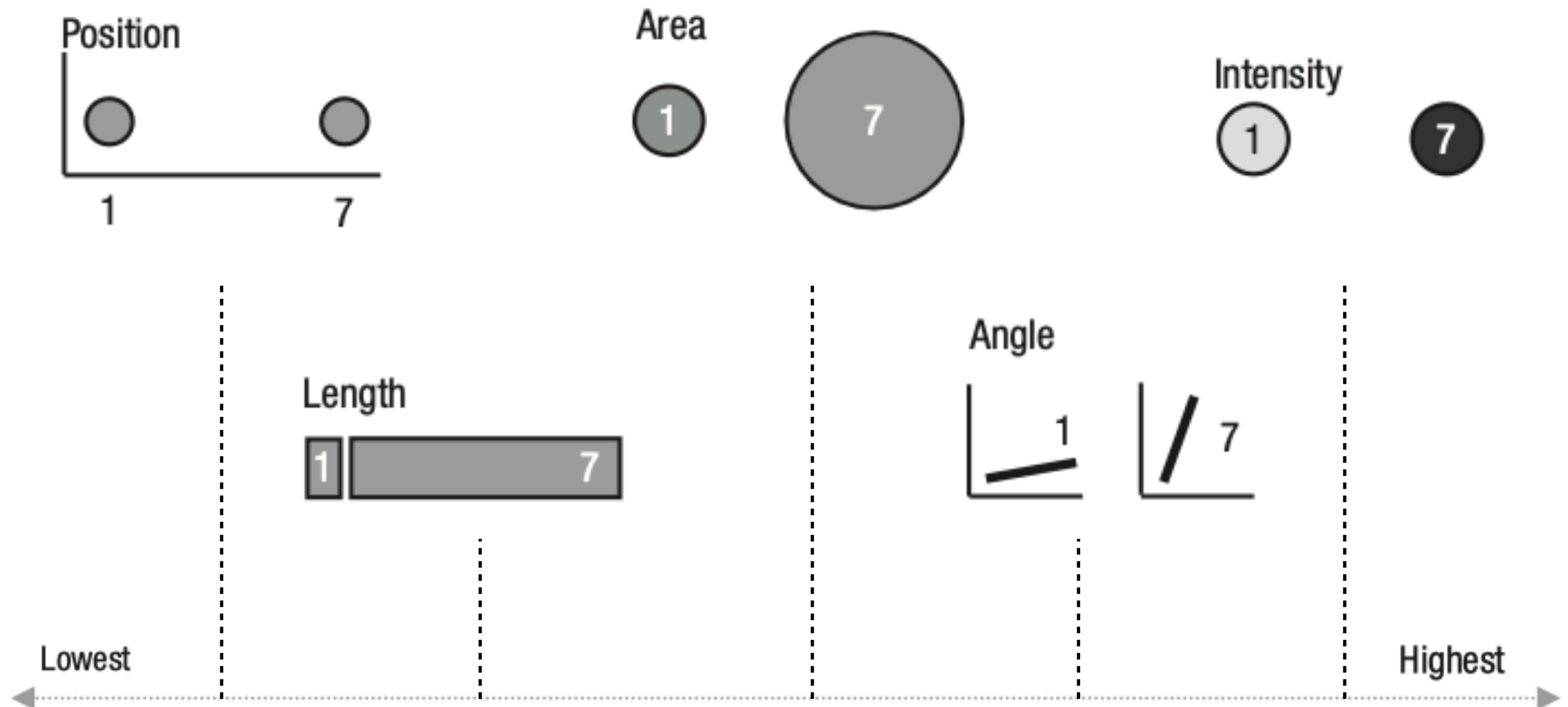
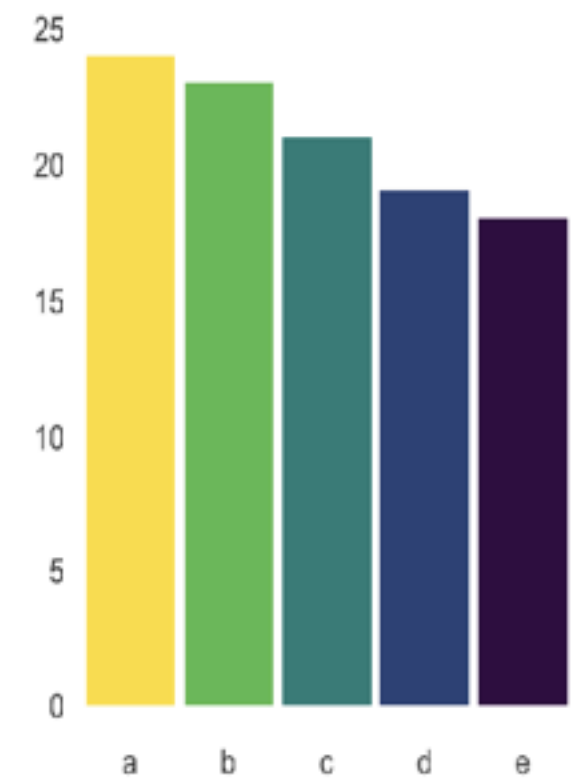
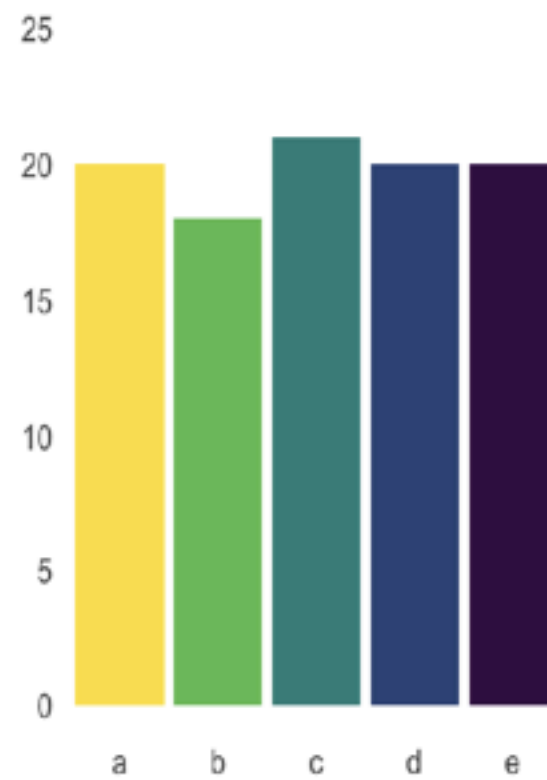
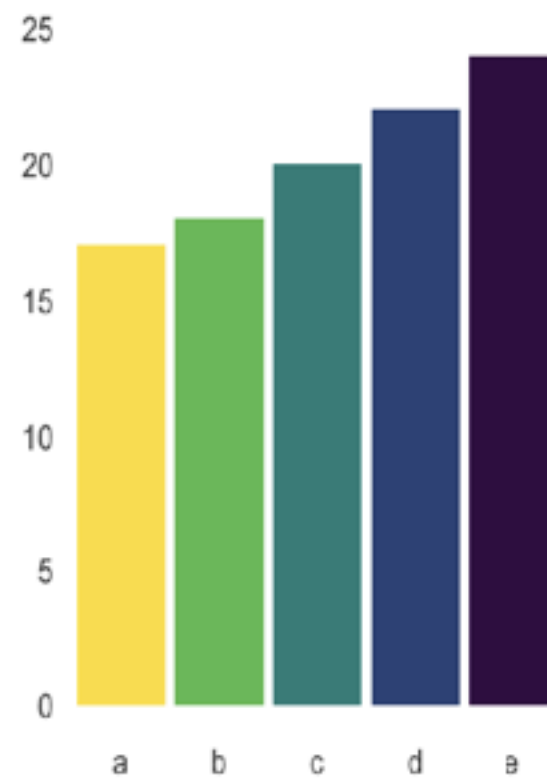


Fig. 2. A summary of the power and limits of visual data processing. The two columns on the left show a quick reference guide to channels that can depict data visually and common illusions for each channel. The column in the center presents a summary of how visual statistics are powerful. The two columns on the right illustrate how comparisons are severely limited and present a set of design techniques that focus viewers on the "right" ones.

Absolute Precision Ranking for Seeing a Single Ratio

Visual estimation of the 1:7
ratio is noisier toward bottom

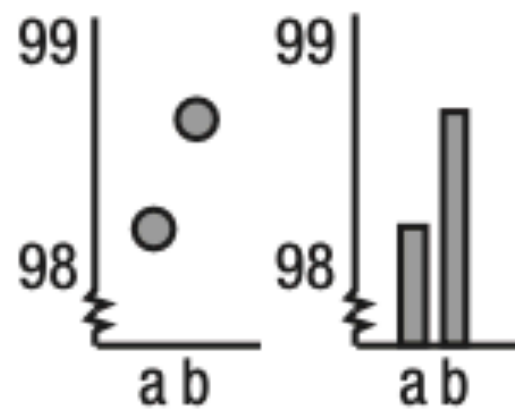




Source: "The Issue with Pie Chart,"
Data to Viz

Common Illusions That Distort Data

Caveats for the visual encoding in each row



Use caution with nonzero axes: Viewers tend to overestimate differences... even when the nonzero base is marked, as in the examples at left.

Stacked bar: Bars on baseline are position-coded = more precise perception.



The black & dark gray bars have the same value differences among them, but the differences are only visible across the black bars.

Common Illusions That Distort Data

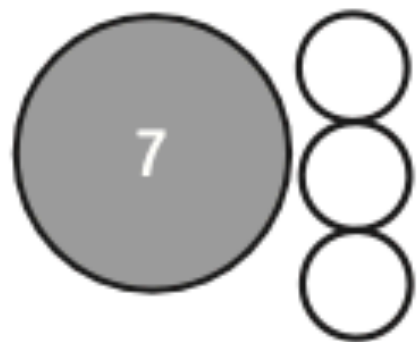
Caveats for the visual encoding in each row



Sure, looks like a ~1:7 area ratio.



But beware: if the number is actually mapped to *Length*, the gray circles depict ~1:2.5 ratio.



The difference is larger for the lighter segments compared with the darker ones, right? That is an illusion—the differences are identical.



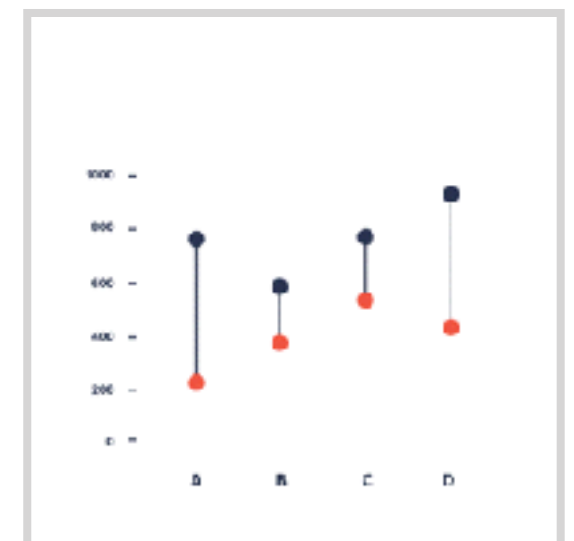
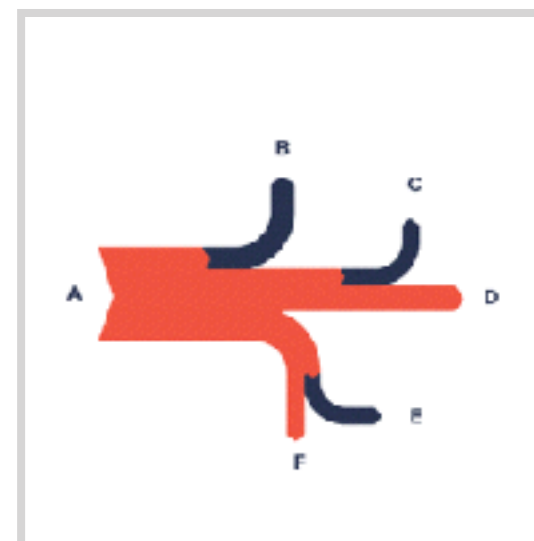
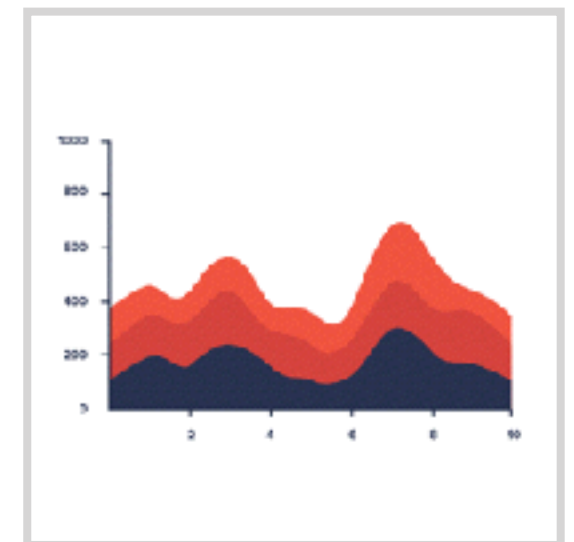
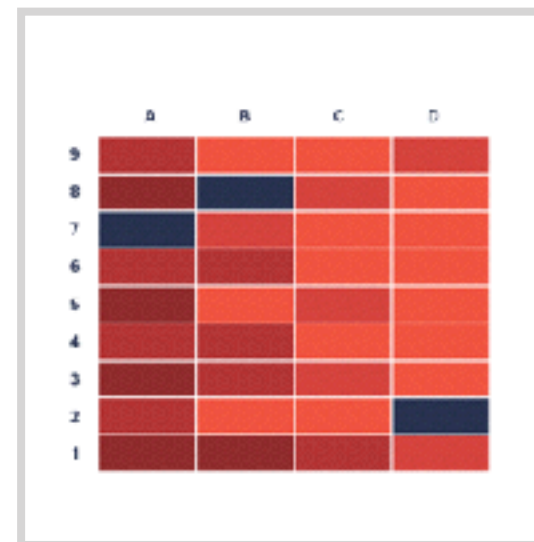
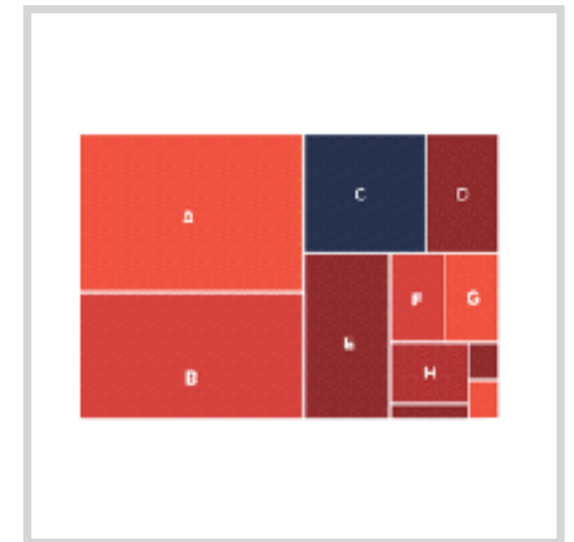
Intensity values can look different depending their backgrounds.

Do not plot intensities on intensities.

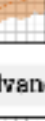
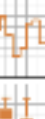
And...

- ▶ Pictorial charts
- ▶ Treemaps
- ▶ 2-D histogram / heat map
- ▶ Area chart
- ▶ Alluvial/Sankey diagram
- ▶ Dumbbell plot

For more, see: datavizproject.com



Source: Data Viz Project

Basic plots		
	plot (<i>X</i> , <i>Y</i> , [<i>fmt</i>], ...) <i>X</i> , <i>Y</i> , <i>fmt</i> , <i>color</i> , <i>marker</i> , <i>linestyle</i>	API
	scatter (<i>X</i> , <i>Y</i> , ...) <i>X</i> , <i>Y</i> , (<i>sizes</i> , <i>cmap</i>), <i>markers</i> , <i>cmap</i>	API
	bar (<i>h</i>)(<i>x</i> , <i>height</i> , ...) <i>x</i> , <i>height</i> , <i>width</i> , <i>bottom</i> , <i>align</i> , <i>color</i>	API
	imshow (<i>Z</i> , [<i>cmap</i>], ...) <i>Z</i> , <i>cmap</i> , <i>interpolation</i> , <i>extent</i> , <i>origin</i>	API
	contour (<i>f</i>)(<i>[X]</i> , <i>[Y]</i> , <i>Z</i> , ...) <i>X</i> , <i>Y</i> , <i>Z</i> , <i>levels</i> , <i>colors</i> , <i>extent</i> , <i>origin</i>	API
	quiver (<i>[X]</i> , <i>[Y]</i> , <i>U</i> , <i>V</i> , ...) <i>X</i> , <i>Y</i> , <i>U</i> , <i>V</i> , <i>l</i> , <i>units</i> , <i>angles</i>	API
	pie (<i>X</i> , [<i>explode</i>], ...) <i>Z</i> , <i>explode</i> , <i>labels</i> , <i>colors</i> , <i>radius</i>	API
	text (<i>x</i> , <i>y</i> , <i>text</i> , ...) <i>x</i> , <i>y</i> , <i>text</i> , <i>va</i> , <i>ha</i> , <i>size</i> , <i>weight</i> , <i>transform</i>	API
	fill (<i>[_between]</i>)(<i>x</i>)(...) <i>X</i> , <i>Y1</i> , <i>Y2</i> , <i>color</i> , <i>where</i>	API
Advanced plots		
	step (<i>X</i> , <i>Y</i> , [<i>fmt</i>], ...) <i>X</i> , <i>Y</i> , <i>fmt</i> , <i>color</i> , <i>marker</i> , <i>where</i>	API
	boxplot (<i>X</i> , ...) <i>X</i> , <i>notch</i> , <i>sym</i> , <i>boxcarap</i> , <i>widths</i>	API
	errorbar (<i>X</i> , <i>Y</i> , <i>xerr</i> , <i>yerr</i> , ...) <i>X</i> , <i>Y</i> , <i>xerr</i> , <i>yerr</i> , <i>fmt</i>	API
	hist (<i>X</i> , <i>bins</i> , ...) <i>X</i> , <i>bins</i> , <i>range</i> , <i>density</i> , <i>weights</i>	API
	violinplot (<i>D</i> , ...) <i>D</i> , <i>positions</i> , <i>widths</i> , <i>ver</i>	API
	barbs (<i>[X]</i> , <i>[Y]</i> , <i>U</i> , <i>V</i> , ...) <i>X</i> , <i>Y</i> , <i>U</i> , <i>V</i> , <i>l</i> , <i>length</i> , <i>pivot</i> , <i>sizes</i>	API
	eventplot (<i>positions</i> , ...) <i>positions</i> , <i>orientation</i> , <i>lineoffsets</i>	API
	hexbin (<i>X</i> , <i>Y</i> , <i>C</i> , ...) <i>X</i> , <i>Y</i> , <i>C</i> , <i>gridsize</i> , <i>bins</i>	API
	xcorr (<i>X</i> , <i>Y</i> , ...) <i>X</i> , <i>Y</i> , <i>normed</i> , <i>dettrend</i>	API

Scales

API

```
ax.set_[xy]scale(scale,...)
```

Linear
any values

log
values > 0

symlog
any values

logit
0 < values < 1

Projections

API

```
subplot(..., projection=p)
```

p='polar'

p='3d'

p=Orthographic()
from cartopy.crs import Cartographic

Lines

API

linestyle or ls

capstyle or dash_capstyle

marker

Markers

API

marker

markerproperty

Colors

API

color

colorbar

colorbar

colorbar

colorbar

Colormaps

API

```
plt.get_cmap(name)
```

Uniform

Sequential

Diverging

Qualitative

Cyclic

[illegible]

Animation

```

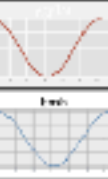
import matplotlib.animation as animation

T = np.linspace(0, 2*np.pi, 100)
S = np.sin(T)
line, = plt.plot(T, S)
def animate(i):
    line.set_ydata(np.sin(-i/53))
anim = animation.FuncAnimation(
    plt.gcf(), animate, interval=5)
plt.show()

```

Styles

```
plt.style.use('style')
```



Quick reminder

```

ax.grid()
ax.patch.set_alpha(0)
ax.set_[xy]lim(vmin, vmax)
ax.set_[xy]label('label')
ax.set_[xy]ticks(list)
ax.set_[xy]ticklabels(list)
ax.set_[sup]title('title')
ax.tick_params(width=10, ...)
ax.set_axis_[on/off]()

ax.tight_layout()
plt.gcf(), plt.gca()
ml.rc('axes', linewidth=1, ...)
fig.patch.set_alpha(0)
text=r'$\frac{1}{2}\pi$'

```

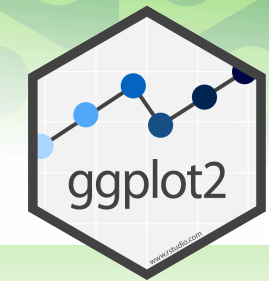
Keyboard shortcuts

Ctrl + S Save	Ctrl + W Close plot
F Reset view	F Fullscreen on/off
F View forward	H View back
P Pan view	C Zoom to rect
X X pan/zoom	Y Y pan/zoom
G Minor grid on/off	G Major grid on/off
L X axis log/linear	L Y axis log/linear

Ten Simple Rules

1. Know Your Audience
2. Lead by Your Message
3. Adapt the Figure
4. Captions Are Not Optional
5. Do Not Trust the Defaults
6. Use Color Effectively
7. Do Not Mislead the Reader
8. Avoid "Chartjunk"
9. Message Trumps Beauty
10. Get the Right Tool

Data Visualization with ggplot2 : : CHEAT SHEET

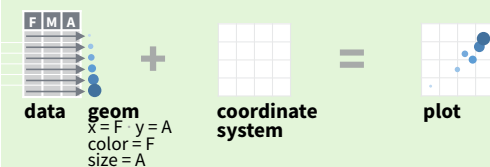


Basics

ggplot2 is based on the **grammar of graphics**, the idea that you can build every graph from the same components: a **data** set, a **coordinate system**, and **geoms**—visual marks that represent data points.



To display values, map variables in the data to visual properties of the geom (**aesthetics**) like **size**, **color**, and **x** and **y** locations.



Complete the template below to build a graph.

```
ggplot( data = <DATA> ) +  
  <GEOM_FUNCTION> (mapping = aes( <MAPPINGS> ),  
  stat = <STAT>, position = <POSITION> ) +  
  <COORDINATE_FUNCTION> +  
  <FACET_FUNCTION> +  
  <SCALE_FUNCTION> +  
  <THEME_FUNCTION>
```

required

Not required, sensible defaults supplied

ggplot(data = mpg, aes(x = cty, y = hwy)) Begins a plot that you finish by adding layers to. Add one geom function per layer.

aesthetic mappings data geom

qplot(x = cty, y = hwy, data = mpg, geom = "point") Creates a complete plot with given data, geom, and mappings. Supplies many useful defaults.

last_plot() Returns the last plot

ggsave("plot.png", width = 5, height = 5) Saves last plot as 5' x 5' file named "plot.png" in working directory. Matches file type to file extension.

Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

GRAPHICAL PRIMITIVES

```
a <- ggplot(economics, aes(date, unemployment))  
b <- ggplot(seals, aes(x = long, y = lat))
```

a + geom_blank()
(Useful for expanding limits)

b + geom_curve(aes(yend = lat + 1, xend = long + 1, curvature = 1) - x, xend, y, yend, alpha, angle, color, curvature, linetype, size)

a + geom_path(lineend = "butt", linejoin = "round", linemitre = 1)
x, y, alpha, color, group, linetype, size

a + geom_polygon(aes(group = group))
x, y, alpha, color, fill, group, linetype, size

b + geom_rect(aes(xmin = long, ymin = lat, xmax = long + 1, ymax = lat + 1)) - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size

a + geom_ribbon(aes(ymin = unemployment - 900, ymax = unemployment + 900)) - x, ymax, ymin, alpha, color, fill, group, linetype, size

LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

b + geom_abline(aes(intercept = 0, slope = 1))
b + geom_hline(aes(yintercept = lat))
b + geom_vline(aes(xintercept = long))

b + geom_segment(aes(yend = lat + 1, xend = long + 1))
b + geom_spoke(aes(angle = 1:1155, radius = 1))

ONE VARIABLE continuous

```
c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)
```

c + geom_area(stat = "bin")
x, y, alpha, color, fill, linetype, size

c + geom_density(kernel = "gaussian")
x, y, alpha, color, fill, group, linetype, size, weight

c + geom_dotplot()
x, y, alpha, color, fill

c + geom_freqpoly() x, y, alpha, color, group, linetype, size

c + geom_histogram(binwidth = 5) x, y, alpha, color, fill, linetype, size, weight

c2 + geom_qq(aes(sample = hwy)) x, y, alpha, color, fill, linetype, size, weight

discrete

```
d <- ggplot(mpg, aes(fl))
```

d + geom_bar()
x, alpha, color, fill, linetype, size, weight

TWO VARIABLES

continuous x, continuous y

```
e <- ggplot(mpg, aes(cty, hwy))
```

e + geom_label(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE) x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

e + geom_jitter(height = 2, width = 2)
x, y, alpha, color, fill, shape, size

e + geom_point() x, y, alpha, color, fill, shape, size, stroke

e + geom_quantile() x, y, alpha, color, group, linetype, size, weight

e + geom_rug(sides = "bl") x, y, alpha, color, linetype, size

e + geom_smooth(method = lm) x, y, alpha, color, fill, group, linetype, size, weight

e + geom_text(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE) x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

discrete x, continuous y

```
f <- ggplot(mpg, aes(class, hwy))
```

f + geom_col() x, y, alpha, color, fill, group, linetype, size

f + geom_boxplot() x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight

f + geom_dotplot(binaxis = "y", stackdir = "center") x, y, alpha, color, fill, group

f + geom_violin(scale = "area") x, y, alpha, color, fill, group, linetype, size, weight

discrete x, discrete y

```
g <- ggplot(diamonds, aes(cut, color))
```

g + geom_count() x, y, alpha, color, fill, shape, size, stroke

THREE VARIABLES

```
seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2)); l <- ggplot(seals, aes(long, lat))
```

l + geom_contour(aes(z = z))
x, y, z, alpha, colour, group, linetype, size, weight

l + geom_raster(aes(fill = z), hjust = 0.5, vjust = 0.5, interpolate = FALSE)
x, y, alpha, fill

l + geom_tile(aes(fill = z)) x, y, alpha, color, fill, linetype, size, width

continuous bivariate distribution

```
h <- ggplot(diamonds, aes(carat, price))
```

h + geom_bin2d(binwidth = c(0.25, 500))
x, y, alpha, color, fill, linetype, size, weight

h + geom_density2d()
x, y, alpha, colour, group, linetype, size

h + geom_hex()
x, y, alpha, colour, fill, size

continuous function

```
i <- ggplot(economics, aes(date, unemployment))
```

i + geom_area()
x, y, alpha, color, fill, linetype, size

i + geom_line()
x, y, alpha, color, group, linetype, size

i + geom_step(direction = "hv")
x, y, alpha, color, group, linetype, size

visualizing error

```
df <- data.frame(grp = c("A", "B"), fit = 4:5, se = 1:2)  
j <- ggplot(df, aes(grp, fit, ymin = fit - se, ymax = fit + se))
```

j + geom_crossbar(fatten = 2)
x, y, ymax, ymin, alpha, color, fill, group, linetype, size

j + geom_errorbar() x, ymax, ymin, alpha, color, group, linetype, size, width (also **geom_errorbarh**())

j + geom_linerange()
x, ymin, ymax, alpha, color, group, linetype, size

j + geom_pointrange()
x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

maps

```
data <- data.frame(murder = USArrests$Murder, state = tolower(rownames(USArrests)))  
map <- map_data("state")  
k <- ggplot(data, aes(fill = murder))
```

k + geom_map(aes(map_id = state), map = map) + **expand_limits**(x = map\$long, y = map\$lat), map_id, alpha, color, fill, linetype, size

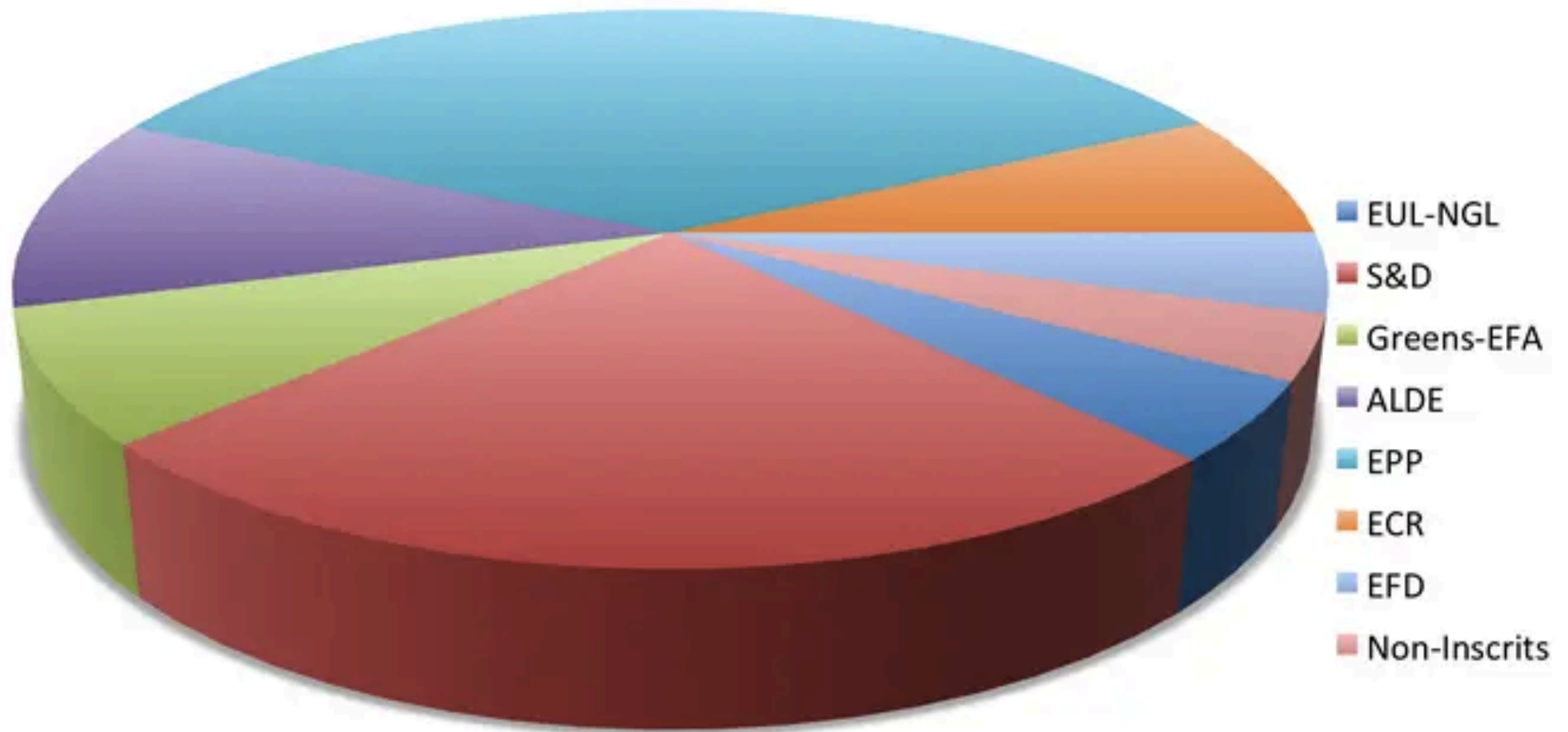


RStudio® is a trademark of RStudio, Inc. • CC BY SA RStudio • info@rstudio.com • 844-448-1212 • rstudio.com • Learn more at <http://ggplot2.tidyverse.org> • ggplot2 3.1.0 • Updated: 2018-12

Roast! This! Viz!

What could be improved?

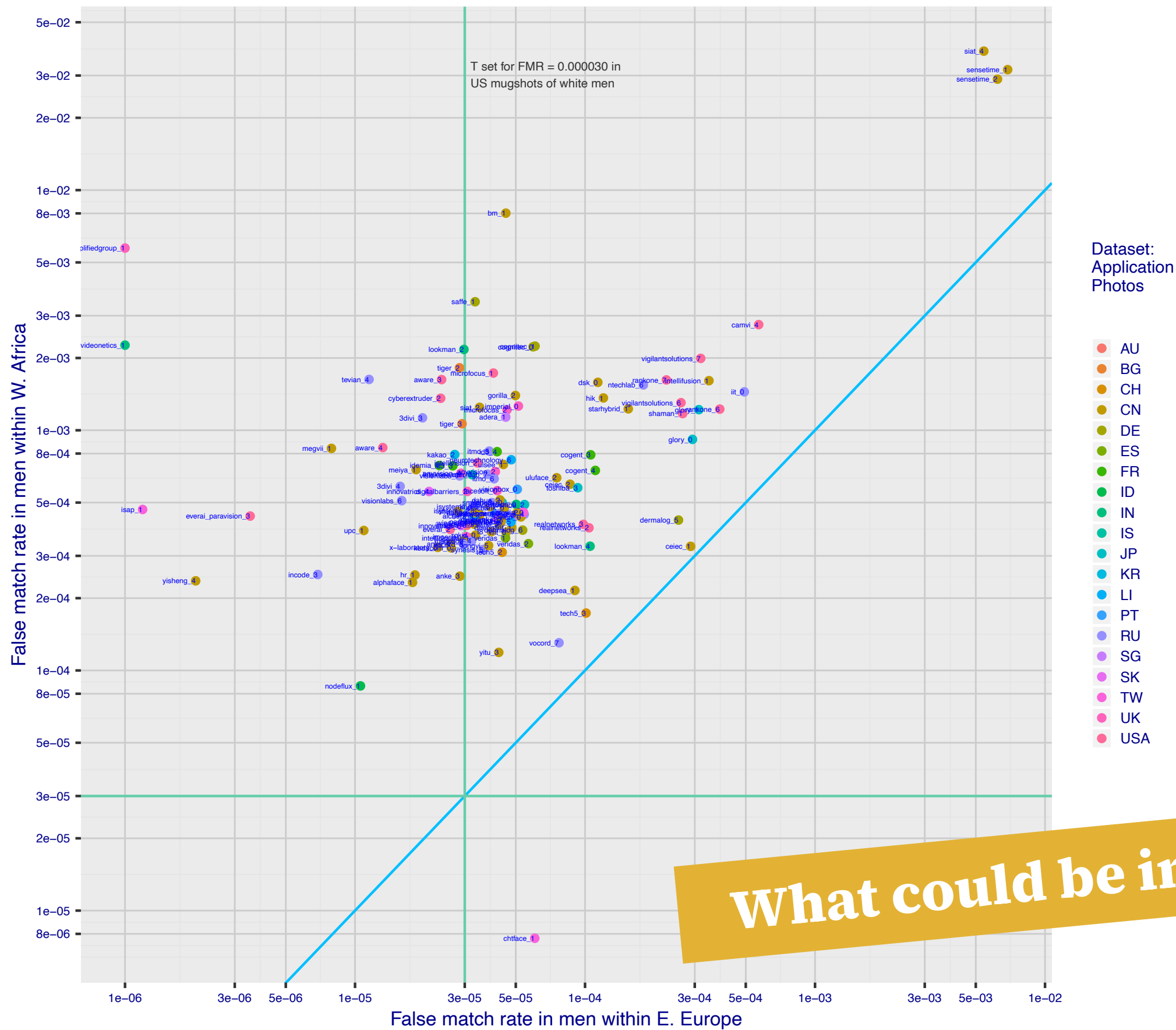
European Parliament Party Breakdown



Source: Walter Hickey, Business Insider

**What could
be improved?**





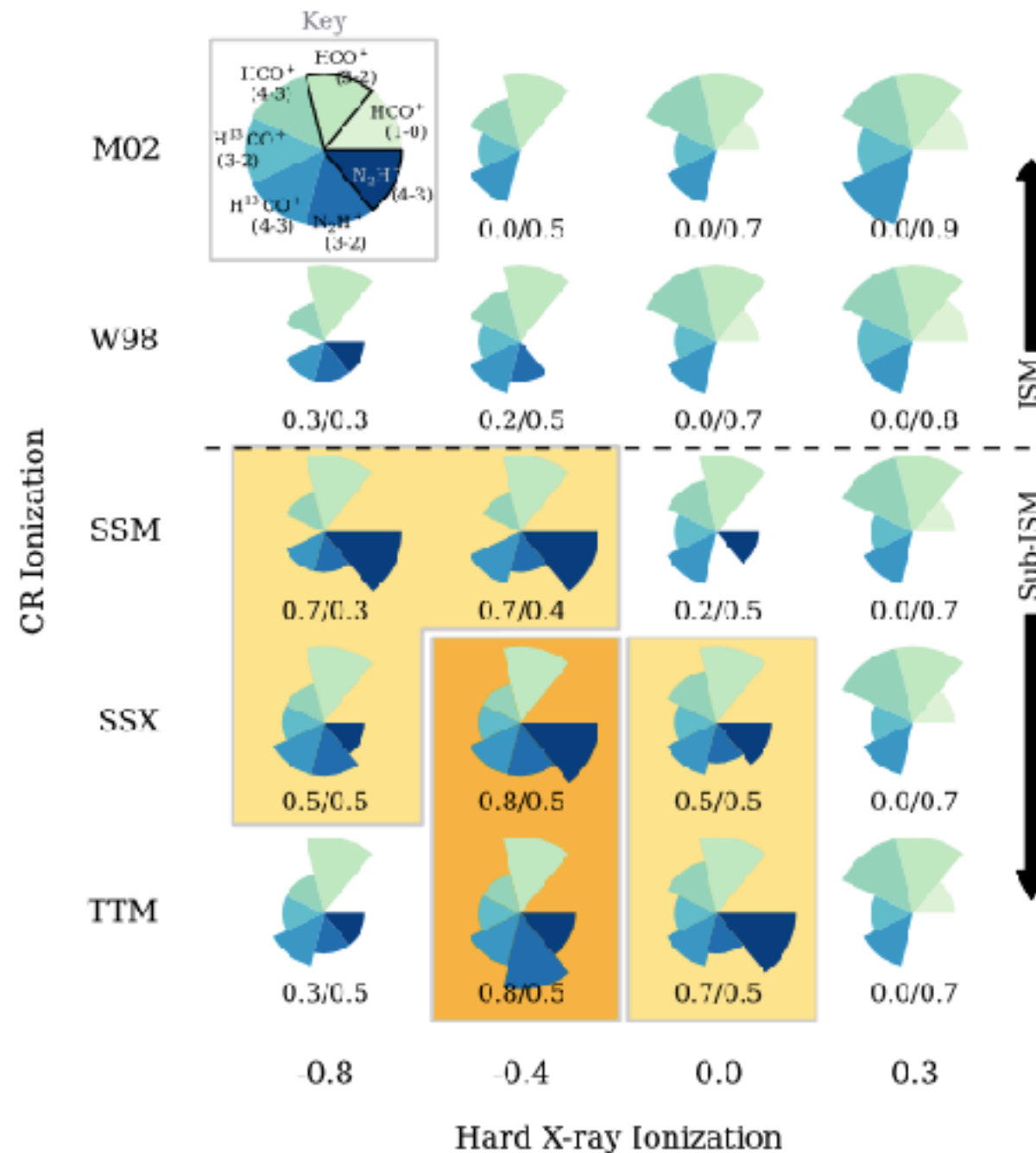
Source: NIST.IR.8280

10 Best Practices for Effective Data Visualization

1. **Know your audience!**
 - For public vizzes, this usually requires bar, line, and scatter plots.
 - Basically never use undefined acronyms or symbols.
2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.
3. **Use descriptive titles:** *tell* your viewer what they should see, and don't be redundant.
4. **Be selective in data and design.** Consider your data-to-ink ratio and remove distractions.
5. **Use color sparingly, intentionally, and consistently.**
 - Keep branding in mind
 - Remember colorblindness
6. **Use reasonable axis tick** formats and frequencies, not just the default.
7. **Consider creative alternatives to legends.**
8. **Don't be afraid of annotations.** Use whatever software you prefer to add them.
9. **Make your text bigger** than you'd think. Nobody likes to squint!
10. **Use the blink test.**

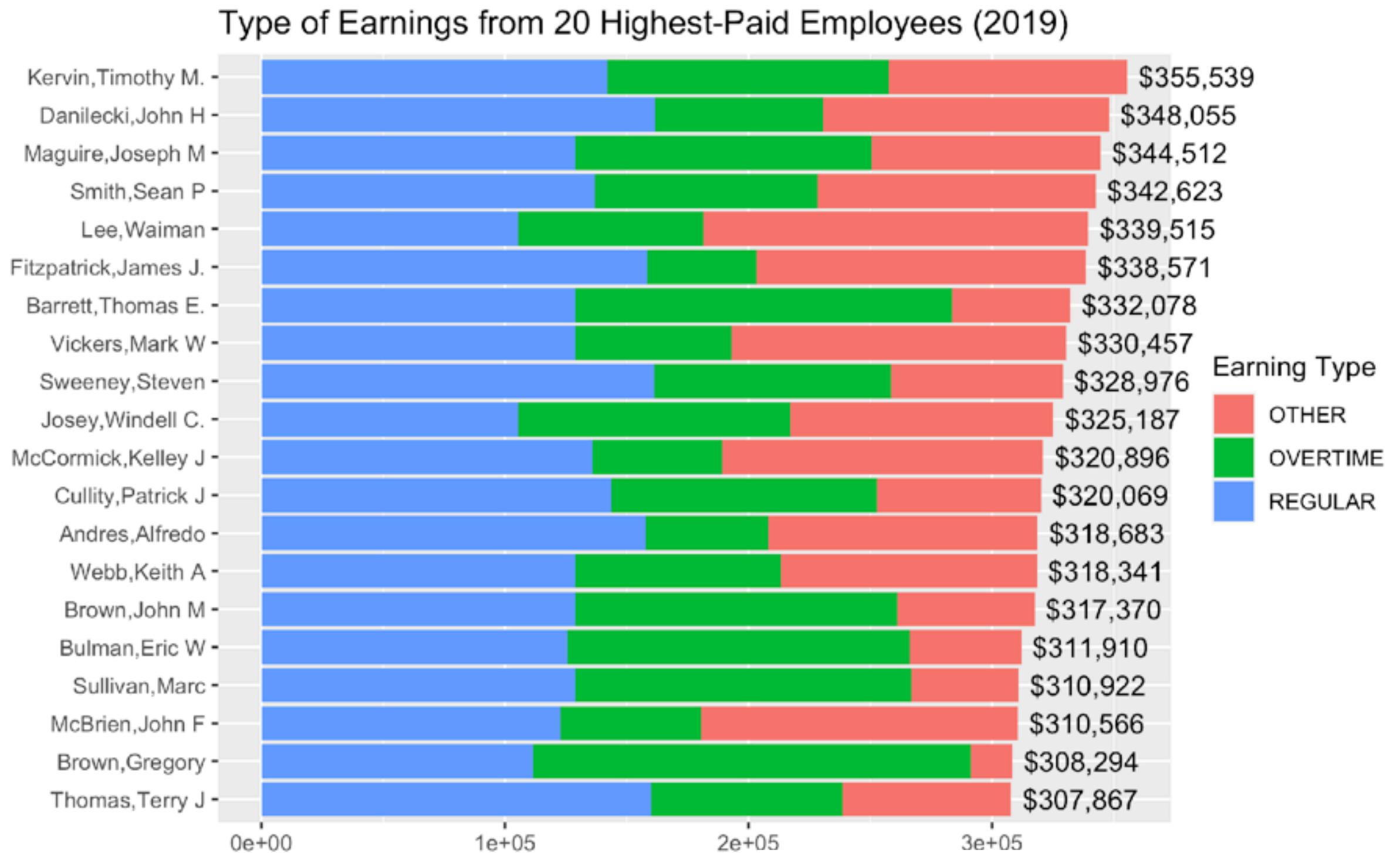
1. Know your audience!

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- Basically never use undefined acronyms or symbols.



Source: "Constraining the X-ray and Cosmic Ray Ionization Chemistry of the TW Hya Protoplanetary Disk: Evidence for a Sub-interstellar Cosmic Ray Rate,"
Cleeves et al., 2015

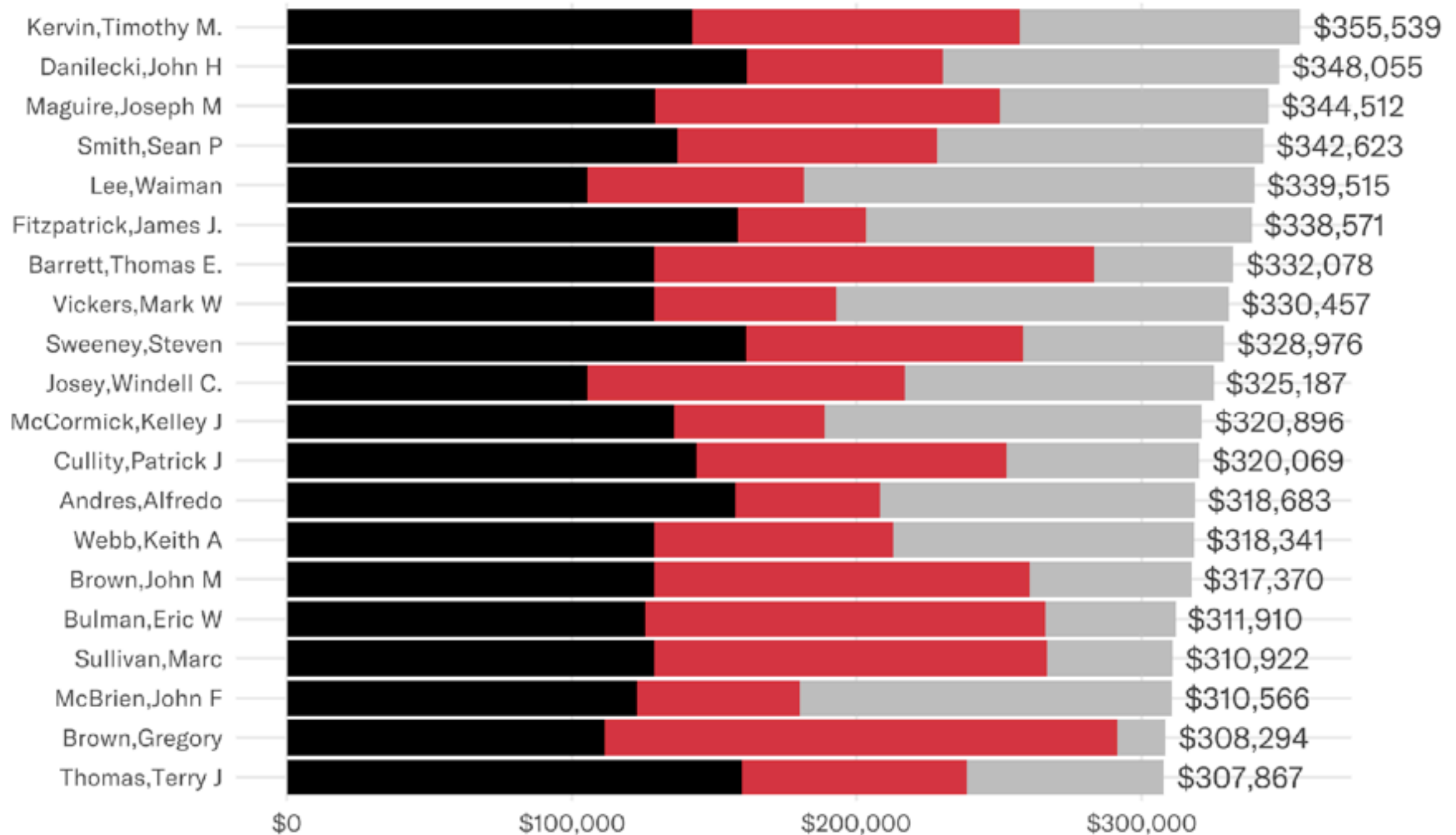
2. Question the defaults. This means fonts, colors, sizes, shapes, and more.



Source: "Unpacking the Boston Police Budget,"
ACLU of Massachusetts Data for Justice Project

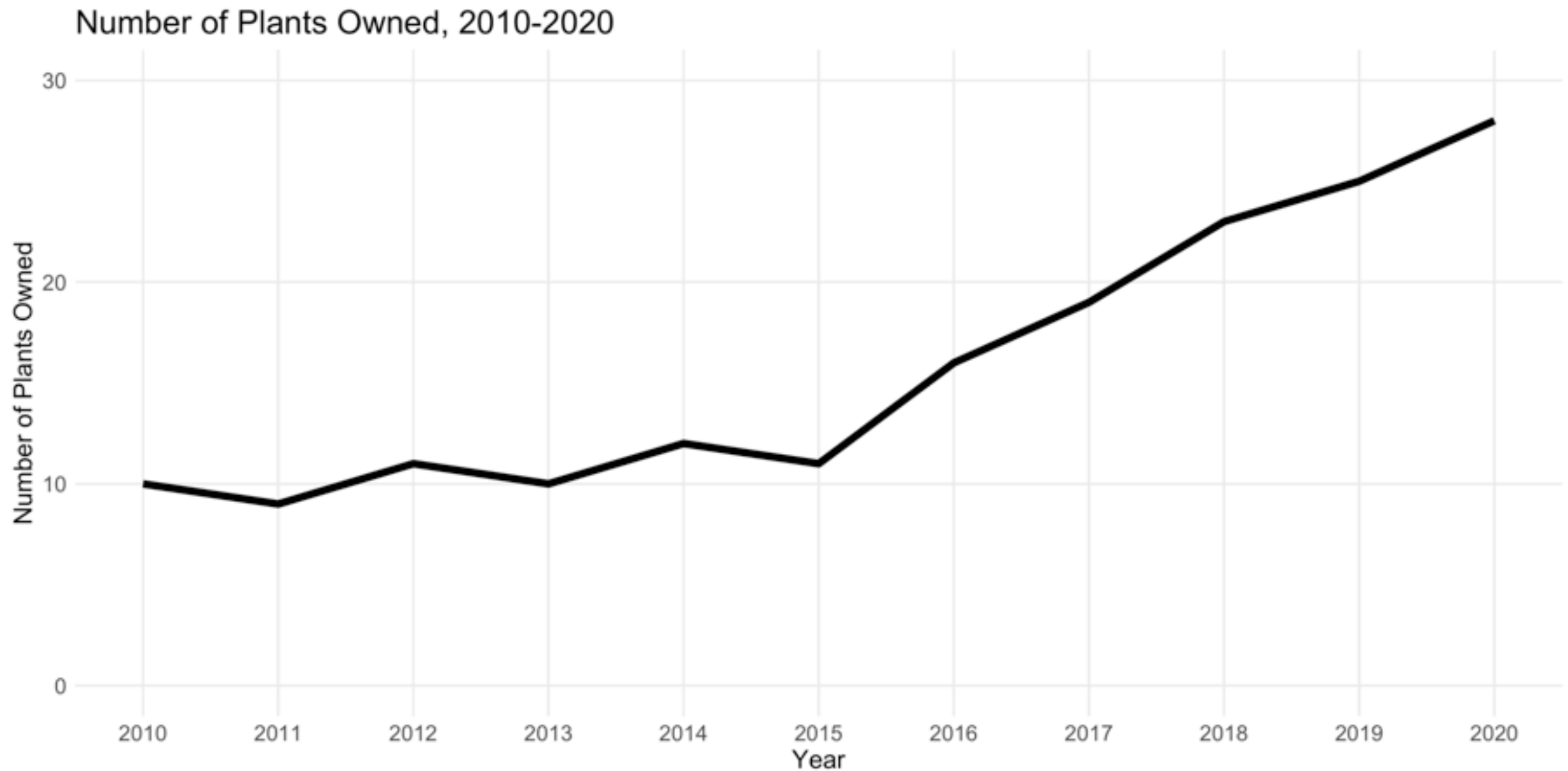
2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.

Type of Earnings from 20 Highest-Paid Employees (2019)

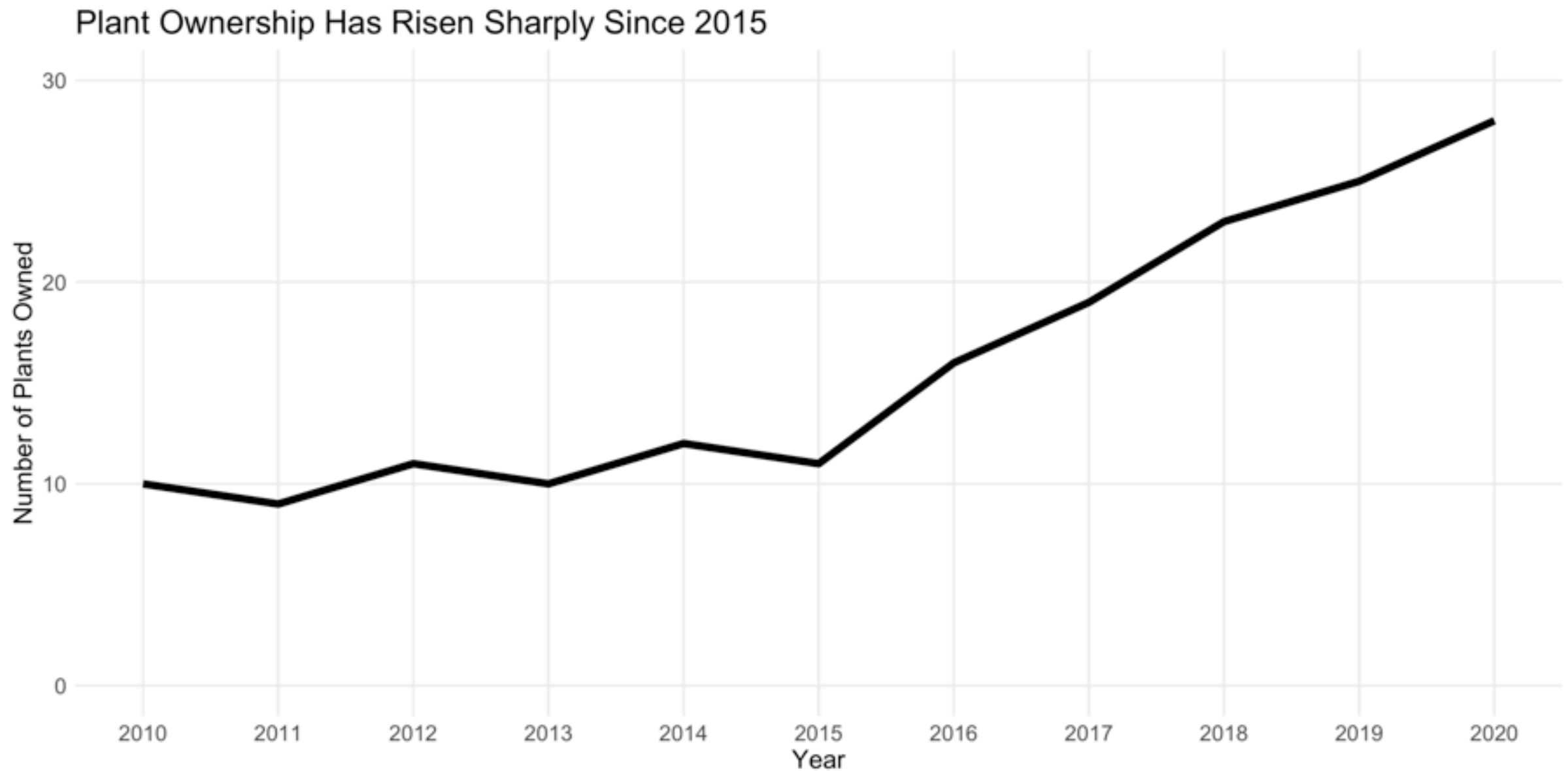


Source: "Unpacking the Boston Police Budget,"
ACLU of Massachusetts Data for Justice Project

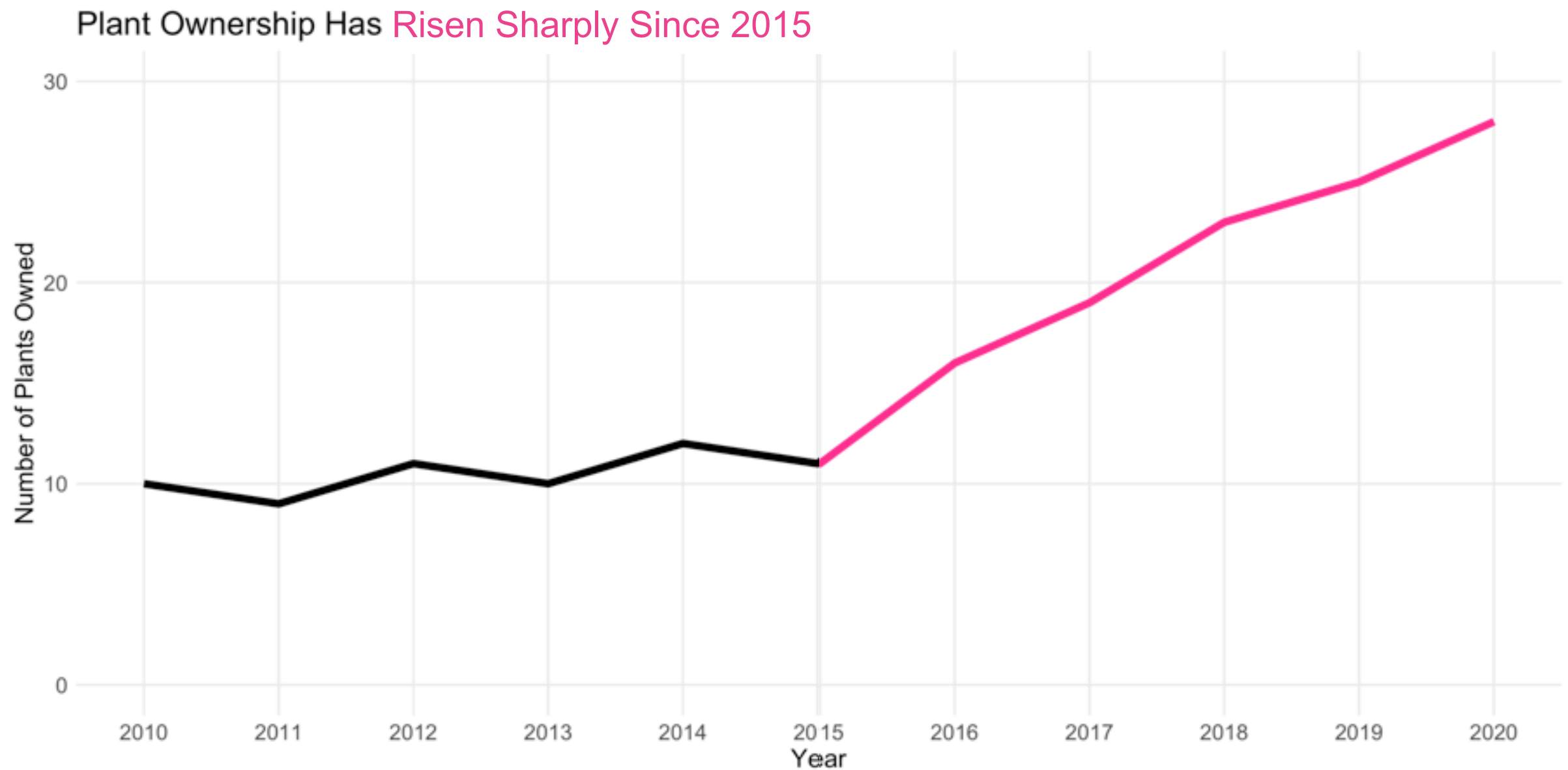
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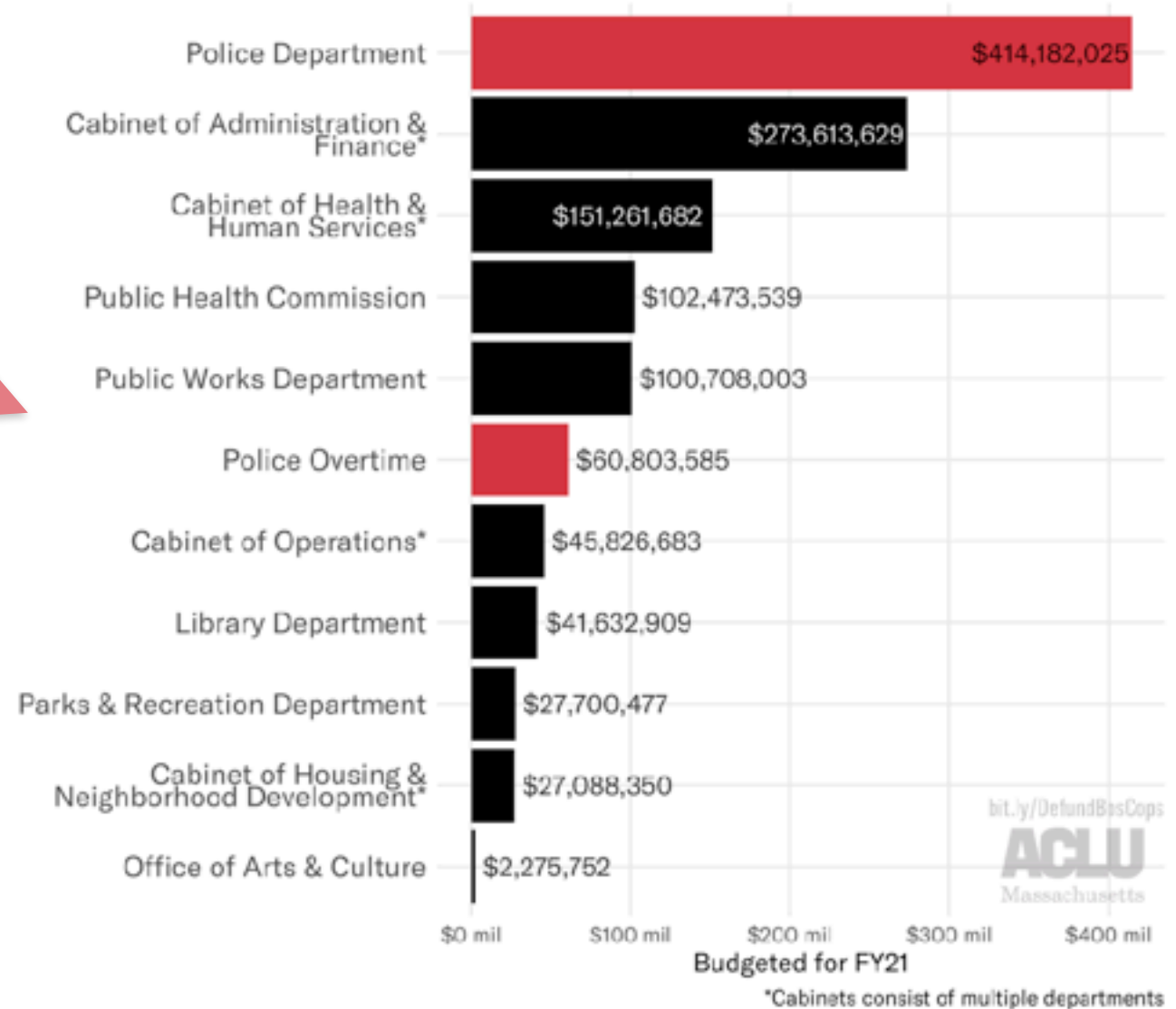


3. **Use descriptive titles:** *tell* your viewer what they should see, and don't be redundant.



4. Be selective in data and design. Consider your data-to-ink ratio and remove distractions.

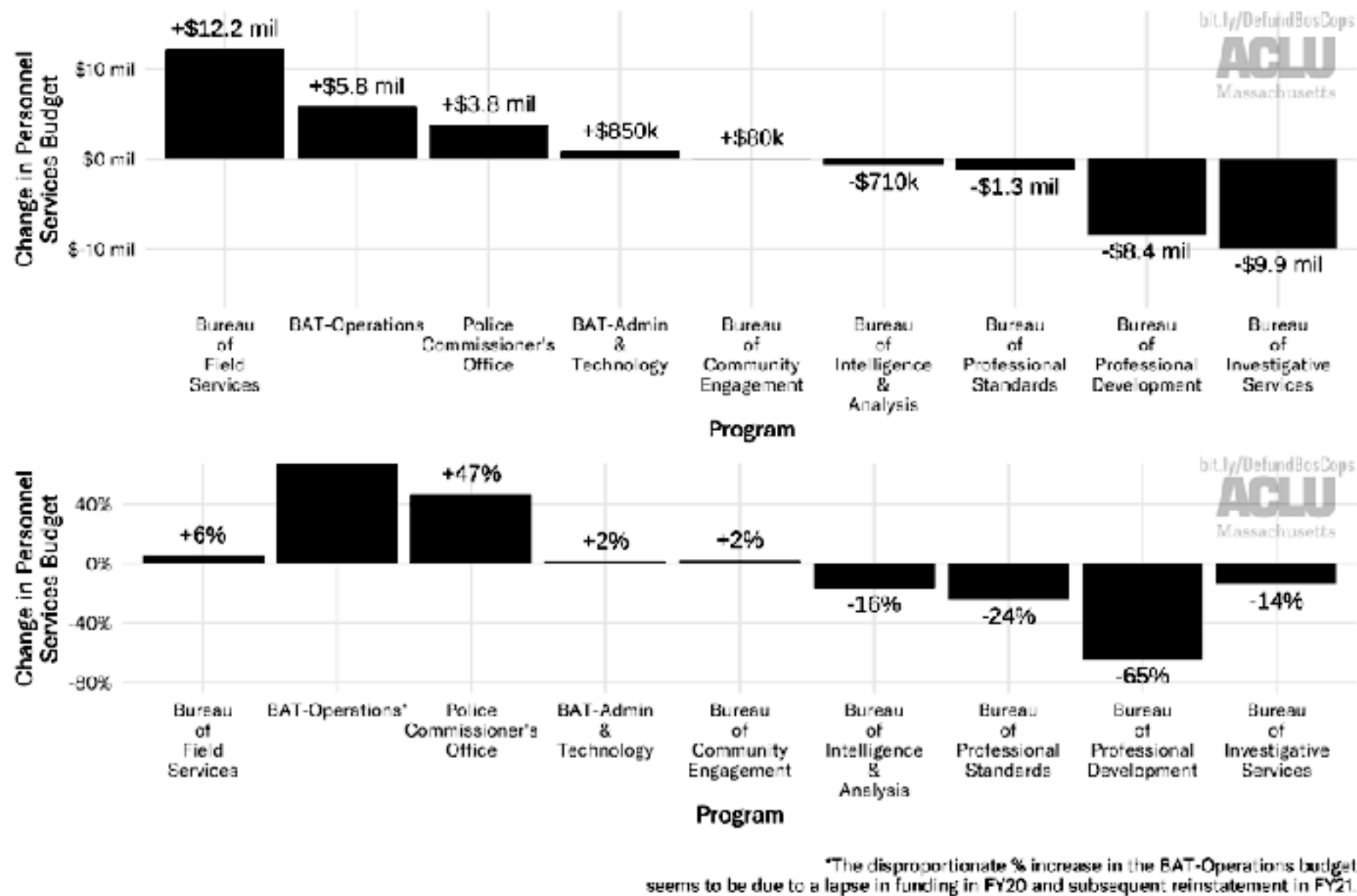
Cabinet	Department	FY 17 Expenditure	FY 18 Expenditure	FY 19 Appropriation	FY 20 Appropriation	FY 21 Total
Mayor's Office	Mayor's Office	4,738,332	4,738,332	5,338,434	5,437,532	30,218
	Deputy Mayor's Office	3,985,807	4,105,889	5,124,590	5,282,814	-112,805
	Intergovernmental Relations	3,197,436	3,798,237	3,548,000	3,587,617	-457,573
	Human Rights Commission	6,532,124	6,175,023	3,172,316	3,531,008	351,212
	Law Department	2,427,700	2,572,236	3,481,133	4,024,817	1,682,754
Total		18,022,548	18,793,835	21,014,073	21,463,781	30,218
Operations	Property Management Department	12,827,880	13,115,217	13,785,900	13,880,170	-1,285,730
	Public Facilities Department	5,449,174	6,534,378	7,119,819	7,261,344	70,655
	Inspectional Services Department	38,568,195	39,179,458	40,348,115	40,999,889	2,058,014
Total		56,845,249	58,829,053	61,253,834	62,141,393	500,989
City Engagement	Neighborhood Services	3,068,183	3,044,743	3,783,138	3,880,589	86,257
Total		3,068,183	3,044,743	3,783,138	3,880,589	86,257
Arts & Culture	Office of Arts & Culture	3,140,305	3,380,505	3,632,616	3,735,752	214,846
	Library Department	33,258,480	33,613,829	40,514,907	41,617,109	1,098,882
	Total	36,406,785	37,004,334	44,147,523	45,352,861	1,313,738
Economic Development	Office of Economic Development	2,301,330	3,106,139	3,446,780	3,881,480	-47,880
	Consumer Affairs & Licensing	1,133,755	1,221,188	1,353,313	1,513,705	163,112
	Office of Tourism	1,314,180	1,312,830	1,313,852	1,613,872	300,120
Total		4,749,265	5,639,157	6,113,945	6,999,057	315,352
Education	Boston Public Schools	1,013,288,330	1,013,428,079	1,013,428,079	1,013,428,079	80,018,860
	Total	1,013,288,330	1,013,428,079	1,013,428,079	1,013,428,079	80,018,860
Environment, Energy & Open Space	Environment Department	2,179,112	4,776,357	3,871,816	3,383,839	183,913
	Parks & Recreation Department	21,327,847	21,388,147	21,388,147	21,388,147	1,813,138
Total		23,506,959	26,164,504	25,260,000	24,771,986	1,997,048
Administration & Finance	Administration & Finance	1,780,988	2,013,117	1,731,937	1,781,276	-347,618
	Accounting Department	1,980,185	2,118,514	2,118,514	2,118,514	137,005
	Budget Management	2,680,326	2,735,380	2,947,289	2,947,289	-1,006
	Capital Management	2,841,663	2,792,454	3,541,745	3,629,877	83,412
	Comptroller of the City	39,533,248	38,233,139	3,000,000	3,000,000	-1,000,000
	Health Insurance	288,946,788	292,078,308	292,078,308	292,078,308	-1,000,000
	Human Resources	4,275,968	4,684,940	3,848,283	3,780,489	178,565
	Information	2,515,432	89,171,357	1,280,000	1,280,000	0
	Office of Labor Relations	2,250,330	1,122,845	1,115,705	1,115,705	32,668
	Procurement & Supplies - City	3,814,282	3,814,282	4,300,000	4,300,000	0
	Procurement & Supplies - County	33,289	13,147	300,000	300,000	0
	Procurement	1,618,940	1,236,775	1,800,000	1,800,000	0
	Registry Division	952,164	1,061,384	1,061,384	1,061,384	32,927
	Treasury Department	5,236,098	28,119,845	13,800,176	4,722,245	-15,077,900
	Unemployment Compensation	0	0	350,000	350,000	0
	Workers Compensation Fund	1,585,148	1,585,148	2,300,000	2,300,000	0
Total		777,548,642	583,473,878	780,418,119	779,113,879	-15,884,880
Health & Human Services	Office of Health & Human Services	0	0	0	1,455,788	1,455,788
	Boston Center for Youth & Families	29,132,568	27,468,542	27,468,542	28,941,636	-1,004,122
	Commission for Persons W/Disabilities	478,158	480,491	480,540	581,686	23,036
	Age Strong	3,568,638	3,713,812	3,915,615	4,116,817	233,437
	Anti-Human Trafficking	238,164	298,204	3,311,116	3,711,116	852
	Office of Immigrant Advancement	413,804	428,134	677,387	875,519	247,782
	Public Health Commission	59,540,119	60,987,402	61,485,418	62,471,529	9,068,305
	Senior YOUTH	3,713,167	2,137,058	4,780,639	4,881,805	-107,804
	South Engagement & Employment	6,136,700	6,098,114	6,098,580	7,188,376	1,075,448
	Total	170,984,250	178,780,549	180,868,814	180,868,814	18,854,818
Housing & Neighborhood Development	Neighborhood Development	33,645,148	14,780,112	20,527,114	22,048,150	6,543,736
	Total	33,645,148	14,780,112	20,527,114	22,048,150	6,543,736
Information & Technology	Department of Innovation and Technology	33,811,196	28,588,363	34,310,387	35,251,549	990,262
	Total	33,811,196	28,588,363	34,310,387	35,251,549	990,262
Public Safety	Emergency Management	180,754	664,212	882,112	915,843	300,217
	Fire Department	231,483,638	288,889,015	271,688,013	278,818,189	5,813,146
	Police Department	209,812,438	181,712,168	181,849,878	181,152,028	-628,850
Total		421,966,830	471,265,395	454,419,903	460,886,060	5,484,513
Streets	Control Fleet Management	2,825,369	2,193,232	3,110,444	3,047,585	-63,119
	Office of Streets	1,788,117	2,138,889	2,777,117	2,827,874	-44,763
	Public Works Department	47,913,493	45,763,039	800,767,513	800,767,513	-480,130
	Street Removal	28,368,776	22,712,894	24,788,144	25,066,583	278,440
Total		80,905,755	72,808,054	830,743,218	829,708,515	-1,034,703
Non-Mayoral Departments	City Clerk	1,234,257	1,234,257	1,234,257	1,234,257	0
	City Council	3,248,445	3,442,478	3,728,399	3,728,399	0
	Finance Commission	274,638	274,638	274,638	274,638	0
Total		4,757,340	4,951,373	5,237,294	5,237,294	0
Grand Total		2,459,990,368	2,378,078,213	2,438,437,077	2,384,966,827	-54,470,250



Source: "Unpacking the Boston Police Budget,"
ACLU of Massachusetts Data for Justice Project

5. Use color sparingly, intentionally, and consistently.

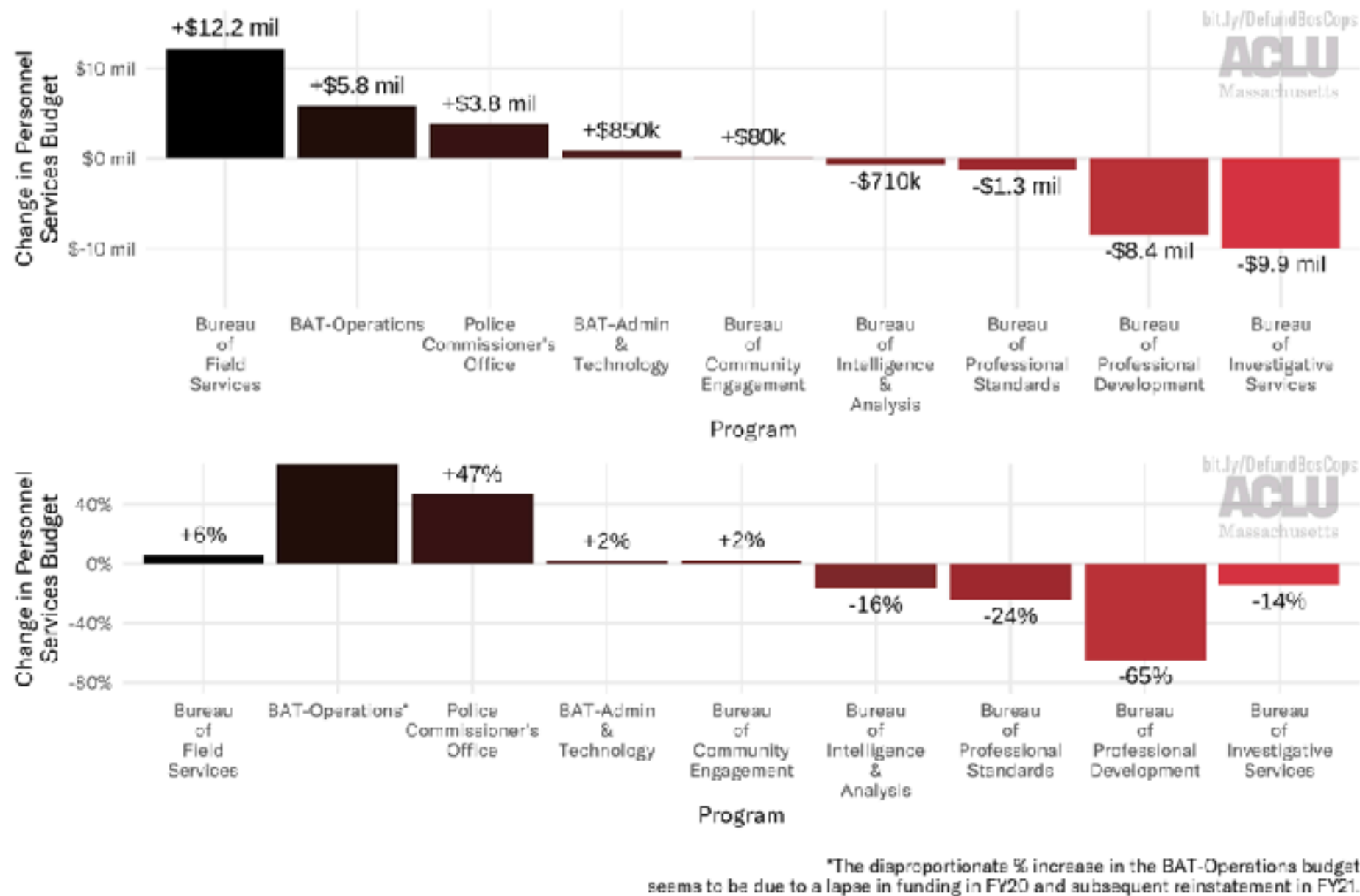
- Keep branding in mind
- Remember colorblindness



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GT America
Century Schoolbook

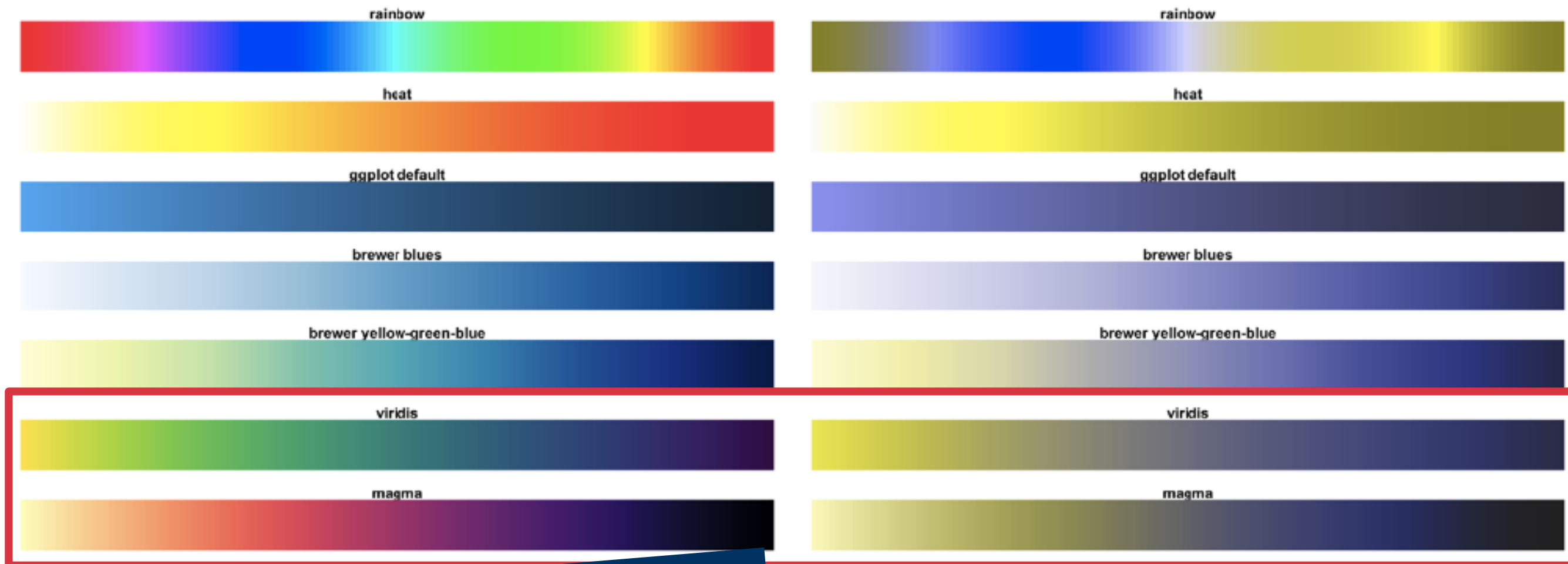


Source Serif
Inter



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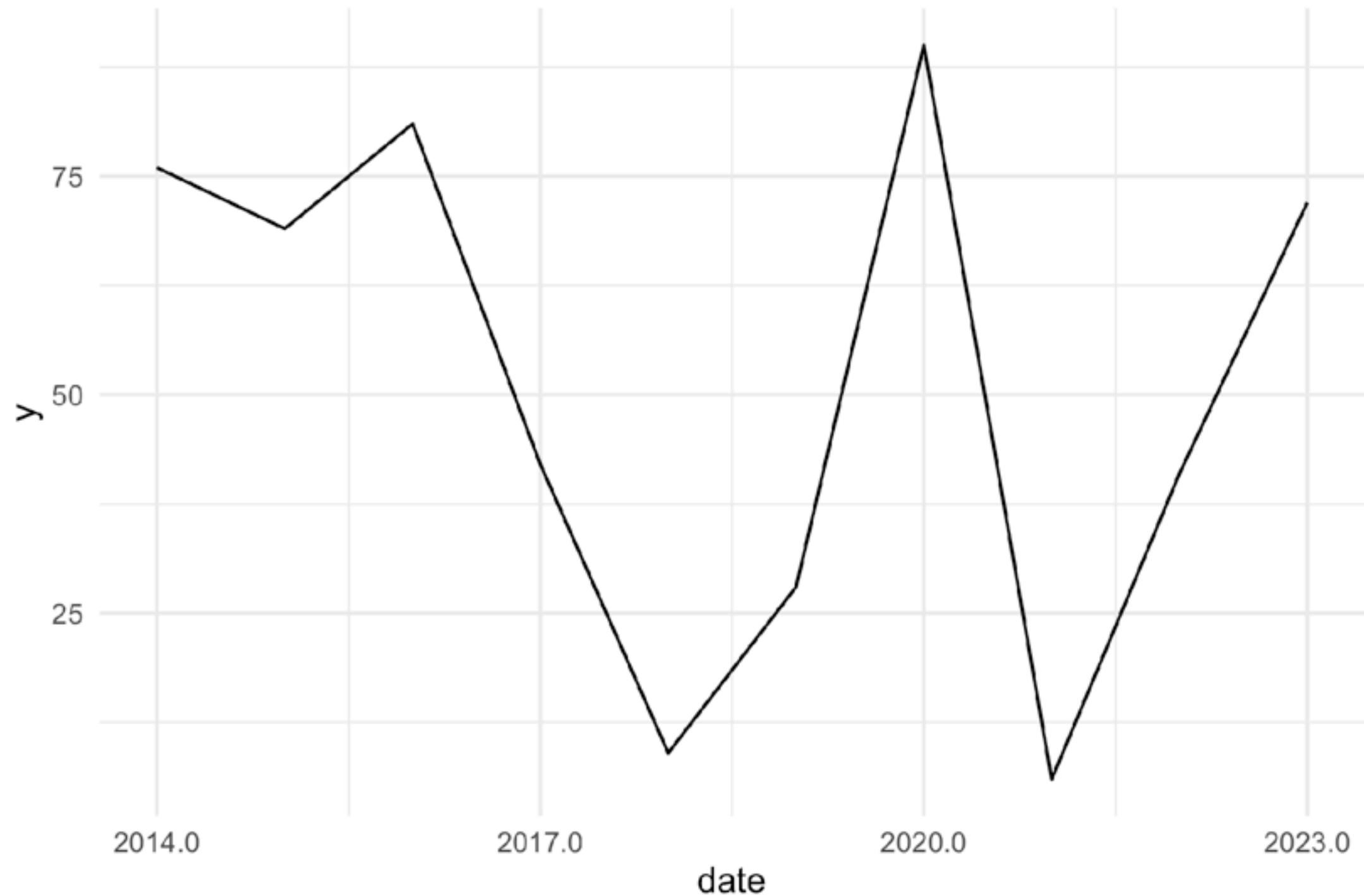


Paul Tol's Palettes

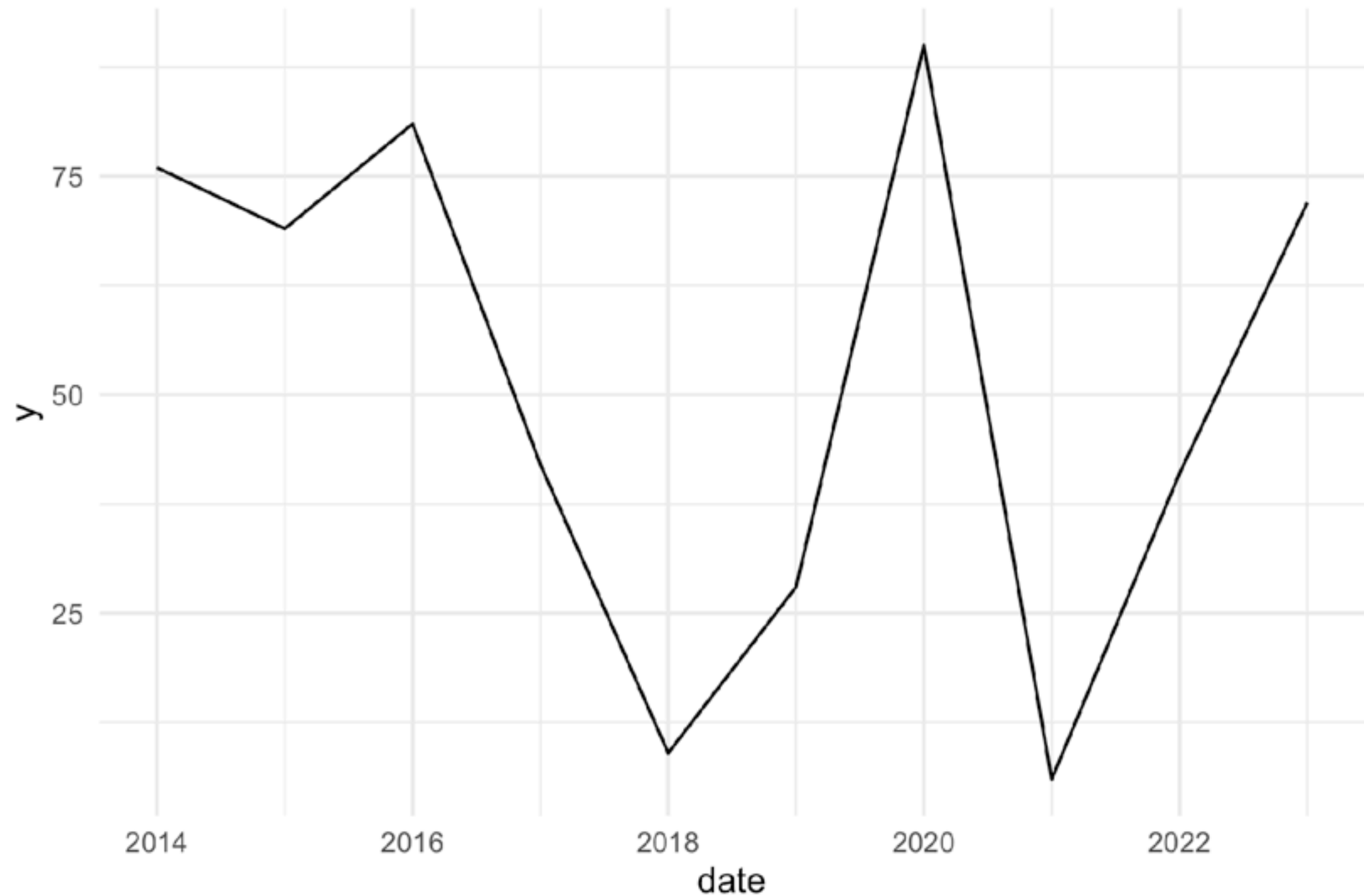
Deuteranopia

Source: "Introduction to the viridis color maps," Bob Rudis, Noam Ross and Simon Garnier, CRAN

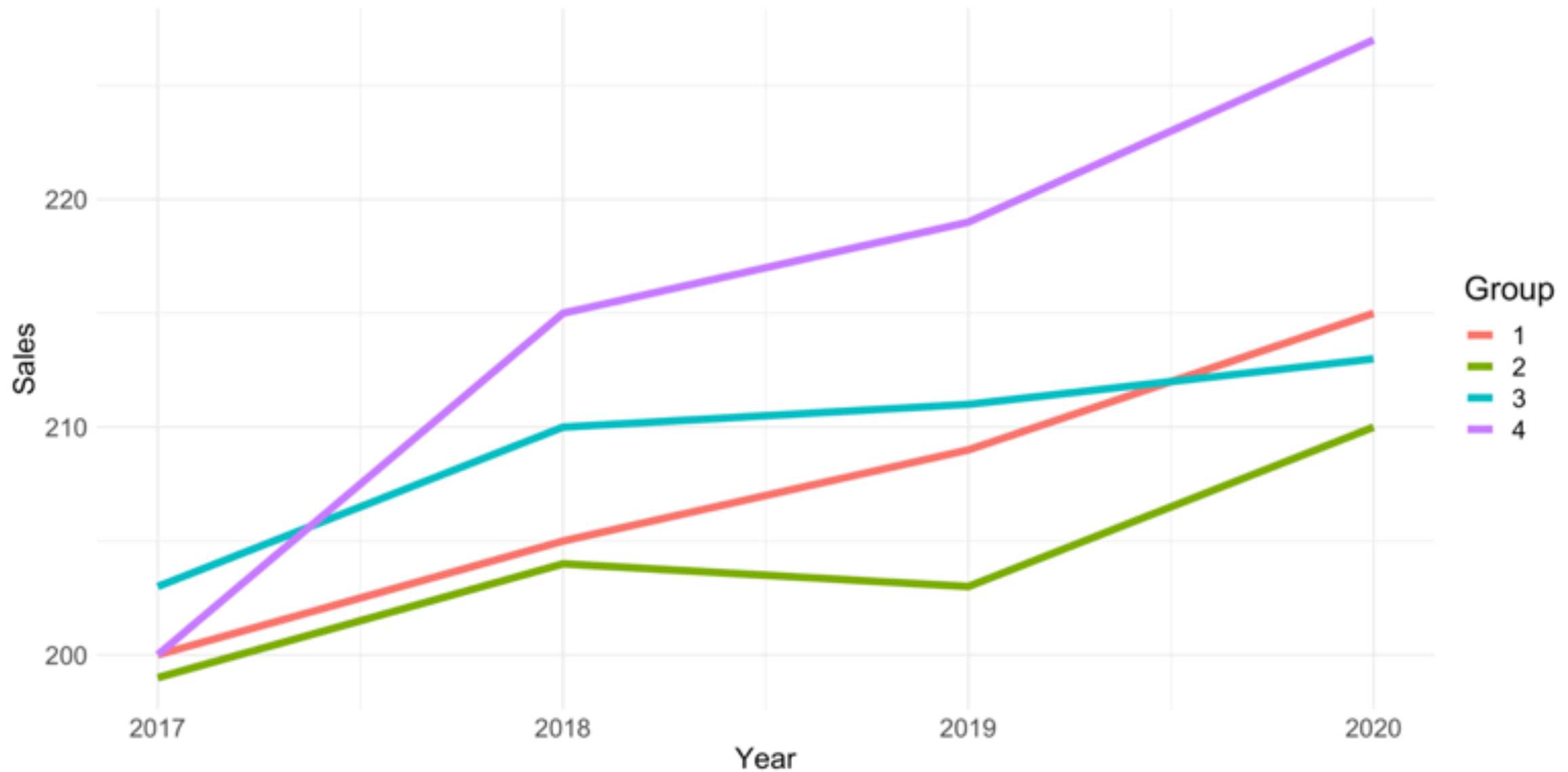
6. Use reasonable axis tick formats and frequencies, not just the default.



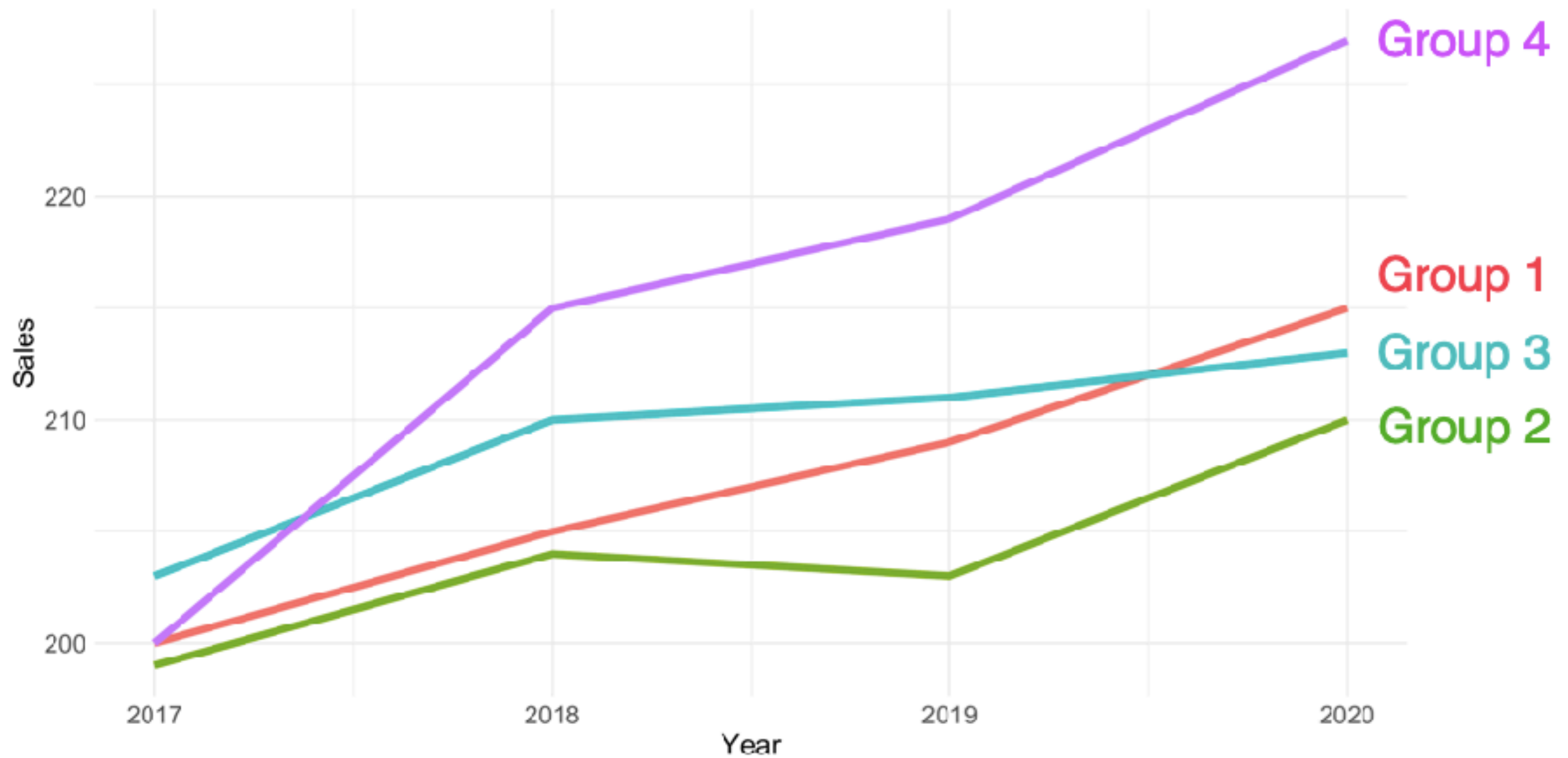
6. Use reasonable axis tick formats and frequencies, not just the default.



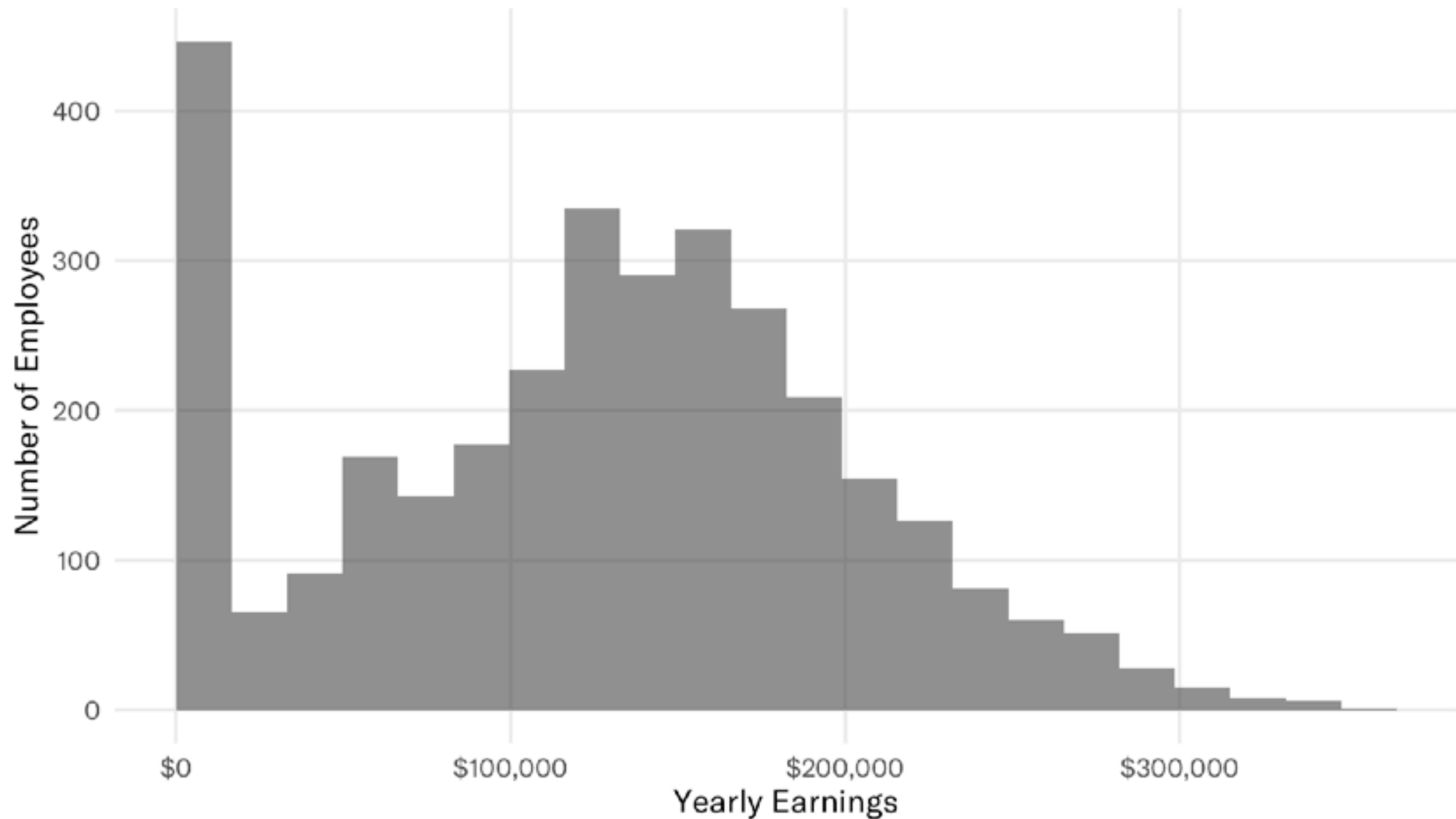
7. Consider creative alternatives to legends.



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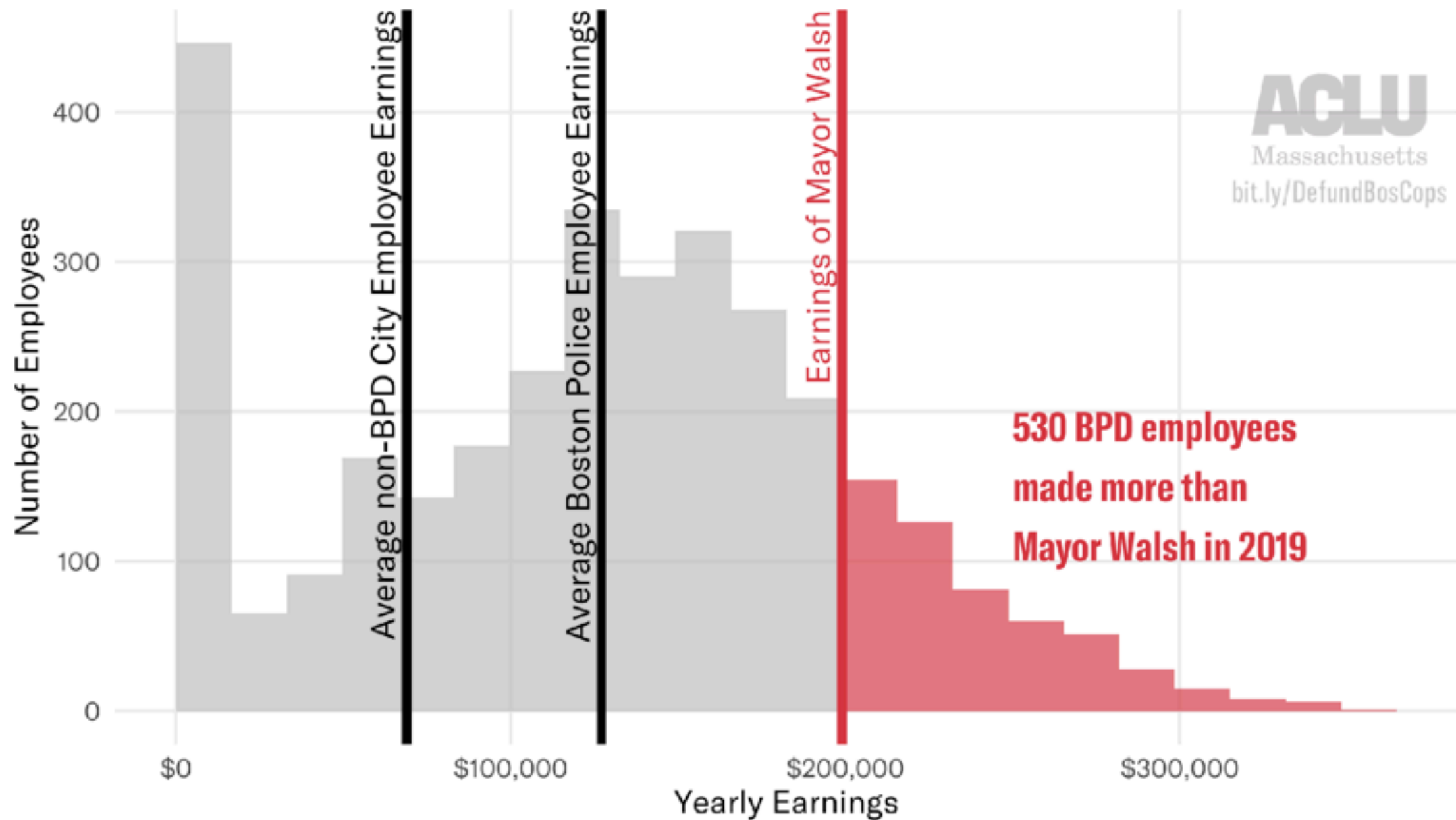


8. Don't be afraid of annotations. Use whatever software you prefer to add them.



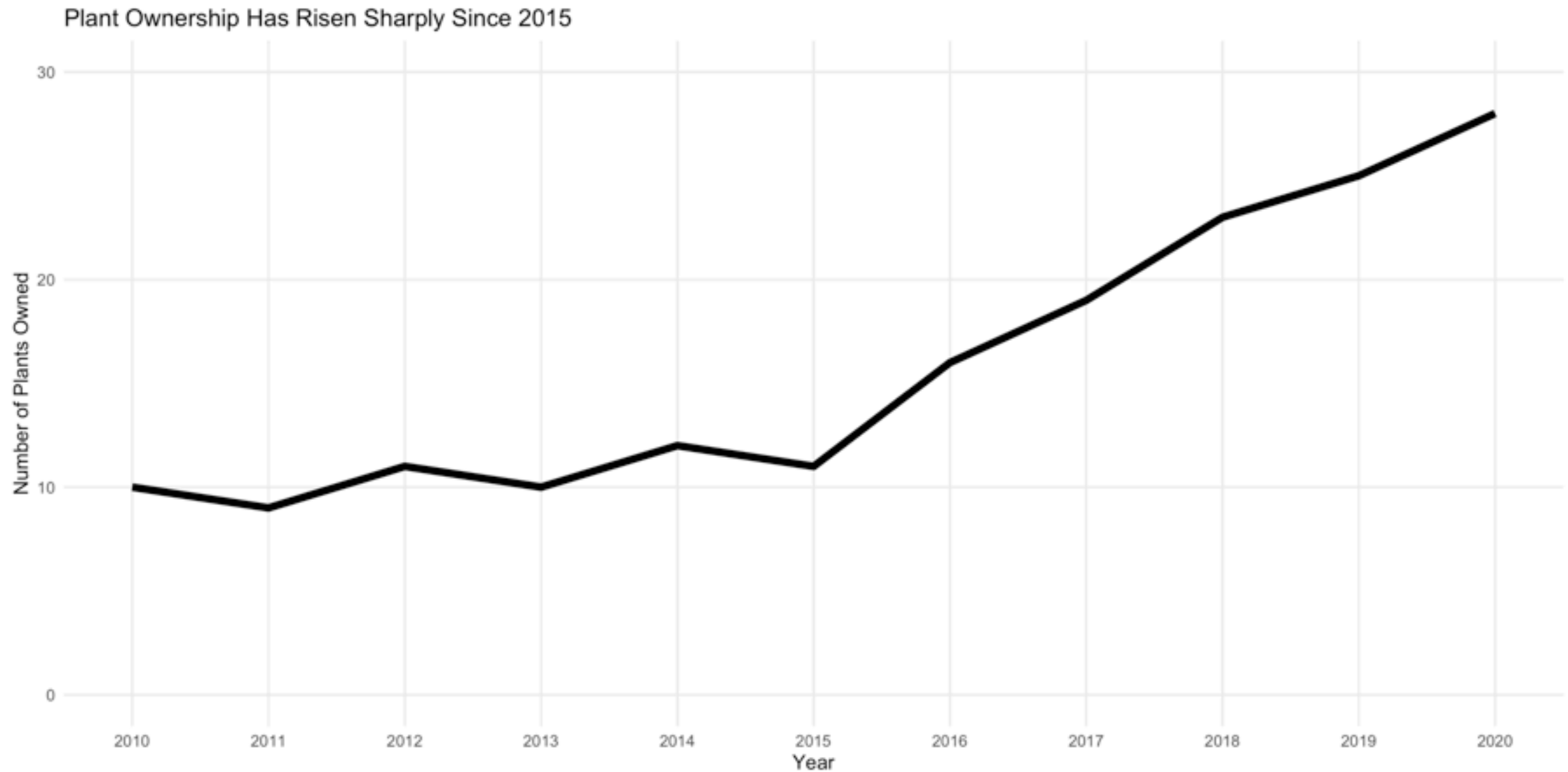
Source: "Unpacking the Boston Police Budget,"
ACLU of Massachusetts Data for Justice Project

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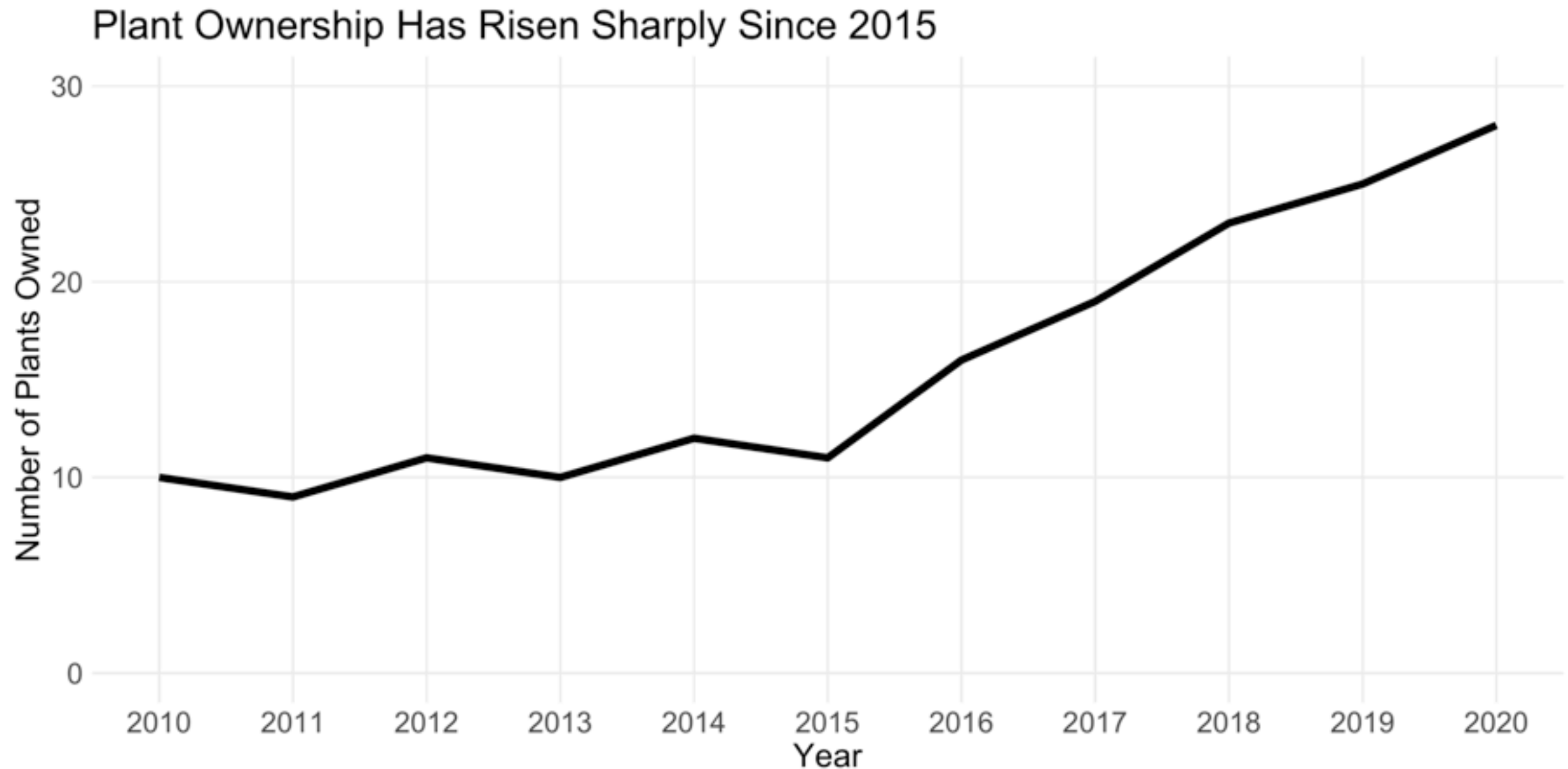


Source: "Unpacking the Boston Police Budget,"
ACLU of Massachusetts Data for Justice Project

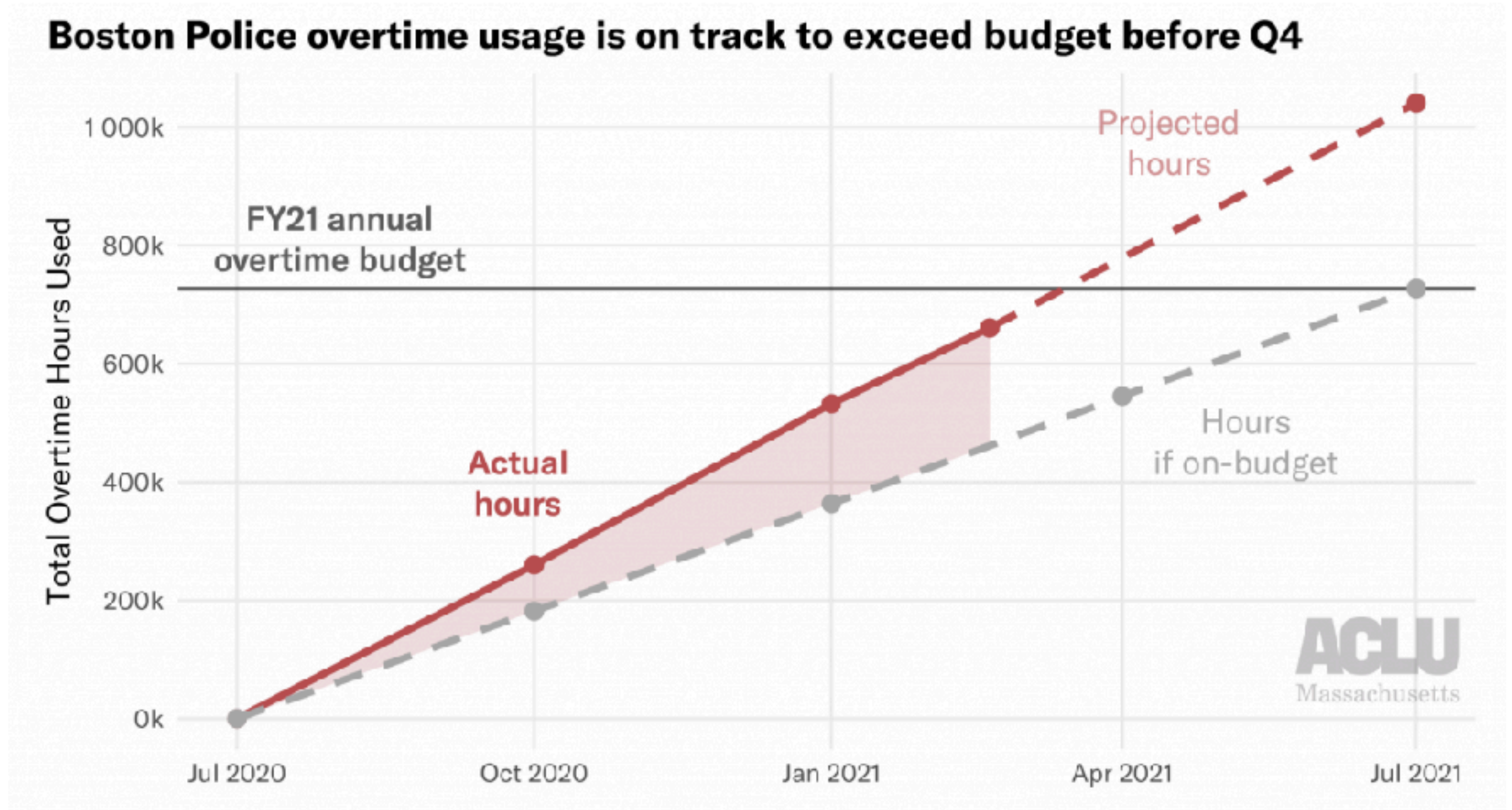
9. **Make your text bigger** than you'd think. Nobody likes to squint!



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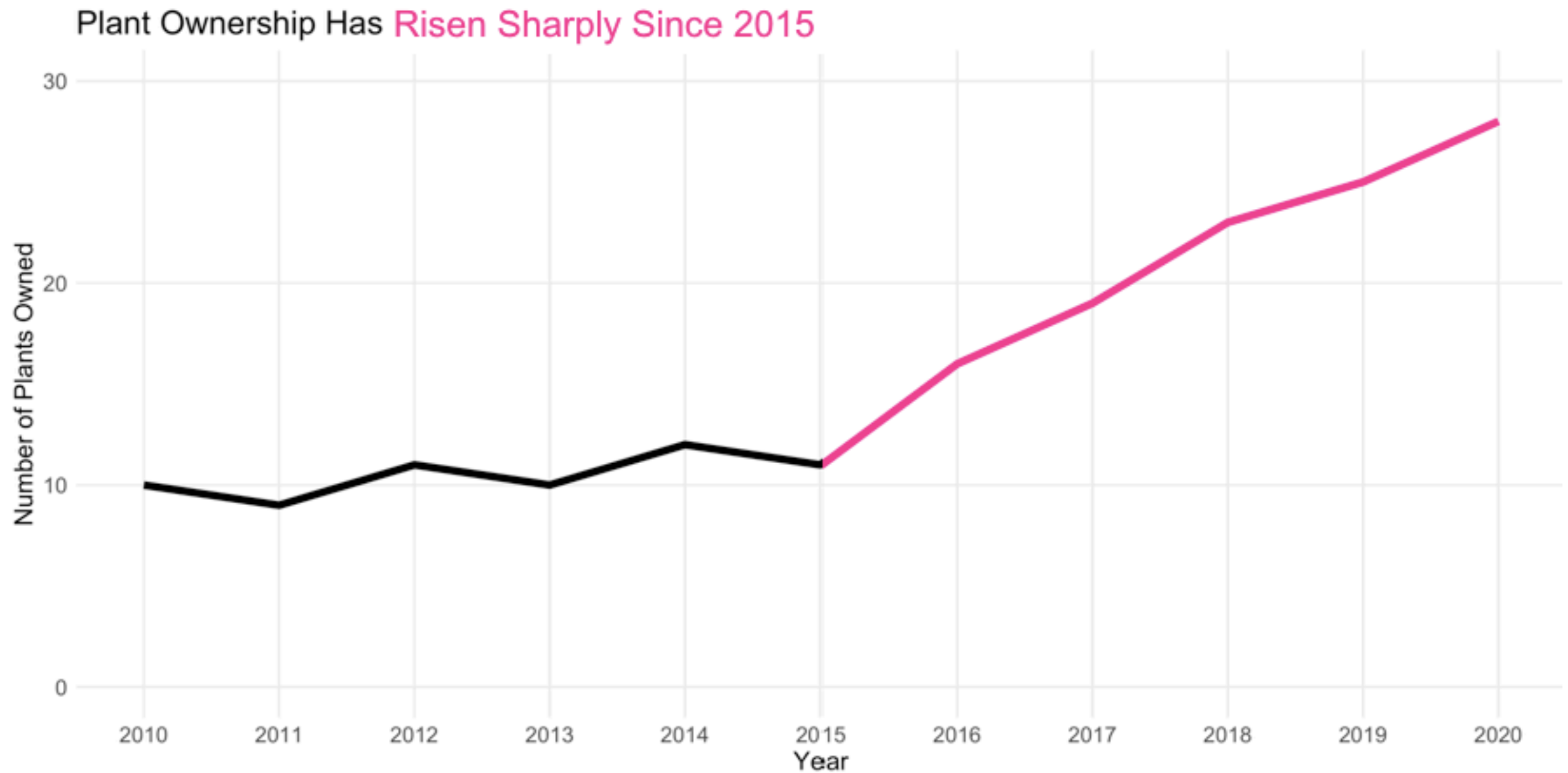


10. Use the blink test.



Source: "A Mayor's Roadmap to Curb Boston Police Overtime,"
ACLU of Massachusetts Data for Justice Project

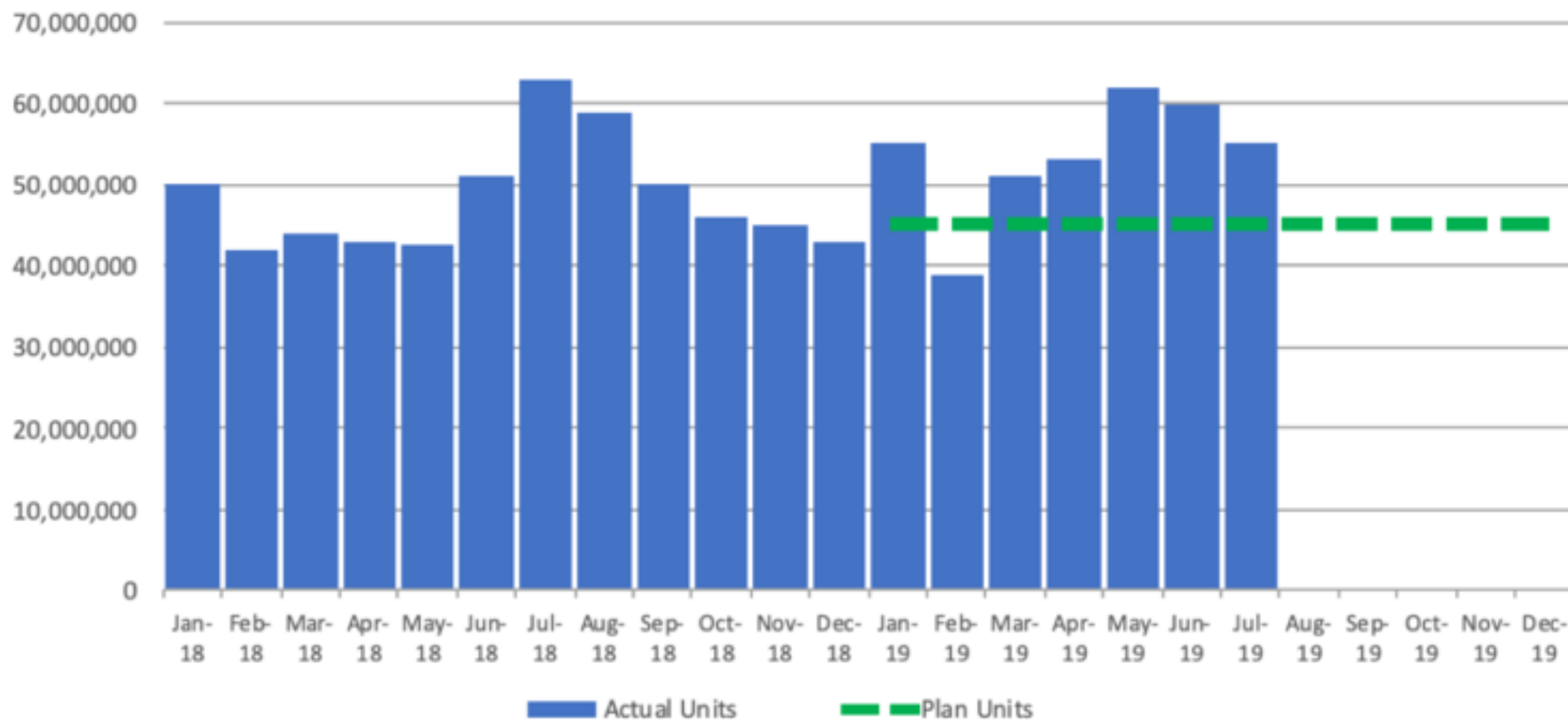
10. Use the blink test.



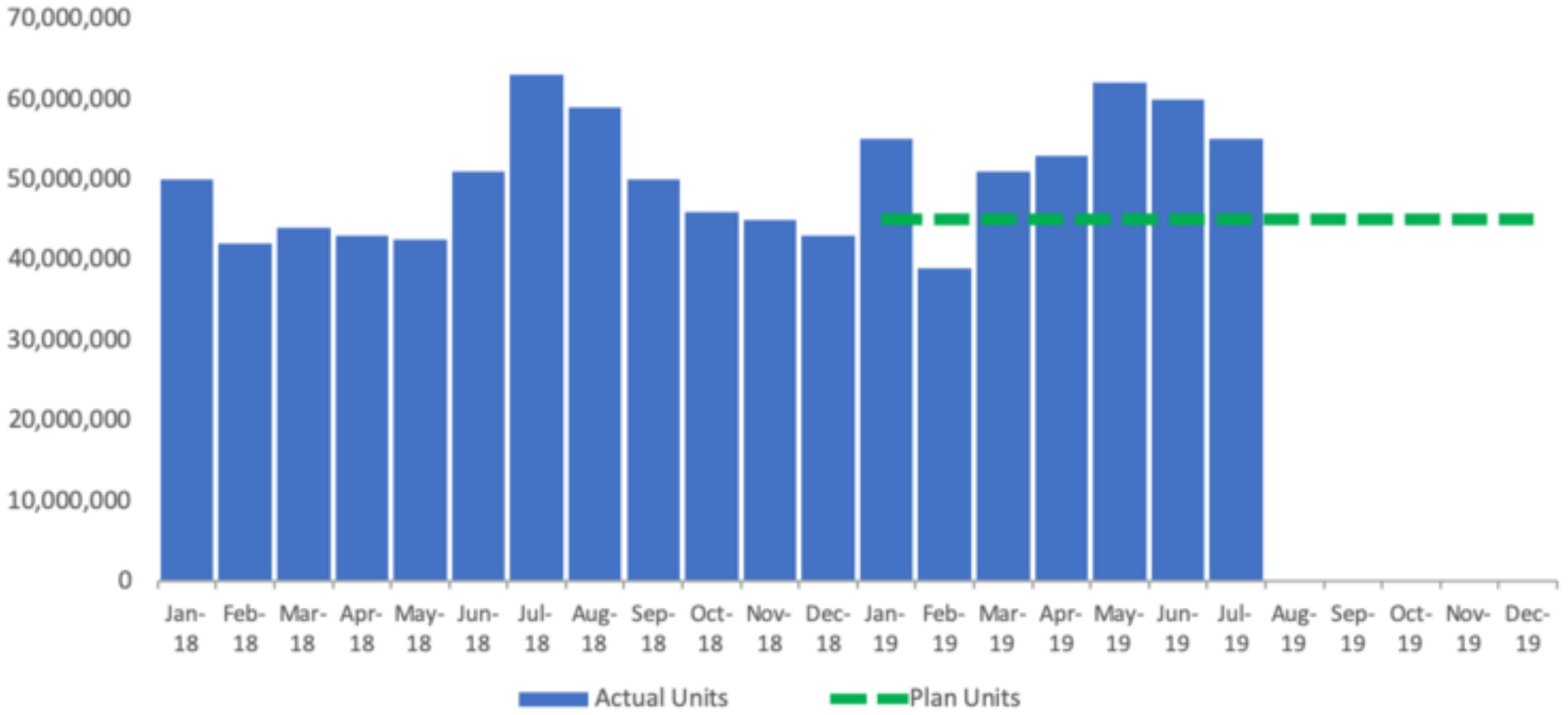
1. **Know your audience!**
 - For public vizzes, this usually requires bar, line, and scatter plots.
 - Basically never use undefined acronyms or symbols.
2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.
3. **Use descriptive titles:** *tell* your viewer what they should see, and don't be redundant.
4. **Be selective in data and design.** Consider your data-to-ink ratio and remove distractions.
5. **Use color sparingly, intentionally, and consistently.**
 - Keep branding in mind
 - Remember colorblindness
6. **Use reasonable axis tick** formats and frequencies, not just the default.
7. **Consider creative alternatives to legends.**
8. **Don't be afraid of annotations.** Use whatever software you prefer to add them.
9. **Make your text bigger** than you'd think. Nobody likes to squint!
10. **Use the blink test.**

Improve! This! Viz!

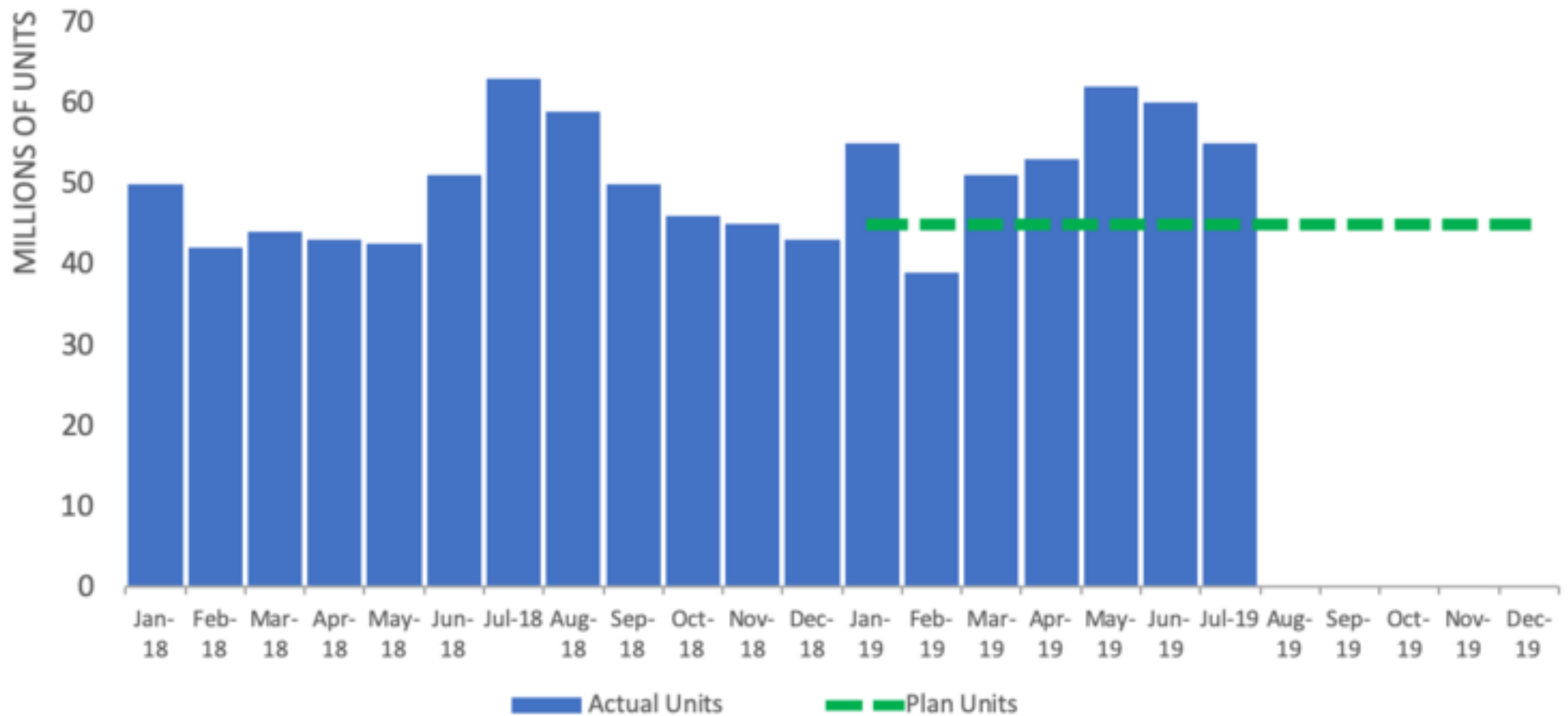
PRIMARY PRODUCT Units



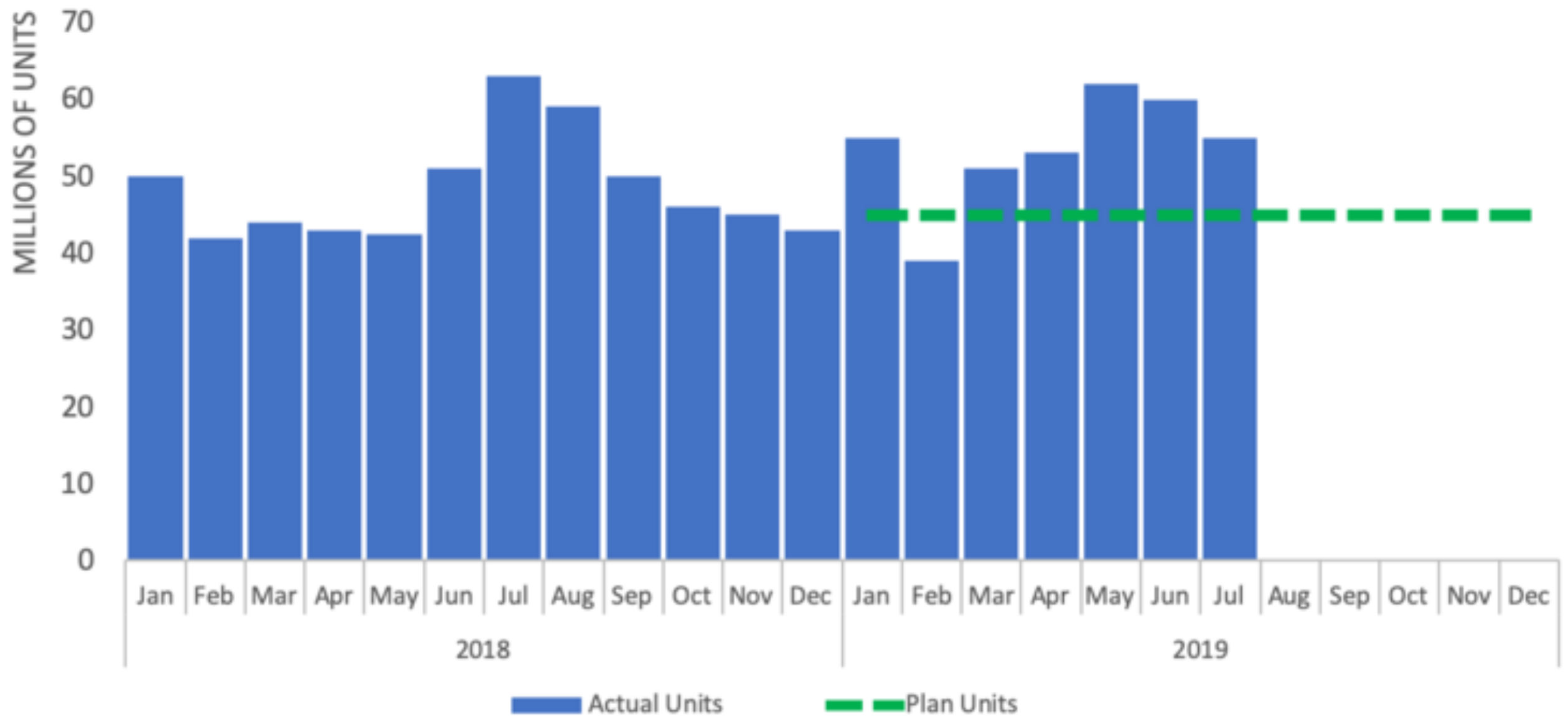
PRIMARY PRODUCT Units



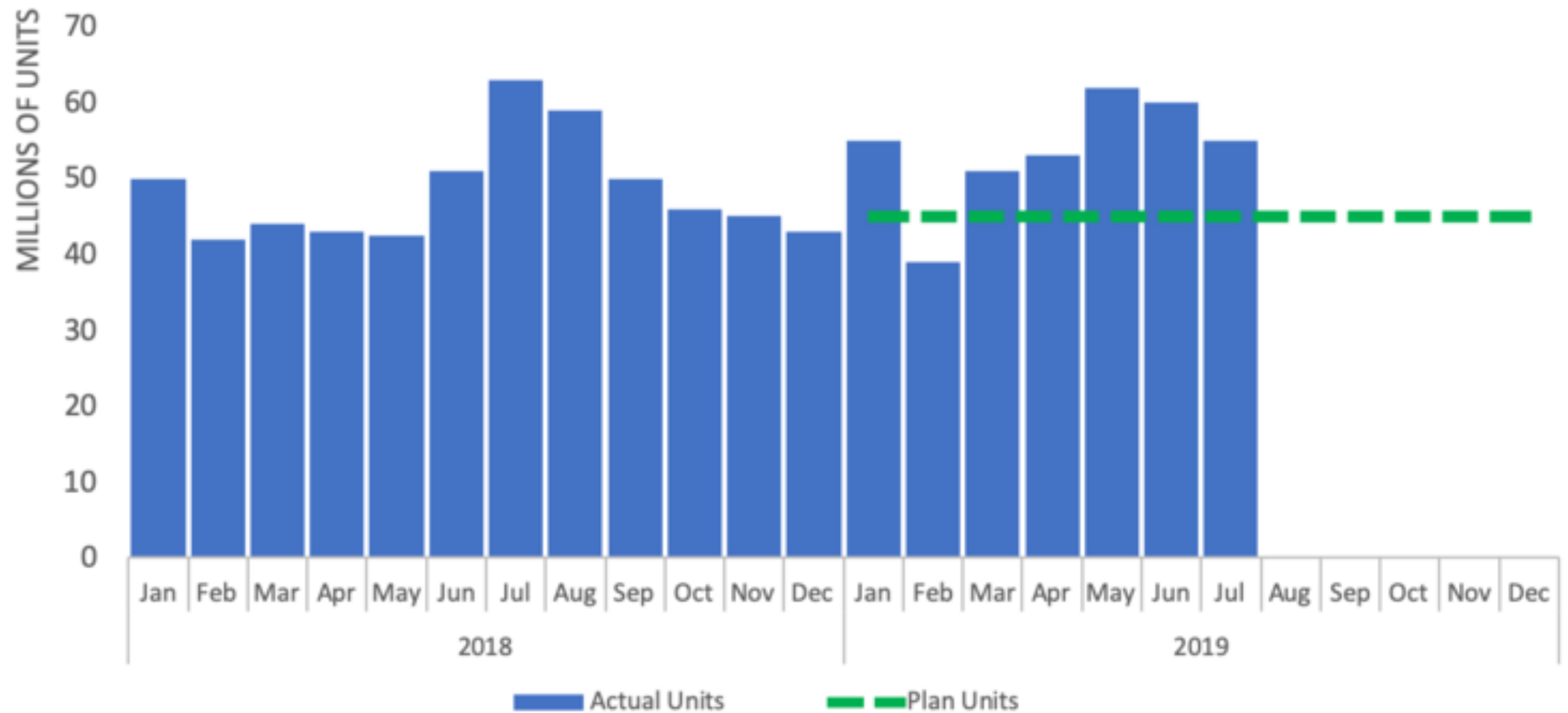
PRIMARY PRODUCT Units



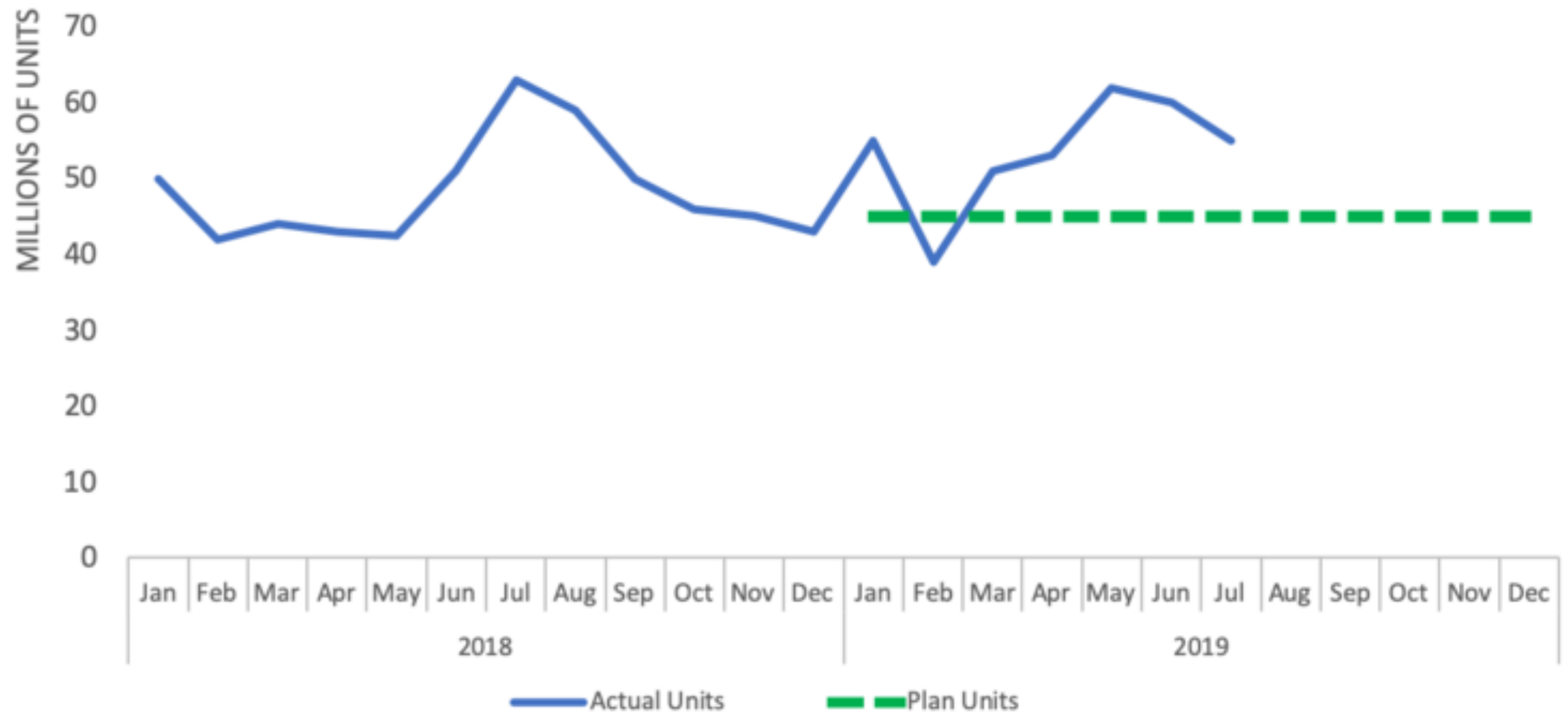
PRIMARY PRODUCT Units



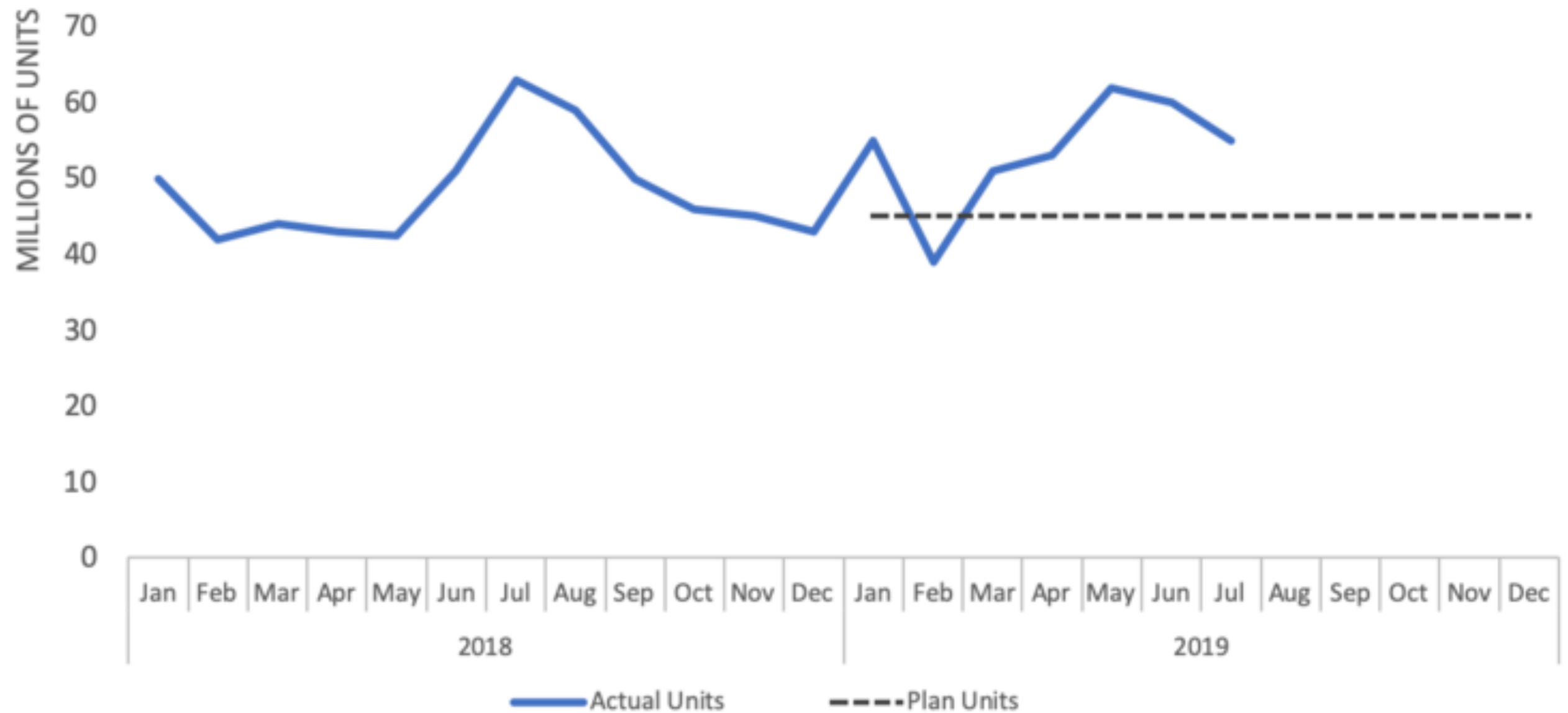
PRIMARY PRODUCT: unit sales



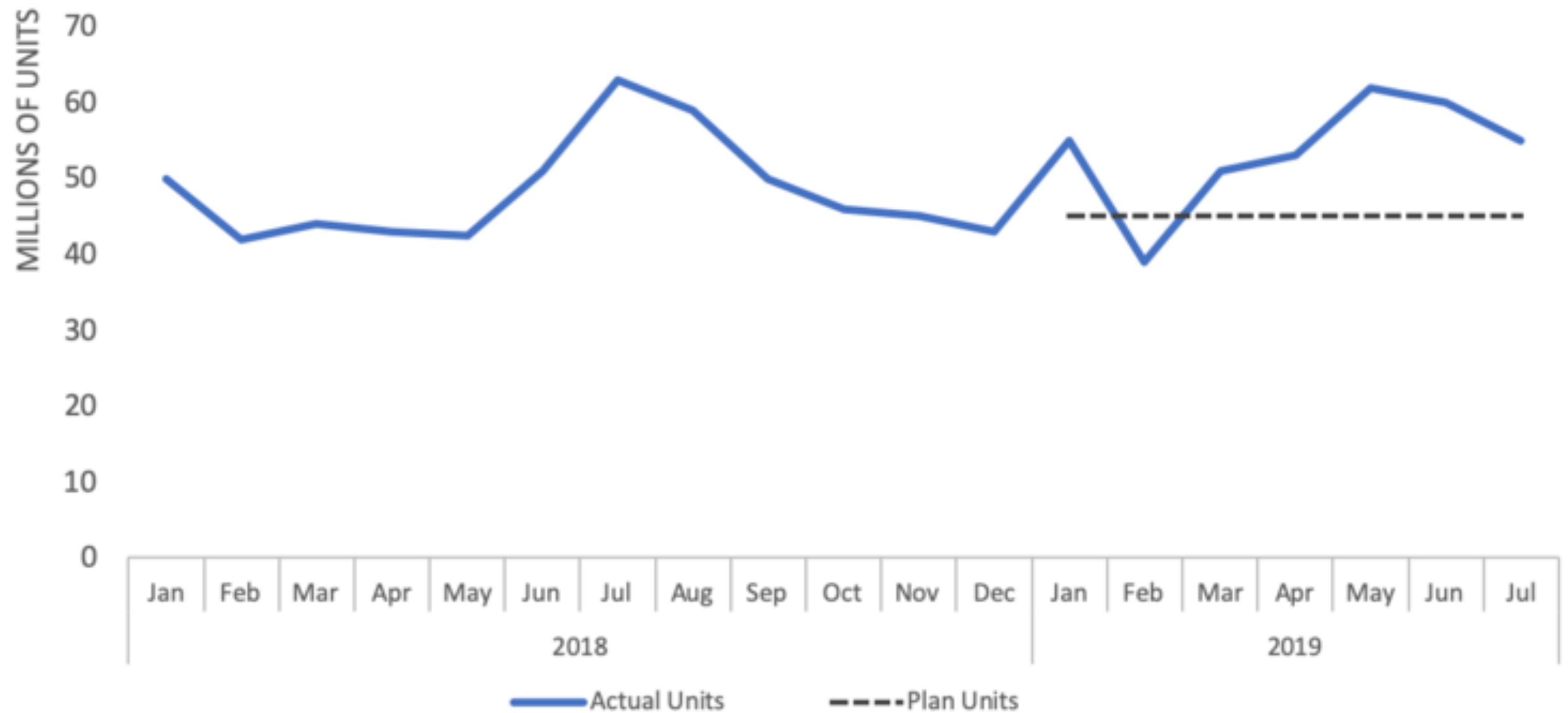
PRIMARY PRODUCT: unit sales



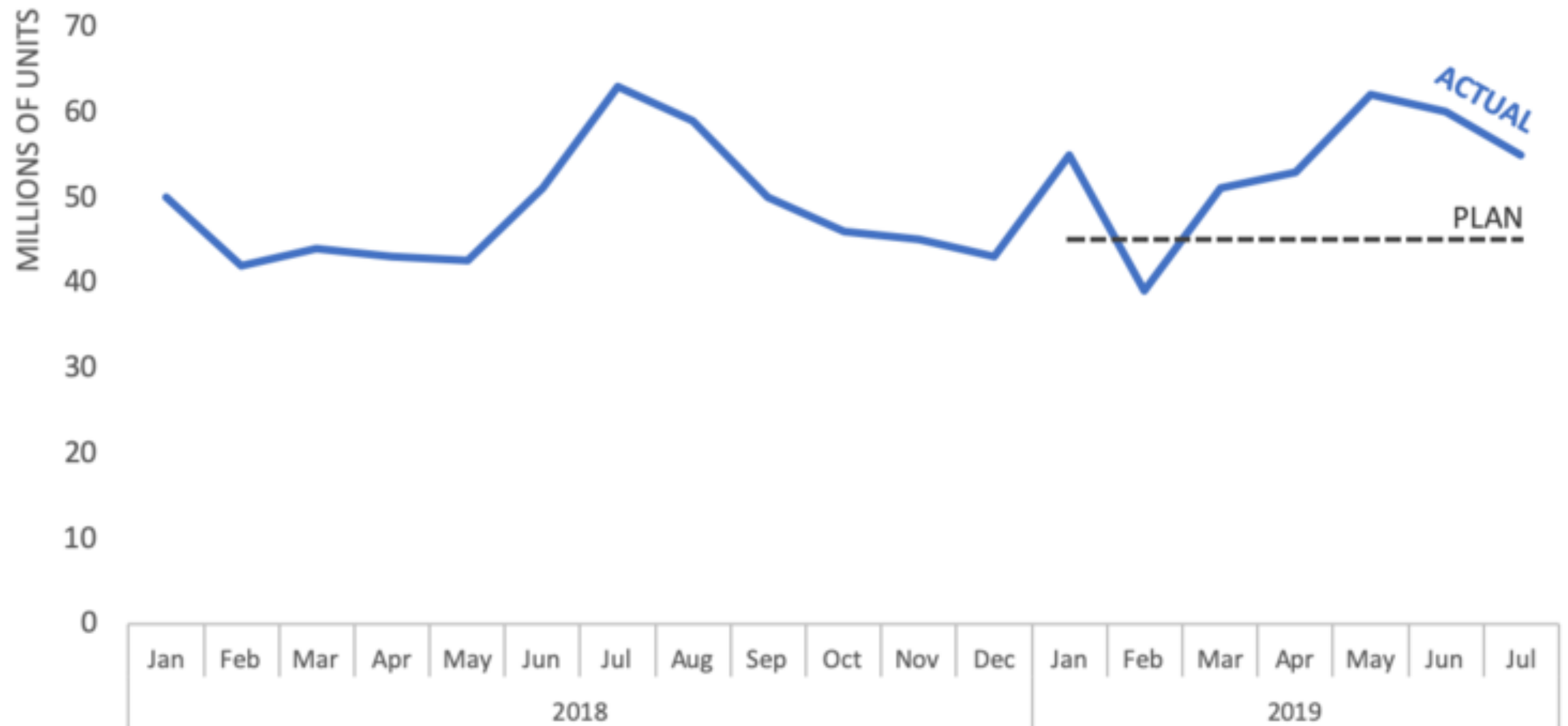
PRIMARY PRODUCT: unit sales



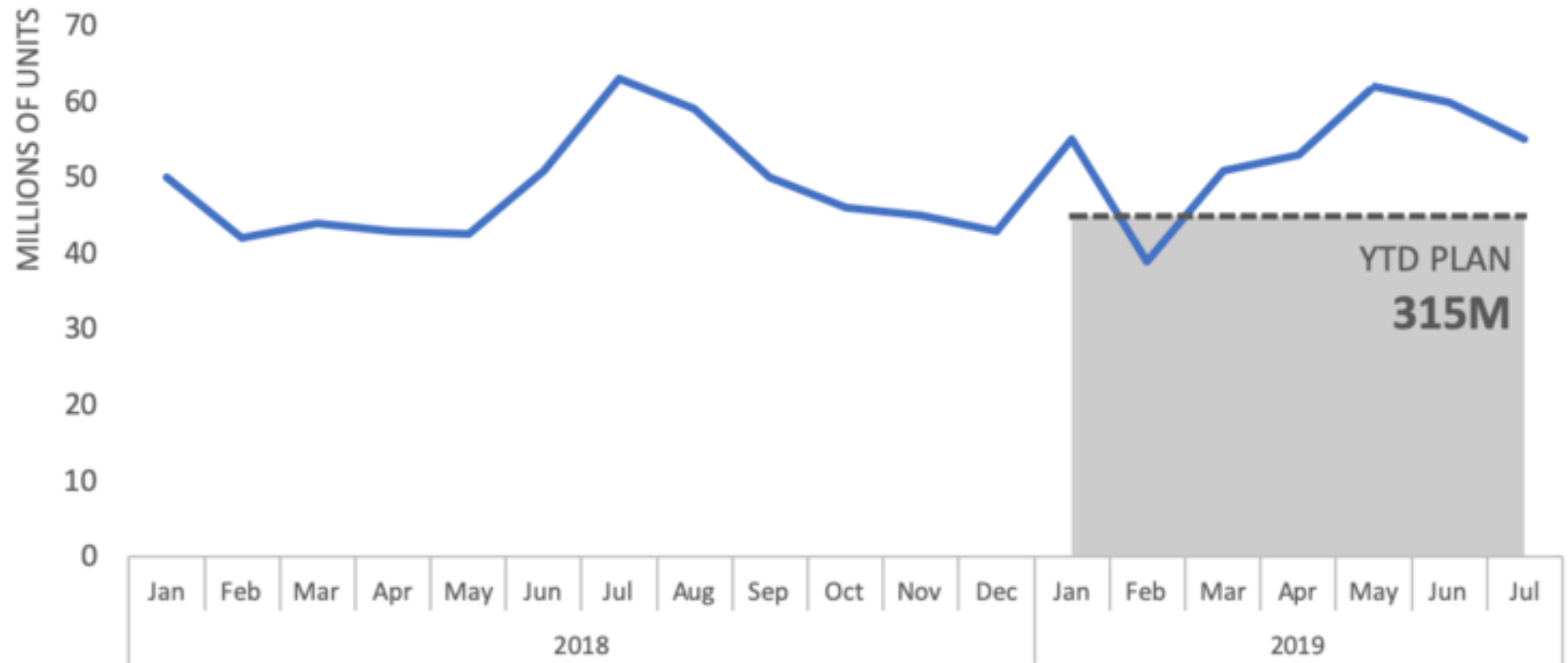
PRIMARY PRODUCT: unit sales



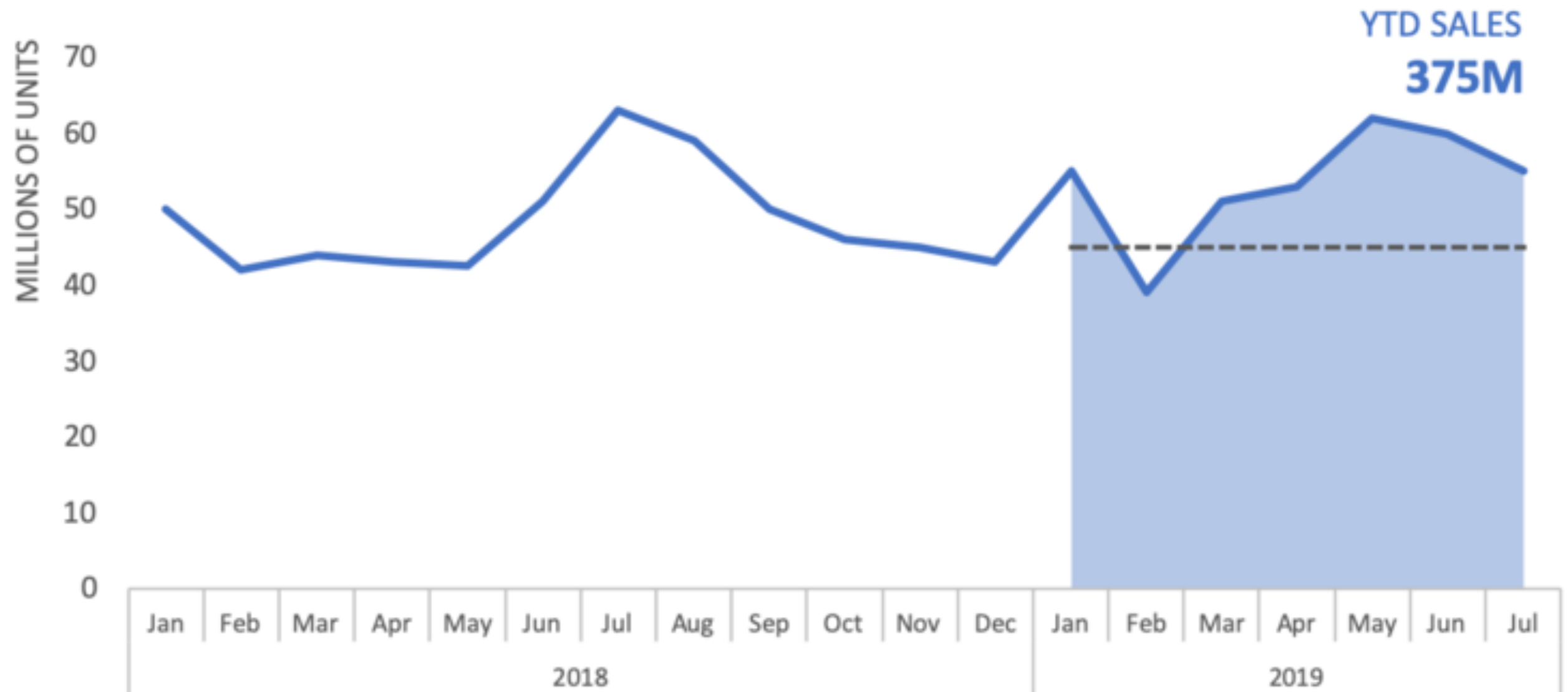
PRIMARY PRODUCT: unit sales



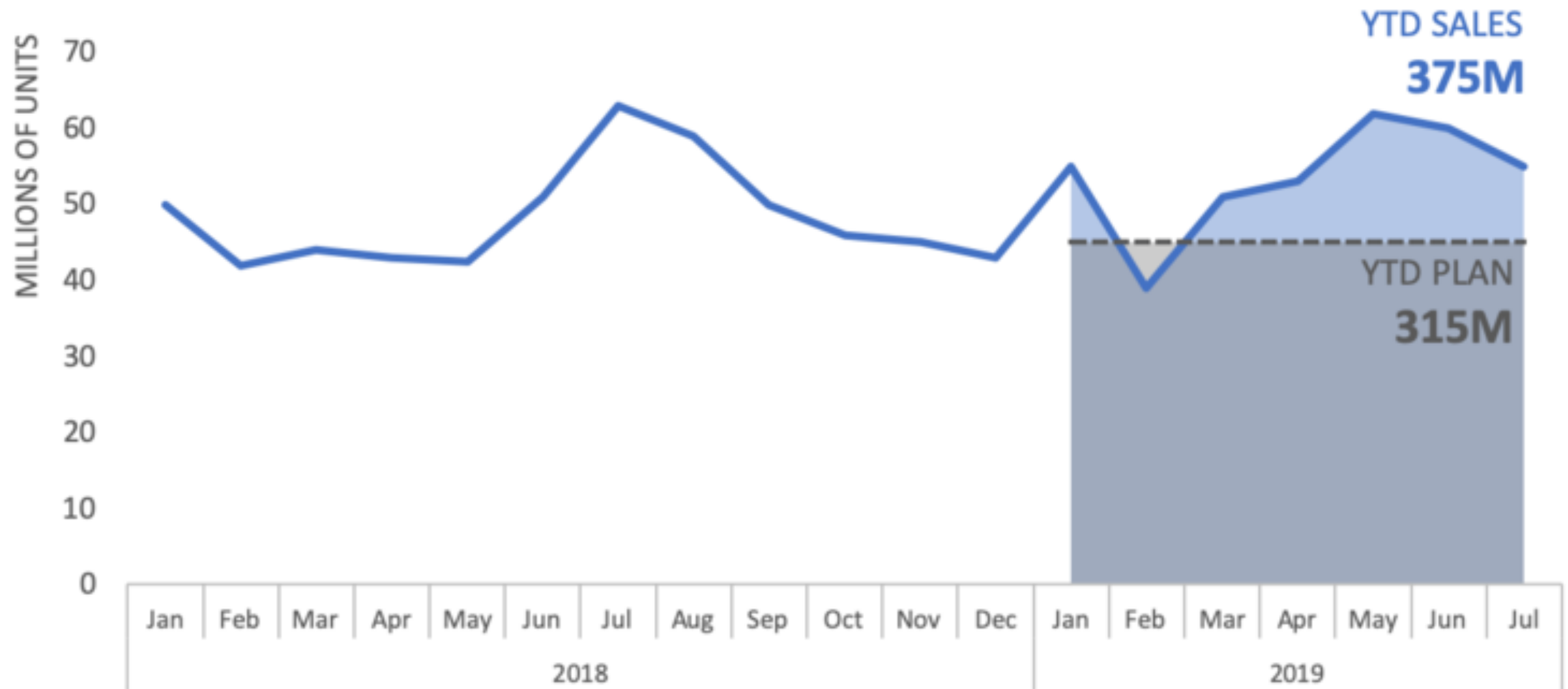
PRIMARY PRODUCT: unit sales



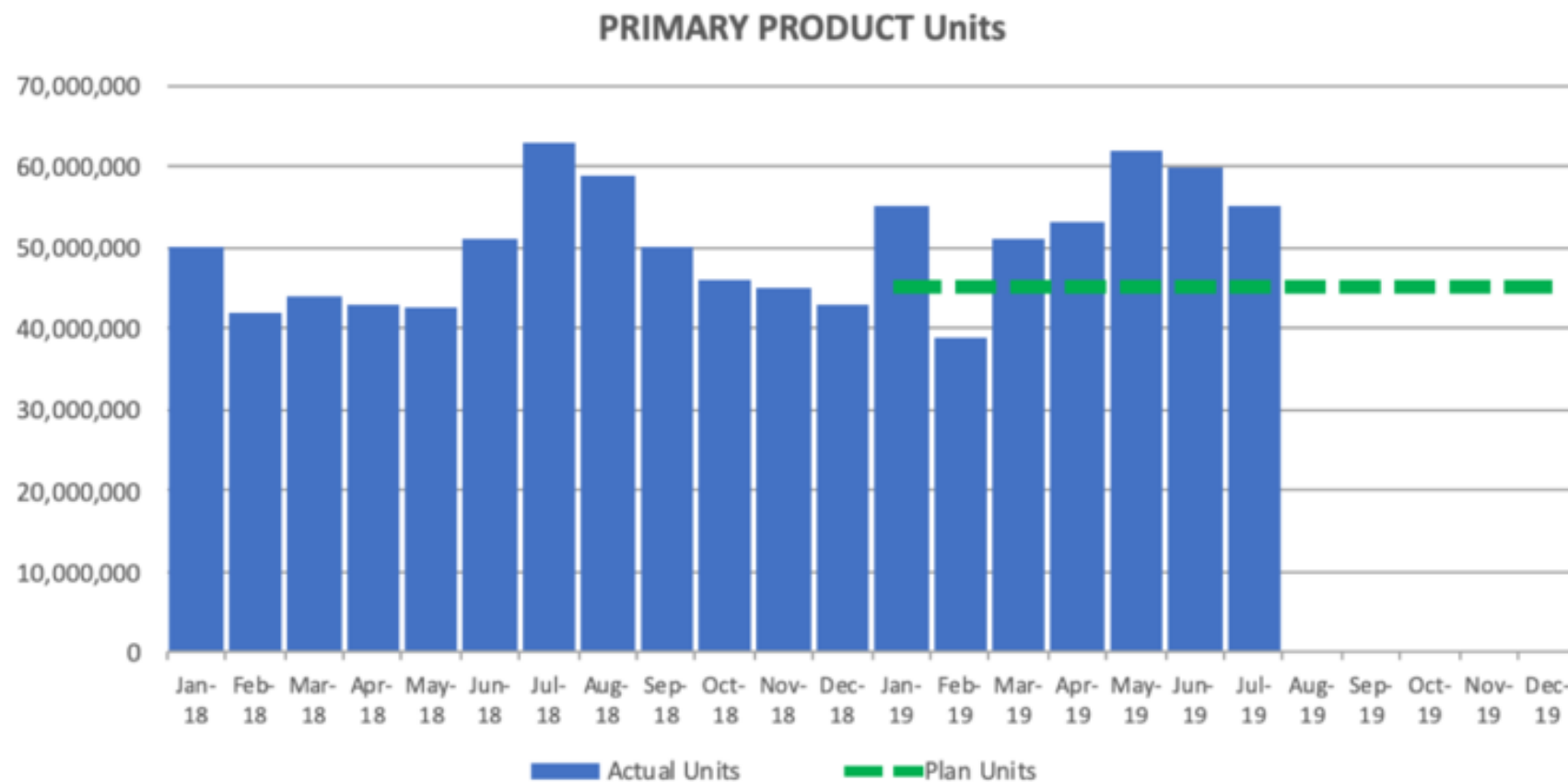
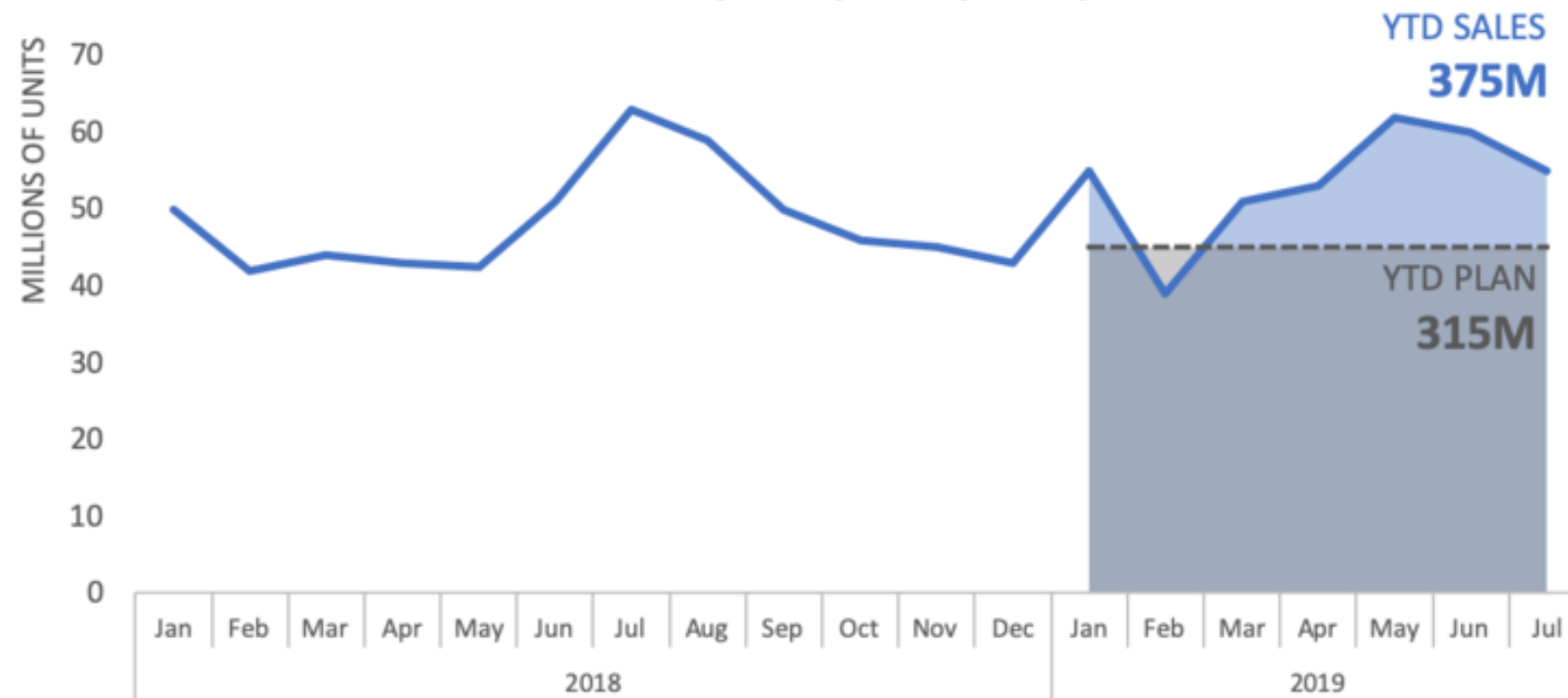
PRIMARY PRODUCT: unit sales



PRIMARY PRODUCT: **unit sales exceed plan by nearly 20% year to date**



PRIMARY PRODUCT: **unit sales exceed plan by nearly 20% year to date**



***Group
Exercise:***

Write the headline

*[http://tiny.cc/
INFO290_headlines](http://tiny.cc/INFO290_headlines)*

***Individual
Exercise:***

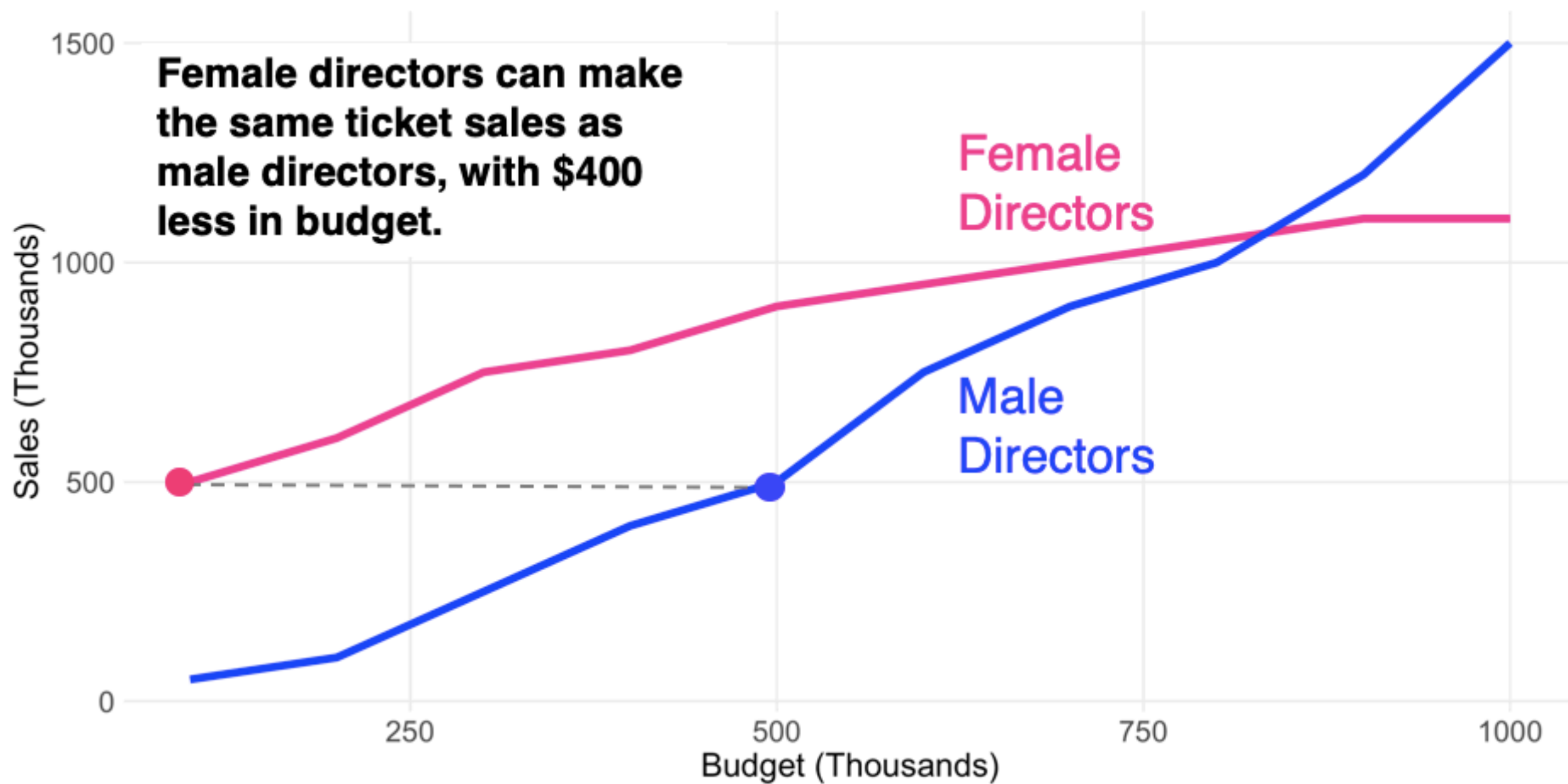
*Visualizing
tabular data by
hand*

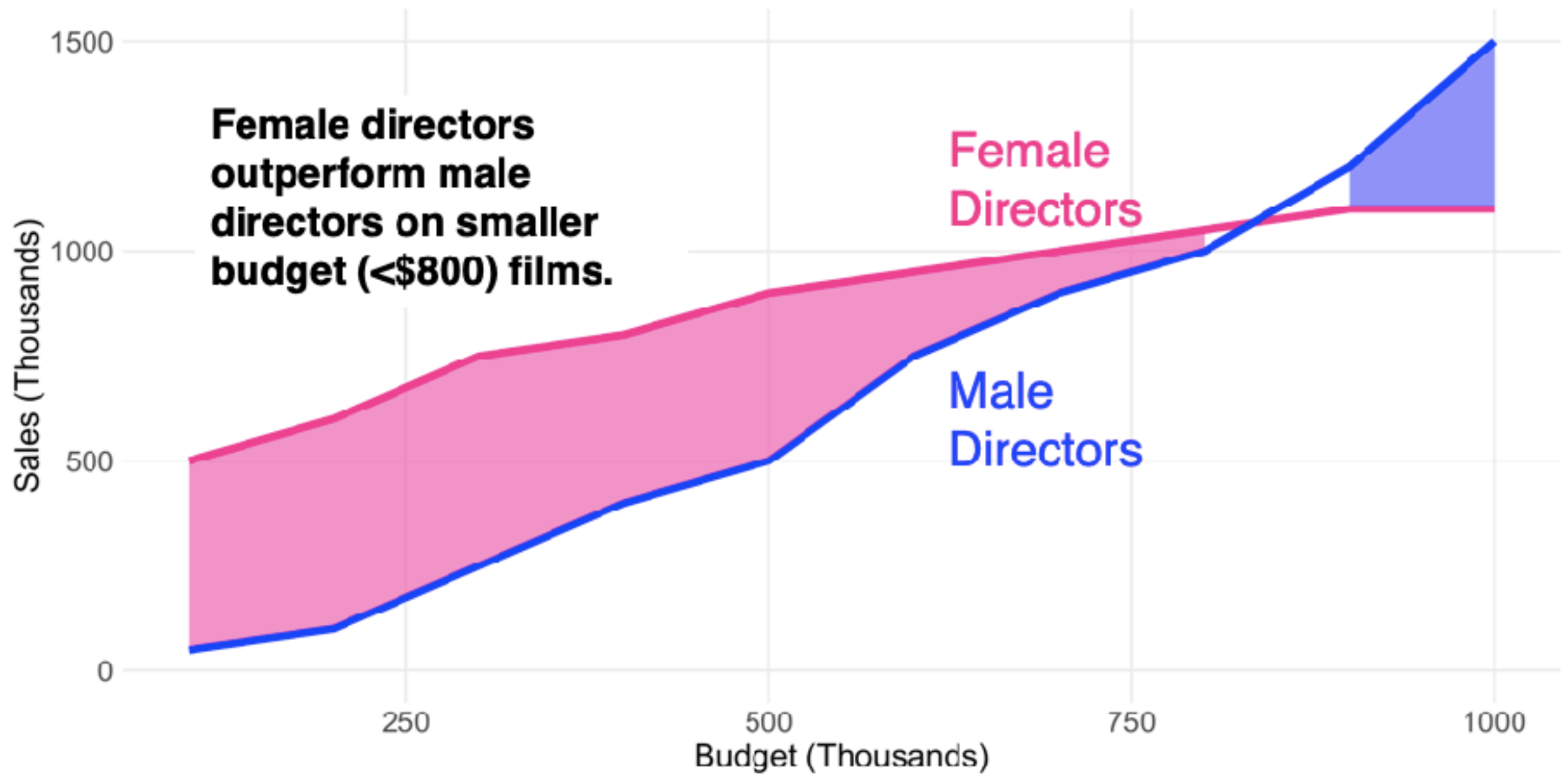
Individual Exercise:
*Visualizing tabular
data by hand*

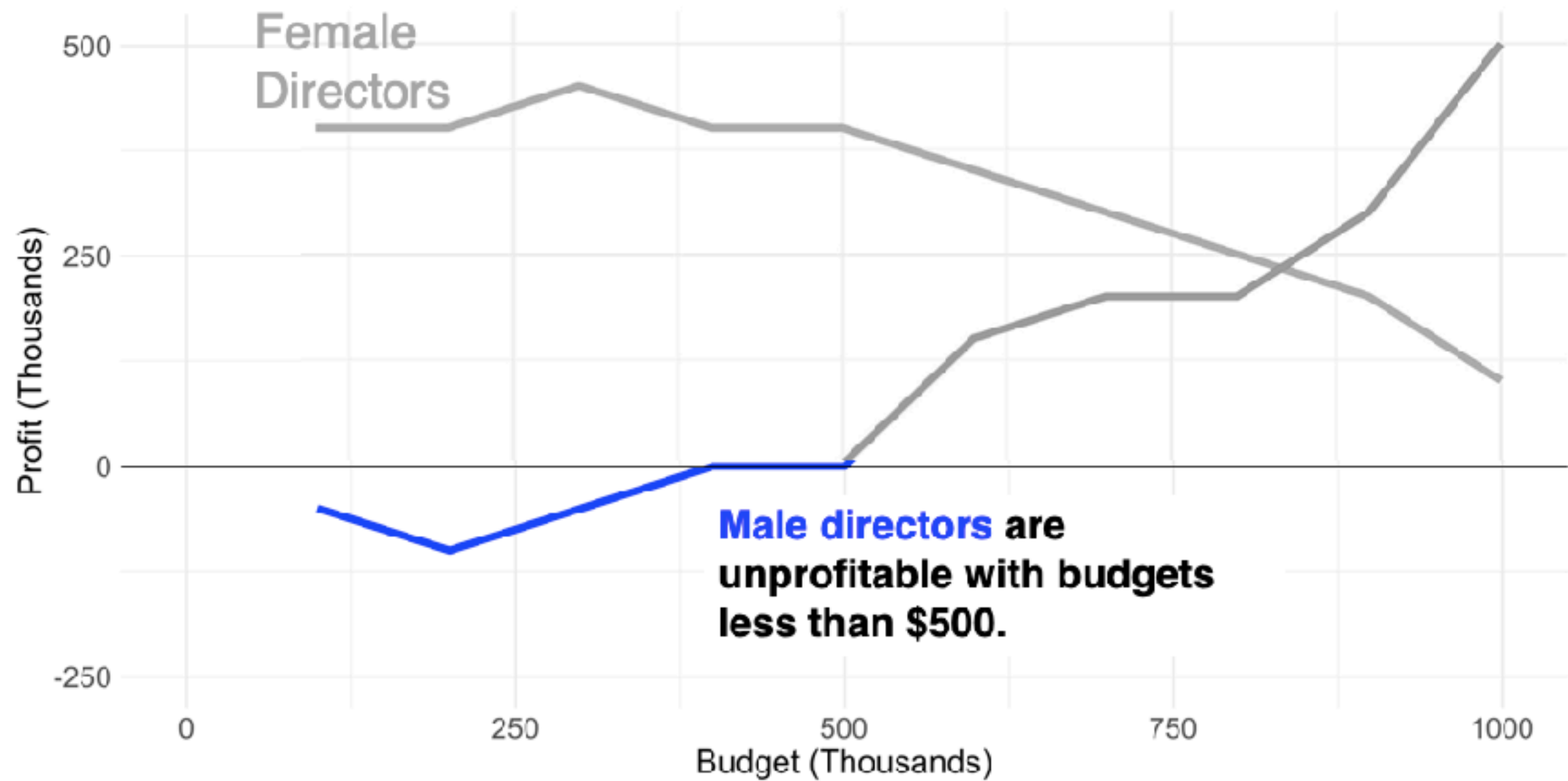
Take 7 minutes to draw, by hand, a visualization of the below data:

Budget (thousands)	Director Gender	Ticket Sales (thousands)
100	Female	500
200	Female	600
300	Female	750
400	Female	800
500	Female	900
600	Female	950
700	Female	1,000
800	Female	1,050
900	Female	1,100
1,000	Female	1,100

Budget (thousands)	Director Gender	Ticket Sales (thousands)
100	Male	50
200	Male	100
300	Male	250
400	Male	400
500	Male	500
600	Male	750
700	Male	900
800	Male	1,000
900	Male	1,200
1,000	Male	1,500







*Going interactive
and beyond*



Examples

- ▶ Tableau: [Massachusetts Trial Court](#)
- ▶ Leaflet: [Police Violence Happens Here](#)
- ▶ Plotly: [Tracking COVID-19 in MA Prisons](#)
- ▶ Bokeh: [COVID-19 in Portugal](#)
- ▶ D3: [NASA Spacecraft](#)

Where to find viz inspiration

► NYT's *The Upshot*

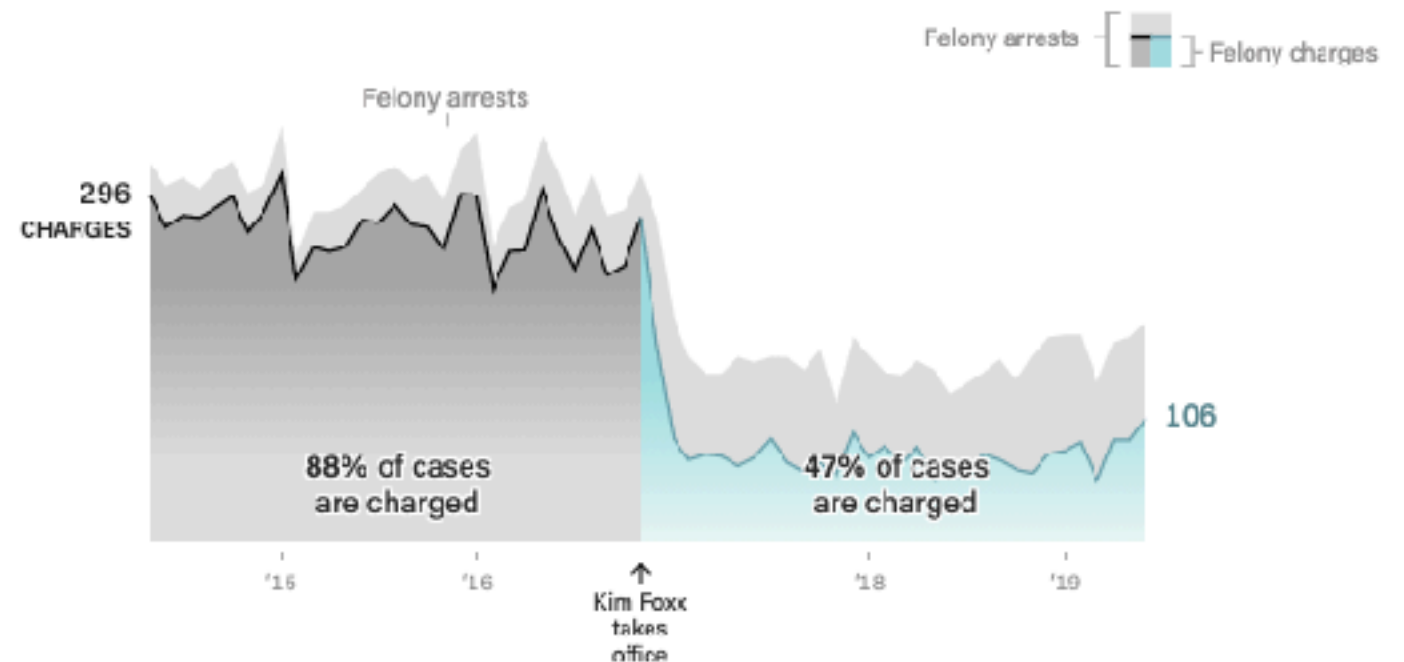


From June 24 to June 30. Source: NOAA / [Global Historical Climatological Network](#).

Where to find viz inspiration

- ▶ NYT's *The Upshot*
- ▶ *The Pudding*

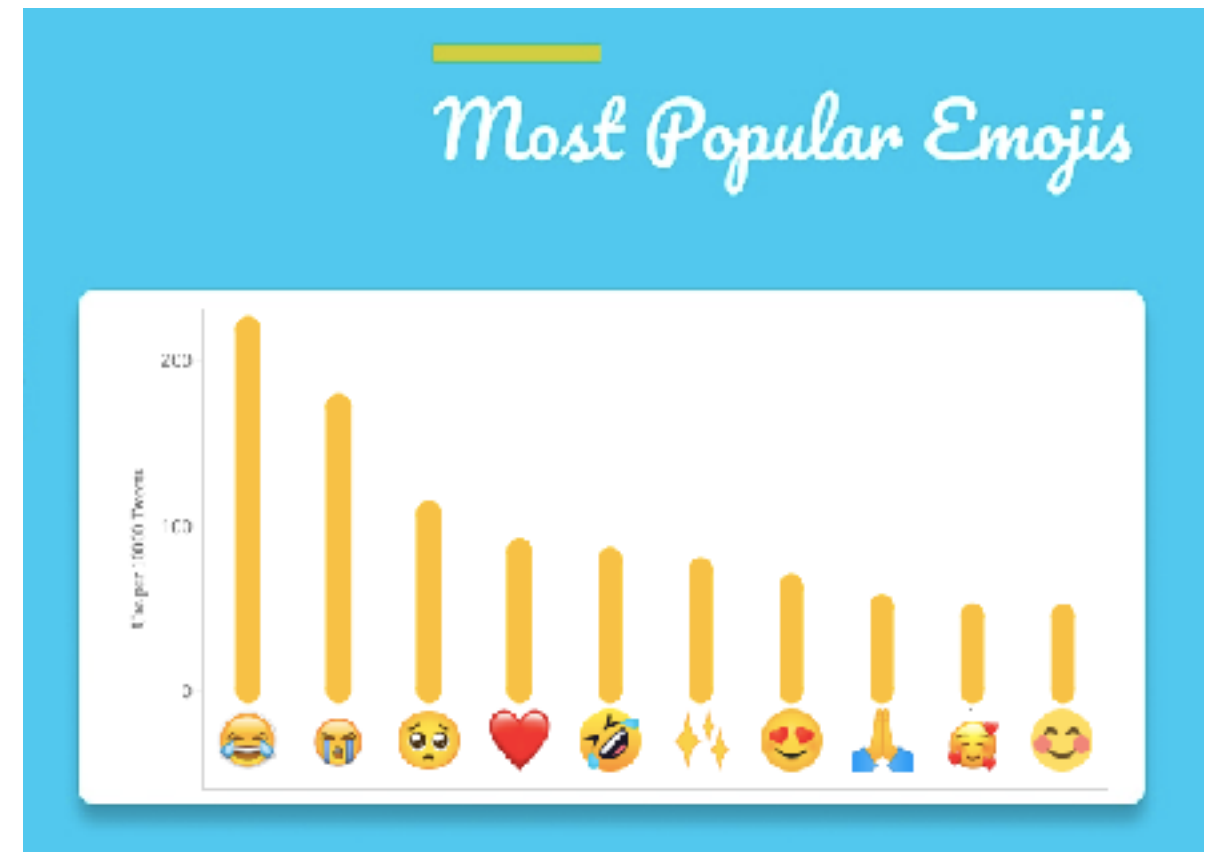
Foxx's policy to prosecute **felony shoplifting** only for thefts of more than \$1,000 led to a steep decline in charges, from 300 per month to about 70.



Source: "The Kim Foxx Effect: How Prosecutions Have Changed in Cook County," *The Pudding*, Matt Daniels

Where to find viz inspiration

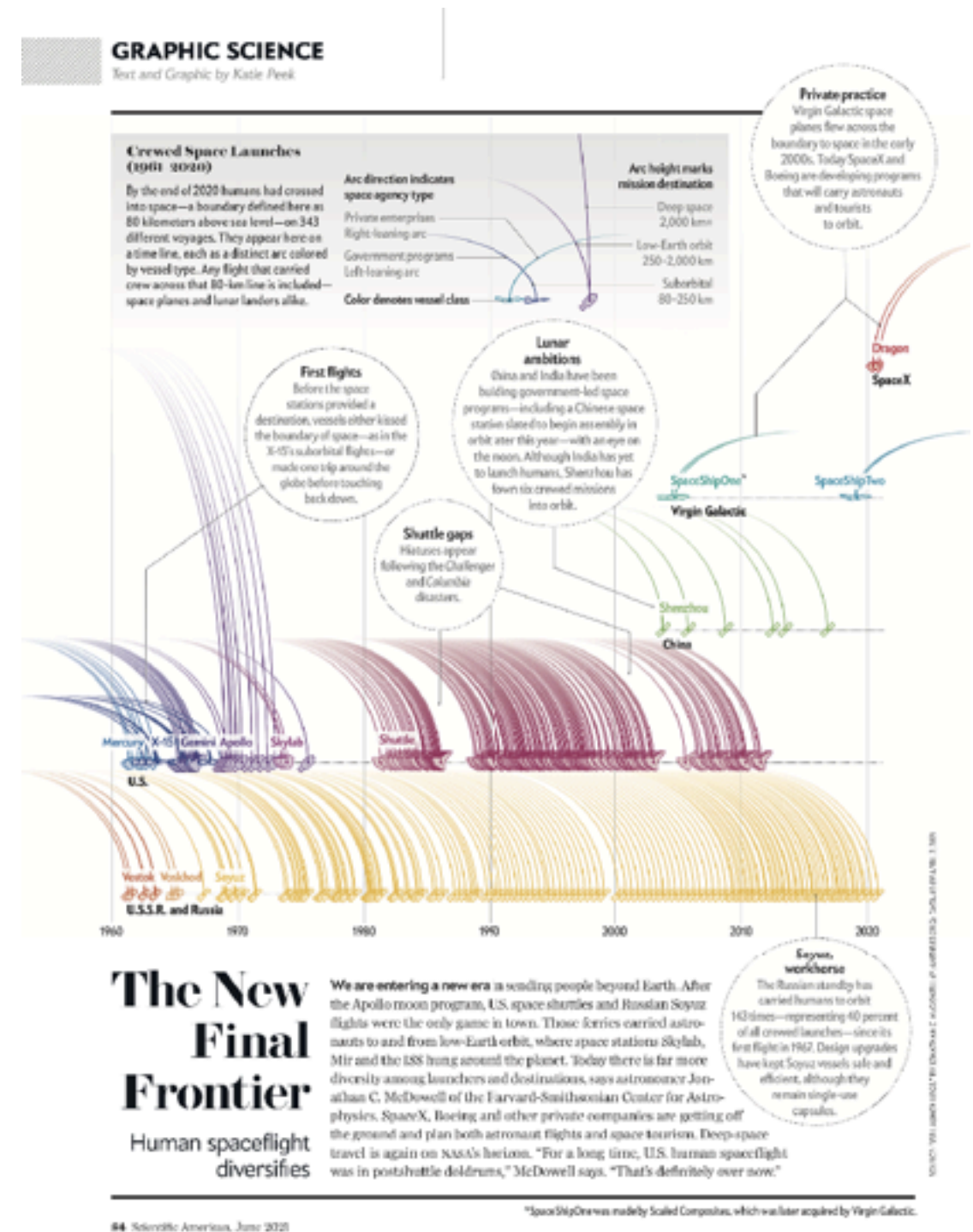
- ▶ NYT's *The Upshot*
- ▶ *The Pudding*
- ▶ Tableau's public gallery



Source: "For the Love of Emoji", Pawan Sachdeva

Where to find viz inspiration

- ▶ NYT's *The Upshot*
- ▶ *The Pudding*
- ▶ Tableau's public gallery
- ▶ Graphic science in *Scientific American*



Helpful resources

- ▶ Paul Tol's [colorblind-friendly categorical color palettes](#)
- ▶ [Data Viz Project](#): catalog of different viz types
- ▶ Gina Reynolds' [ggplot flipbook](#)
- ▶ Kira Tebbe's [Effective Data Visualization workshop](#)
- ▶ Free online book: [Fundamentals of Data Visualization](#)
- ▶ *Storytelling with Data* [books](#)

Find these resources:

*[https://github.com/laurenmarietta/
effective-data-viz-2025](https://github.com/laurenmarietta/effective-data-viz-2025)*

lauren@ischool.berkeley.edu